



PCI Express® Base Specification Revision 6.4

June 05, 2025

Copyright© 2002-2025 PCI-SIG

PCI-SIG disclaims all warranties and liability for the use of this document and the information contained herein and assumes no responsibility for any errors that may appear in this document, nor does PCI-SIG make a commitment to update the information contained herein.

This PCI Specification is provided “as is” without any warranties of any kind, including any warranty of merchantability, non-infringement, fitness for any particular purpose, or any warranty otherwise arising out of any proposal, specification, or sample. PCI-SIG disclaims all liability for infringement of proprietary rights, relating to use of information in this specification. This document itself may not be modified in any way, including by removing the copyright notice or references to PCI-SIG. No license, express or implied, by estoppel or otherwise, to any intellectual property rights is granted herein. PCI, PCI Express, PCIe, and PCI-SIG are trademarks or registered trademarks of PCI-SIG. All other product names are trademarks, registered trademarks, or servicemarks of their respective owners.

Table of Contents

1.	Introduction.....	137
1.1	An Evolving I/O Interconnect	137
1.2	PCI Express Link.....	138
1.3	PCI Express Fabric Topology	139
1.3.1	Root Complex	140
1.3.2	Endpoints	141
1.3.2.1	Legacy Endpoint Rules.....	141
1.3.2.2	PCI Express Endpoint Rules	142
1.3.2.3	Root Complex Integrated Endpoint Rules.....	142
1.3.3	Switch	143
1.3.4	Root Complex Event Collector.....	144
1.3.5	PCI Express to PCI/PCI-X Bridge.....	144
1.4	Hardware/Software Model for Discovery, Configuration and Operation	144
1.5	PCI Express Layering Overview	145
1.5.1	Transaction Layer	146
1.5.2	Data Link Layer.....	146
1.5.3	Physical Layer.....	146
1.5.4	Layer Functions and Services	147
1.5.4.1	Transaction Layer Services	147
1.5.4.2	Data Link Layer Services	148
1.5.4.3	Physical Layer Services	148
1.5.4.4	Inter-Layer Interfaces.....	149
1.5.4.4.1	Transaction/Data Link Interface	149
1.5.4.4.2	Data Link/Physical Interface	149
2.	Transaction Layer Specification	151
2.1	Transaction Layer Overview	151
2.1.1	Address Spaces, Transaction Types, and Usage	151
2.1.1.1	Memory Transactions.....	152
2.1.1.2	I/O Transactions	152
2.1.1.3	Configuration Transactions	153
2.1.1.4	Message Transactions	153
2.1.2	Packet Format Overview	153
2.2	Transaction Layer Protocol - Packet Definition.....	155
2.2.1	Common Packet Header Fields	155
2.2.1.1	Common Packet Header Fields for Non-Flit Mode	155
2.2.1.2	Common Packet Header Fields for Flit Mode.....	158
2.2.2	TLPs with Data Payloads - Rules.....	181
2.2.3	TLP Digest Rules - Non-Flit Mode Only	184
2.2.4	Routing and Addressing Rules.....	185
2.2.4.1	Address Based Routing Rules	185
2.2.4.2	ID Based Routing Rules	188
2.2.5	First/Last DW Byte Enables Rules	192
2.2.5.1	Byte Enable Rules for Non-Flit Mode.....	192
2.2.5.2	Byte Enable Rules for Flit Mode	195
2.2.6	Transaction Descriptor.....	195

2.2.6.1	Overview	195
2.2.6.2	Transaction Descriptor - Transaction ID Field	196
2.2.6.2.1	Tag Rules for Non-UIO Channel Operation	197
2.2.6.2.2	Tag Rules for UIO Channel Operation	203
2.2.6.3	Transaction Descriptor - Attributes Field	203
2.2.6.4	Relaxed Ordering and ID-Based Ordering Attributes	204
2.2.6.5	No Snoop Attribute	205
2.2.6.6	Transaction Descriptor - Traffic Class Field	205
2.2.7	Memory, I/O, and Configuration Request Rules	206
2.2.7.1	Non-Flit Mode	206
2.2.7.1.1	TPH Rules – Non-Flit Mode	209
2.2.7.2	Flit Mode	212
2.2.8	Message Request Rules	214
2.2.8.1	INTx Interrupt Signaling - Rules	216
2.2.8.2	Power Management Messages	220
2.2.8.3	Error Signaling Messages	221
2.2.8.4	Locked Transactions Support	222
2.2.8.5	Slot Power Limit Support	223
2.2.8.6	Vendor-Defined Messages	224
2.2.8.6.1	PCI-SIG Defined VDMs	226
2.2.8.6.2	Device Readiness Status (DRS) Message	227
2.2.8.6.3	Function Readiness Status Message (FRS Message)	228
2.2.8.6.4	Hierarchy ID Message	229
2.2.8.7	Ignored Messages	231
2.2.8.8	Latency Tolerance Reporting (LTR) Message	232
2.2.8.9	Optimized Buffer Flush/Fill (OBFF) Message	233
2.2.8.10	Precision Time Measurement (PTM) Messages	234
2.2.8.11	Integrity and Data Encryption (IDE) Messages	237
2.2.9	Completion Rules	240
2.2.9.1	Completion Rules for Non-Flit Mode	241
2.2.9.2	Completion Rules for Flit Mode	243
2.2.10	TLP Prefix Rules	246
2.2.10.1	TLP Prefix General Rules - Non-Flit Mode	246
2.2.10.2	Local TLP Prefix Processing	246
2.2.10.2.1	Vendor Defined Local TLP Prefix	247
2.2.10.3	Flit Mode Local TLP Prefix	247
2.2.10.4	End-End TLP Prefix Processing - Non-Flit Mode	248
2.2.10.4.1	Vendor Defined End-End TLP Prefix	250
2.2.10.4.2	Root Ports with End-End TLP Prefix Supported	250
2.2.11	OHC-E Rules - Flit Mode	250
2.3	Handling of Received TLPs	252
2.3.1	Request Handling Rules	255
2.3.1.1	Data Return for Non-UIO Read Requests	261
2.3.1.2	UIO Read Completions	266
2.3.1.3	UIO Write Completions	267
2.3.2	Completion Handling Rules	267
2.4	Transaction Ordering	269
2.4.1	Transaction Ordering Rules for TLPs not using UIO or Flow-Through IDE Streams	269
2.4.2	Ordering Rules for UIO	275
2.4.3	Update Ordering and Granularity Observed by a Read Transaction	276
2.4.3.1	Ordering and Granularity for Non-UIO Reads	276

2.4.3.2	Ordering and Granularity for UIO Reads	277
2.4.4	Update Ordering and Granularity Provided by a Write Transaction	277
2.4.4.1	Ordering and Granularity for Non-UIO Writes.....	277
2.4.4.2	Ordering and Granularity for UIO Writes	277
2.5	Virtual Channel (VC) Mechanism	278
2.5.1	Virtual Channel Identification (VC ID).....	282
2.5.2	TC to VC Mapping	282
2.5.3	VC and TC Rules.....	284
2.6	Ordering and Receive Buffer Flow Control.....	285
2.6.1	Flow Control (FC) Rules.....	286
2.6.1.1	FC Information Tracked by Transmitter	294
2.6.1.2	FC Information Tracked by Receiver.....	300
2.7	End-to-End Data Integrity	307
2.7.1	ECRC Rules.....	307
2.7.2	Error Forwarding (Data Poisoning).....	312
2.7.2.1	Rules For Use of Data Poisoning.....	313
2.8	Completion Timeout Mechanism	314
2.9	Link Status Dependencies.....	315
2.9.1	Transaction Layer Behavior in DL_Down Status.....	315
2.9.2	Transaction Layer Behavior in DL_Up Status.....	316
2.9.3	Transaction Layer Behavior During Downstream Port Containment	317
3.	Data Link Layer Specification	319
3.1	Data Link Layer Overview	319
3.2	Data Link Control and Management State Machine	320
3.2.1	Data Link Control and Management State Machine Rules	321
3.3	Data Link Feature Exchange.....	324
3.4	Flow Control Initialization Protocol	326
3.4.1	Flow Control Initialization State Machine Rules	326
3.4.2	Scaled Flow Control	334
3.5	Data Link Layer Packets (DLLPs).....	334
3.5.1	Data Link Layer Packet Rules.....	335
3.6	Data Integrity Mechanisms	344
3.6.1	Introduction	344
3.6.2	LCRC, TLP Sequence Number, and Retry Management (TLP Transmitter)	345
3.6.2.1	LCRC and TLP Sequence Number Rules (TLP Transmitter).....	345
3.6.2.2	Handling of Received DLLPs (Non-Flit Mode)	353
3.6.2.3	Handling of Received DLLPs (Flit Mode)	354
3.6.3	LCRC and TLP Sequence Number (TLP Receiver) (Non-Flit Mode)	355
3.6.3.1	LCRC and TLP Sequence Number Rules (TLP Receiver)	355
4.	Physical Layer Logical Block.....	361
4.1	Introduction.....	361
4.2	Logical Sub-block.....	361
4.2.1	8b/10b Encoding for 2.5 GT/s and 5.0 GT/s Data Rates	363
4.2.1.1	Symbol Encoding	363
4.2.1.1.1	Serialization and De-serialization of Data.....	364
4.2.1.1.2	Special Symbols for Framing and Link Management (K Codes)	365
4.2.1.1.3	8b/10b Decode Rules	366
4.2.1.2	Framing and Application of Symbols to Lanes	367
4.2.1.2.1	Framing and Application of Symbols to Lanes for TLPs and DLLPs in Non-Flit Mode	367

4.2.1.3	Data Scrambling.....	370
4.2.2	128b/130b Encoding for 8.0 GT/s, 16.0 GT/s, and 32.0 GT/s Data Rates	371
4.2.2.1	Lane Level Encoding	372
4.2.2.2	Ordered Set Blocks.....	373
4.2.2.2.1	Block Alignment	373
4.2.2.3	Data Blocks.....	374
4.2.2.3.1	Framing Tokens in Non-Flit Mode.....	374
4.2.2.3.2	Transmitter Framing Requirements in Non-Flit Mode.....	379
4.2.2.3.3	Receiver Framing Requirements in Non-Flit Mode	380
4.2.2.3.4	Receiver Framing Requirements in Flit Mode	382
4.2.2.3.5	Recovery from Framing Errors in Non-Flit Mode and Flit Mode	383
4.2.2.4	Scrambling in Non-Flit Mode and Flit Mode	383
4.2.2.5	Precoding.....	389
4.2.2.5.1	Precoding at 32.0 GT/s Data Rate.....	390
4.2.2.6	Loopback with 128b/130b Code in Non-Flit Mode and Flit Mode.....	392
4.2.3	Flit Mode Operation	392
4.2.3.1	1b/1b Encoding for 64.0 GT/s and higher Data Rates	392
4.2.3.1.1	PAM4 Signaling.....	394
4.2.3.1.2	1b/1b Scrambling.....	395
4.2.3.1.3	Gray Coding at 64.0 GT/s and Higher Data Rates	396
4.2.3.1.4	Precoding at 64.0 GT/s and Higher Data Rates.....	397
4.2.3.1.5	Ordered Set Blocks at 64.0 GT/s and Higher Data Rates.....	400
4.2.3.1.6	Alignment at Block/ Flit Level for 1b/1b Encoding	401
4.2.3.2	Processing of Ordered Sets During Flit Mode Data Stream	402
4.2.3.3	Data Stream in Flit Mode.....	404
4.2.3.4	Bytes in Flit Layout.....	410
4.2.3.4.1	TLP Bytes in Flit	410
4.2.3.4.2	DLP Bytes in Flit.....	413
4.2.3.4.2.1	Flit Sequence Number and Retry Mechanism	417
4.2.3.4.2.1.1	IDLE Flit Handshake Phase	423
4.2.3.4.2.1.2	Sequence Number Handshake Phase.....	424
4.2.3.4.2.1.3	Normal Flit Exchange Phase	425
4.2.3.4.2.1.4	Received Ack and Nak Processing.....	427
4.2.3.4.2.1.5	Ack, Nak, and Discard Rules	428
4.2.3.4.2.1.6	Flit Replay Scheduling	435
4.2.3.4.2.1.7	Flit Replay Transmit Rules	436
4.2.3.4.3	NOP Flit Payload.....	443
4.2.3.4.3.1	NOP.Empty Flit	443
4.2.3.4.3.2	NOP.Debug Flit	443
4.2.3.4.3.2.1	PCI-SIG Defined Debug Chunk Opcode Values	446
4.2.3.4.3.2.2	Empty Debug Chunk	447
4.2.3.4.3.2.3	Start Capture Trigger Debug Chunk	448
4.2.3.4.3.2.4	Stop Capture Trigger Debug Chunk.....	448
4.2.3.4.3.2.5	FC Information Tracked by Transmitter Debug Chunk.....	449
4.2.3.4.3.2.6	FC Information Tracked by Receiver Debug Chunk	451
4.2.3.4.3.2.7	Flit Mode Transmitter Retry Flags and Counters Debug Chunk.....	452
4.2.3.4.3.2.8	Flit Mode Receiver Retry Flags and Counters Debug Chunk	453
4.2.3.4.3.2.9	Buffer Occupancy Debug Chunk	454
4.2.3.4.3.2.10	Link Debug Request Debug Chunk.....	454
4.2.3.4.3.3	NOP.Vendor Flit	455
4.2.3.4.4	CRC Bytes in Flit	455

4.2.3.4.5	ECC Bytes in Flit.....	456
4.2.3.4.6	Ordered Set insertion in Data Stream in Flit Mode	462
4.2.4	Link Equalization Procedure for 8.0 GT/s and Higher Data Rates	463
4.2.4.1	Rules for Transmitter Coefficients	475
4.2.4.2	Encoding of Presets.....	476
4.2.5	Link Initialization and Training	477
4.2.5.1	Training Sequences.....	478
4.2.5.2	Alternate Protocol Negotiation.....	499
4.2.5.3	Electrical Idle Sequences (EIOS and EIEOS).....	502
4.2.5.4	Inferring Electrical Idle.....	507
4.2.5.5	Lane Polarity Inversion	508
4.2.5.6	Fast Training Sequence (FTS).....	509
4.2.5.7	Start of Data Stream Ordered Set (SDS Ordered Set)	511
4.2.5.8	Link Error Recovery	511
4.2.5.9	Reset	512
4.2.5.9.1	Fundamental Reset	512
4.2.5.9.2	Hot Reset	513
4.2.5.10	Link Data Rate Negotiation	513
4.2.5.11	Link Width and Lane Sequence Negotiation.....	513
4.2.5.11.1	Required and Optional Port Behavior	513
4.2.5.12	Lane-to-Lane De-skew	514
4.2.5.13	Lane vs. Link Training.....	515
4.2.6	Link Training and Status State Machine (LTSSM) Descriptions	515
4.2.6.1	Detect Overview	516
4.2.6.2	Polling Overview	516
4.2.6.3	Configuration Overview	516
4.2.6.4	Recovery Overview.....	516
4.2.6.5	L0 Overview	517
4.2.6.6	L0s Overview	517
4.2.6.7	L0p Overview.....	517
4.2.6.7.1	Link Management DLLP	522
4.2.6.8	L1 Overview	525
4.2.6.9	L2 Overview	525
4.2.6.10	Disabled Overview	526
4.2.6.11	Loopback Overview	526
4.2.6.12	Hot Reset Overview.....	527
4.2.7	Link Training and Status State Rules.....	527
4.2.7.1	Detect.....	529
4.2.7.1.1	Detect.Quiet	529
4.2.7.1.2	Detect.Active	530
4.2.7.2	Polling.....	531
4.2.7.2.1	Polling.Active.....	531
4.2.7.2.2	Polling.Compliance	533
4.2.7.2.3	Polling.Configuration.....	538
4.2.7.2.4	Polling.Speed	538
4.2.7.3	Configuration	539
4.2.7.3.1	Configuration.Linkwidth.Start.....	539
4.2.7.3.1.1	Downstream Lanes	539
4.2.7.3.1.2	Upstream Lanes	541
4.2.7.3.2	Configuration.Linkwidth.Accept	543
4.2.7.3.2.1	Downstream Lanes	543

4.2.7.3.2.2	Upstream Lanes	544
4.2.7.3.3	Configuration.Lanenum.Accept	547
4.2.7.3.3.1	Downstream Lanes	548
4.2.7.3.3.2	Upstream Lanes	549
4.2.7.3.4	Configuration.Lanenum.Wait	550
4.2.7.3.4.1	Downstream Lanes	551
4.2.7.3.4.2	Upstream Lanes	551
4.2.7.3.5	Configuration.Complete	551
4.2.7.3.5.1	Downstream Lanes	552
4.2.7.3.5.2	Upstream Lanes	553
4.2.7.3.6	Configuration.Idle	555
4.2.7.4	Recovery	559
4.2.7.4.1	Recovery.RcvrLock	559
4.2.7.4.2	Recovery.Equalization	565
4.2.7.4.2.1	Downstream Lanes	566
4.2.7.4.2.1.1	Phase 1 of Transmitter Equalization	567
4.2.7.4.2.1.2	Phase 2 of Transmitter Equalization	569
4.2.7.4.2.1.3	Phase 3 of Transmitter Equalization	570
4.2.7.4.2.2	Upstream Lanes	572
4.2.7.4.2.2.1	Phase 0 of Transmitter Equalization	573
4.2.7.4.2.2.2	Phase 1 of Transmitter Equalization	575
4.2.7.4.2.2.3	Phase 2 of Transmitter Equalization	575
4.2.7.4.2.2.4	Phase 3 of Transmitter Equalization	578
4.2.7.4.3	Recovery.Speed	579
4.2.7.4.4	Recovery.RcvrCfg	580
4.2.7.4.5	Recovery.Idle	587
4.2.7.5	L0	590
4.2.7.6	L0s	592
4.2.7.6.1	Receiver L0s	592
4.2.7.6.1.1	Rx_L0s.Entry	592
4.2.7.6.1.2	Rx_L0s.Idle	592
4.2.7.6.1.3	Rx_L0s.FTS	593
4.2.7.6.2	Transmitter L0s	593
4.2.7.6.2.1	Tx_L0s.Entry	593
4.2.7.6.2.2	Tx_L0s.Idle	594
4.2.7.6.2.3	Tx_L0s.FTS	594
4.2.7.7	L1	595
4.2.7.7.1	L1.Entry	595
4.2.7.7.2	L1.Idle	595
4.2.7.8	L2	597
4.2.7.8.1	L2.Idle	597
4.2.7.8.2	L2.TransmitWake	598
4.2.7.9	Disabled	598
4.2.7.10	Loopback	599
4.2.7.10.1	Loopback.Entry	599
4.2.7.10.2	Loopback.Active	603
4.2.7.10.3	Loopback.Exit	605
4.2.7.11	Hot Reset	606
4.2.8	Clock Tolerance Compensation	607
4.2.8.1	SKP Ordered Set for 8b/10b Encoding	608
4.2.8.2	SKP Ordered Set for 128b/130b Encoding	608

4.2.8.3	SKP Ordered Set for 1b/1b Encoding	612
4.2.8.4	Rules for Transmitters	616
4.2.8.5	Rules for Receivers	619
4.2.8.6	SKP Ordered Set Usage with Optical Retimer Solutions	620
4.2.9	Compliance Pattern in 8b/10b Encoding	622
4.2.10	Modified Compliance Pattern in 8b/10b Encoding	623
4.2.11	Compliance Pattern in 128b/130b Encoding	625
4.2.12	Modified Compliance Pattern in 128b/130b Encoding	627
4.2.13	Jitter Measurement Pattern in 128b/130b	627
4.2.14	Compliance Pattern in 1b/1b Encoding	628
4.2.15	Modified Compliance Pattern in 1b/1b Encoding	629
4.2.16	Jitter Measurement Pattern in 1b/1b Encoding	629
4.2.17	Toggle Patterns in 1b/1b encoding	630
4.2.18	Lane Margining at Receiver	630
4.2.18.1	Receiver Number, Margin Type, Usage Model, and Margin Payload Fields	631
4.2.18.1.1	Step Margin Execution Status	636
4.2.18.1.2	Margin Payload for Step Margin Commands	636
4.2.18.2	Margin Command and Response Flow	637
4.2.18.3	Flit Mode 8.0 GT/s Margining Behavior	640
4.2.18.4	Receiver Margin Testing Requirements	640
4.3	Retimers	645
4.3.1	Retimer Requirements	646
4.3.2	Supported Retimer Topologies	647
4.3.3	Variables	650
4.3.4	Receiver Impedance Propagation Rules	651
4.3.5	Switching Between Modes	651
4.3.6	Activation Rules	651
4.3.7	Forwarding Rules	655
4.3.7.1	Forwarding Type Rules	658
4.3.7.2	Orientation, Lane Numbers, and Data Stream Mode Rules	658
4.3.7.3	Electrical Idle Exit Rules	659
4.3.7.4	Data Rate Change and Determination Rules	662
4.3.7.5	Electrical Idle Entry Rules	663
4.3.7.6	Transmitter Settings Determination Rules	664
4.3.7.7	Ordered Set Modification Rules	667
4.3.7.8	DLLP, TLP, Logical Idle, and Flit Modification Rules	669
4.3.7.9	8b/10b Encoding Rules	669
4.3.7.10	8b/10b Scrambling Rules	670
4.3.7.11	Hot Reset Rules	670
4.3.7.12	Disable Link Rules	671
4.3.7.13	Loopback	671
4.3.7.14	Compliance Receive Rules	672
4.3.7.15	Enter Compliance Rules	674
4.3.8	Execution Mode Rules	676
4.3.8.1	CompLoadBoard Rules	676
4.3.8.1.1	CompLoadBoard.Entry	677
4.3.8.1.2	CompLoadBoard.Pattern	677
4.3.8.1.3	CompLoadBoard.Exit	678
4.3.8.2	Link Equalization Rules	678
4.3.8.2.1	Downstream Lanes	679
4.3.8.2.1.1	Phase 1	679

4.3.8.2.1.2	Phase 2	679
4.3.8.2.1.3	Phase 3 Active	679
4.3.8.2.1.4	Phase 3 Passive	680
4.3.8.2.2	Upstream Lanes	680
4.3.8.2.2.1	Phase 0	680
4.3.8.2.2.2	Phase 1 Active	680
4.3.8.2.2.3	Phase 2 Active	681
4.3.8.2.2.4	Phase 2 Passive	681
4.3.8.2.2.5	Phase 3	681
4.3.8.2.3	Force Timeout	682
4.3.8.3	Follower Loopback.....	682
4.3.8.3.1	Follower Loopback.Entry	682
4.3.8.3.2	Follower Loopback.Active	683
4.3.8.3.3	Follower Loopback.Exit	683
4.3.9	Retimer Latency	683
4.3.9.1	Measurement.....	683
4.3.9.2	Maximum Limit on Retimer Latency	684
4.3.9.3	Impacts on Upstream and Downstream Ports.....	684
4.3.10	SRIS.....	684
4.3.11	L1 PM Substates Support	686
4.3.12	Retimer Configuration Parameters.....	687
4.3.12.1	Global Parameters.....	688
4.3.12.2	Per Physical Pseudo Port Parameters	688
4.3.13	In Band Register Access.....	689
5.	Power Management	691
5.1	Overview.....	691
5.2	Link State Power Management	692
5.3	PCI-PM Software Compatible Mechanisms	696
5.3.1	Device Power Management States (D-States) of a Function	696
5.3.1.1	D0 State.....	697
5.3.1.2	D1 State.....	697
5.3.1.3	D2 State.....	697
5.3.1.4	D3 State.....	698
5.3.1.4.1	D3 _{Hot} State.....	698
5.3.1.4.2	D3 _{Cold} State	700
5.3.2	PM Software Control of the Link Power Management State	701
5.3.2.1	Entry into the L1 State.....	702
5.3.2.2	Exit from L1 State	704
5.3.2.3	Entry into the L2/L3 Ready State	705
5.3.3	Power Management Event Mechanisms	705
5.3.3.1	Motivation.....	705
5.3.3.2	Link Wakeup	706
5.3.3.2.1	PME Synchronization	707
5.3.3.3	PM_PME Messages	708
5.3.3.3.1	PM_PME “Backpressure” Deadlock Avoidance	709
5.3.3.4	PME Rules	709
5.3.3.5	PM_PME Delivery State Machine	710
5.4	Native PCI Express Power Management Mechanisms	711
5.4.1	Active State Power Management (ASPM)	711
5.4.1.1	L0s ASPM State	713

5.4.1.1.1	Entry into the L0s State	715
5.4.1.1.2	Exit from the L0s State	715
5.4.1.2	ASPM L0p State.....	716
5.4.1.3	ASPM L1 State.....	716
5.4.1.3.1	ASPM Entry into the L1 State	717
5.4.1.3.2	Exit from the L1 State	722
5.4.1.4	ASPM Configuration	724
5.4.1.4.1	Software Flow for Enabling or Disabling ASPM	727
5.5	L1 PM Substates	728
5.5.1	Entry conditions for L1 PM Substates and L1.0 Requirements	732
5.5.2	L1.1 Requirements	733
5.5.2.1	Exit from L1.1.....	733
5.5.3	L1.2 Requirements	734
5.5.3.1	L1.2.Entry.....	735
5.5.3.2	L1.2.Idle	736
5.5.3.3	L1.2.Exit	736
5.5.3.3.1	Exit from L1.2.....	737
5.5.4	L1 PM Substates Configuration.....	738
5.5.5	L1 PM Substates Timing Parameters	738
5.5.6	Link Activation	739
5.6	Auxiliary Power Support	740
5.7	Power Management System Messages and DLLPs	741
5.8	PCI Function Power State Transitions	742
5.9	State Transition Recovery Time Requirements.....	742
5.10	SR-IOV Power Management	742
5.10.1	VF Device Power Management States	743
5.10.2	PF Device Power Management States	743
5.11	PCI Bridges and Power Management	744
5.11.1	Switches and PCI Express to PCI Bridges	745
5.12	Power Management Events	745
6.	System Architecture	747
6.1	Interrupt and PME Support.....	747
6.1.1	Rationale for PCI Express Interrupt Model	747
6.1.2	PCI-compatible INTx Emulation	748
6.1.3	INTx Emulation Software Model	748
6.1.4	MSI and MSI-X Operation	748
6.1.4.1	MSI Configuration.....	749
6.1.4.2	MSI-X Configuration	750
6.1.4.3	Enabling Operation	751
6.1.4.4	Sending Messages	752
6.1.4.5	Per-vector Masking and Function Masking	752
6.1.4.6	Hardware/Software Synchronization.....	753
6.1.4.7	Message Transaction Reception and Ordering Requirements.....	755
6.1.5	PME Support.....	755
6.1.6	Native PME Software Model.....	755
6.1.7	Legacy PME Software Model	756
6.1.8	Operating System Power Management Notification	756
6.1.9	PME Routing Between PCI Express and PCI Hierarchies	756
6.2	Error Signaling and Logging	757
6.2.1	Scope	757

6.2.2	Error Classification	757
6.2.2.1	Correctable Errors	758
6.2.2.2	Uncorrectable Errors	759
6.2.2.2.1	Fatal Errors	759
6.2.2.2.2	Non-Fatal Errors	759
6.2.3	Error Signaling.....	759
6.2.3.1	Completion Status	760
6.2.3.2	Error Messages	760
6.2.3.2.1	Uncorrectable Error Severity Programming (Advanced Error Reporting)	761
6.2.3.2.2	Masking Individual Errors	761
6.2.3.2.3	Error Pollution	762
6.2.3.2.4	Advisory Non-Fatal Error Cases	762
6.2.3.2.4.1	Completer Sending a Completion with UR/CA Status.....	763
6.2.3.2.4.2	Intermediate Receiver.....	763
6.2.3.2.4.3	Ultimate PCI Express Receiver of a Poisoned TLP or IDE TLP with PCRC Check Failed	764
6.2.3.2.4.4	Requester with Completion Timeout.....	764
6.2.3.2.4.5	Receiver of an Unexpected Completion.....	765
6.2.3.2.5	Requester Receiving a Completion with UR/CA Status	765
6.2.3.3	Error Forwarding (Data Poisoning).....	765
6.2.3.4	Optional Error Checking	765
6.2.4	Error Logging.....	766
6.2.4.1	Root Complex Considerations (Advanced Error Reporting).....	767
6.2.4.1.1	Error Source Identification	767
6.2.4.1.2	Interrupt Generation	767
6.2.4.1.3	RC ECS Handling.....	768
6.2.4.2	Multiple Error Handling (Advanced Error Reporting Extended Capability)	768
6.2.4.2.1	Multiple Error Handling in VFs	770
6.2.4.3	Advisory Non-Fatal Error Logging	771
6.2.4.4	End-End TLP Prefix Logging - Non-Flit Mode	771
6.2.5	Sequence of Device Error Signaling and Logging Operations.....	772
6.2.6	Error Message Controls	774
6.2.7	Error Listing and Rules	775
6.2.7.1	Conventional PCI Mapping	780
6.2.8	Virtual PCI Bridge Error Handling.....	780
6.2.8.1	Error Message Forwarding and PCI Mapping for Bridge - Rules.....	781
6.2.9	SR-IOV Baseline Error Handling.....	781
6.2.10	Internal Errors	782
6.2.11	Downstream Port Containment (DPC)	783
6.2.11.1	DPC Interrupts	786
6.2.11.2	DPC ERR_COR Signaling	786
6.2.11.3	Root Port Programmed I/O (RP PIO) Error Controls	787
6.2.11.4	Software Triggering of DPC.....	790
6.2.11.5	DL_Active ERR_COR Signaling.....	790
6.3	Virtual Channel Support	791
6.3.1	Introduction and Scope	791
6.3.2	TC/VC Mapping and Example Usage.....	792
6.3.3	VC Arbitration	793
6.3.3.1	Traffic Flow and Switch Arbitration Model.....	794
6.3.3.2	VC Arbitration - Arbitration Between VCs.....	795
6.3.3.2.1	Strict Priority Arbitration Model	796

6.3.3.2.2	Round Robin Arbitration Model.....	796
6.3.3.3	Port Arbitration - Arbitration Within VC.....	797
6.3.3.4	Multi-Function Devices and Function Arbitration.....	797
6.3.4	Isochronous Support	800
6.3.4.1	Rules for Software Configuration	800
6.3.4.2	Rules for Requesters	801
6.3.4.3	Rules for Completers.....	801
6.3.4.4	Rules for Switches and Root Complexes	801
6.3.4.5	Rules for Multi-Function Devices	801
6.3.5	SVC and VC/MFVC Capability Coexistence	802
6.4	Device Synchronization	802
6.5	Locked Transactions	803
6.5.1	Introduction	803
6.5.2	Initiation and Propagation of Locked Transactions - Rules	804
6.5.3	Switches and Lock - Rules.....	804
6.5.4	PCI Express/PCI Bridges and Lock - Rules	805
6.5.5	Root Complex and Lock - Rules	805
6.5.6	Legacy Endpoints	805
6.5.7	PCI Express Endpoints	806
6.6	PCI Express Reset - Rules	806
6.6.1	Conventional Reset	806
6.6.2	Function Level Reset (FLR).....	809
6.7	PCI Express Native Hot-Plug	812
6.7.1	Elements of Hot-Plug	813
6.7.1.1	Indicators.....	813
6.7.1.1.1	Attention Indicator	814
6.7.1.1.2	Power Indicator	814
6.7.1.2	Manually-operated Retention Latch (MRL)	815
6.7.1.3	MRL Sensor	815
6.7.1.4	Electromechanical Interlock.....	816
6.7.1.5	Attention Button.....	816
6.7.1.6	Software User Interface.....	816
6.7.1.7	Slot Numbering	817
6.7.1.8	Power Controller	817
6.7.2	Registers Grouped by Hot-Plug Element Association.....	818
6.7.2.1	Attention Button Registers.....	818
6.7.2.2	Attention Indicator Registers	818
6.7.2.3	Power Indicator Registers	818
6.7.2.4	Power Controller Registers	818
6.7.2.5	Presence Detect Registers.....	819
6.7.2.6	MRL Sensor Registers	819
6.7.2.7	Electromechanical Interlock Registers.....	819
6.7.2.8	Command Completed Registers.....	819
6.7.2.9	Port Capabilities and Slot Information Registers	820
6.7.2.10	Hot-Plug Interrupt Control Register	820
6.7.3	PCI Express Hot-Plug Events	820
6.7.3.1	Slot Events	820
6.7.3.2	Command Completed Events.....	821
6.7.3.3	Data Link Layer State Changed Events.....	821
6.7.3.4	Software Notification of Hot-Plug Events	822
6.7.4	System Firmware Intermediary (SFI) Support.....	823

6.7.4.1	SFI ERR_COR Event Signaling	823
6.7.4.2	SFI Downstream Port Filtering (DPF).....	823
6.7.4.3	SFI CAM	824
6.7.4.4	SFI Interactions with Readiness Notifications	827
6.7.4.5	SFI Suppression of Hot-Plug Surprise Functionality	828
6.7.4.6	SFI Quarantine Mode	829
6.7.4.7	SFI Extended Capability In-Band Hiding	834
6.7.4.8	SFI Extended Capability In-Band Read-Only.....	834
6.7.5	Firmware Support for Hot-Plug	835
6.7.6	Async Removal	835
6.8	Power Budgeting Mechanism	836
6.8.1	System Power Budgeting Process Recommendations.....	838
6.8.2	Device Power Considerations	838
6.8.3	Power Limit Mechanisms	838
6.9	Slot Power Limit Control.....	840
6.10	Root Complex Topology Discovery.....	843
6.11	Link Speed Management	845
6.12	Access Control Services (ACS).....	845
6.12.1	ACS Component Capability Requirements	846
6.12.1.1	ACS Downstream Ports	846
6.12.1.2	ACS Functions in SR-IOV, SIOV, and Multi-Function Devices	849
6.12.1.3	Functions in Single-Function Devices	851
6.12.2	Interoperability	851
6.12.3	ACS Peer-to-Peer Control Interactions	851
6.12.4	ACS Enhanced Capability.....	853
6.12.5	ACS Violation Error Handling.....	854
6.12.6	ACS Redirection Impacts on Ordering Rules.....	855
6.12.6.1	Completions Passing Posted Requests	855
6.12.6.2	Requests Passing Posted Requests	856
6.13	Alternative Routing-ID Interpretation (ARI).....	857
6.14	Multicast Operations	860
6.14.1	Multicast TLP Processing	860
6.14.2	Multicast Ordering.....	862
6.14.3	Multicast Capability Structure Field Updates	863
6.14.4	MC Blocked TLP Processing.....	863
6.14.5	MC_Overlay Mechanism.....	863
6.15	Atomic Operations (AtomicOps)	866
6.15.1	AtomicOp Use Models and Benefits	867
6.15.2	AtomicOp Transaction Protocol Summary	868
6.15.3	Root Complex Support for AtomicOps	869
6.15.3.1	Root Ports with AtomicOp Completer Capabilities	869
6.15.3.2	Root Ports with AtomicOp Routing Capability.....	870
6.15.3.3	RCs with AtomicOp Requester Capabilities	870
6.15.4	Switch Support for AtomicOps	871
6.16	Dynamic Power Allocation (DPA) Capability	871
6.16.1	DPA Capability with Multi-Function Devices.....	872
6.17	TLP Processing Hints (TPH)	872
6.17.1	Processing Hints.....	872
6.17.2	Steering Tags	873
6.17.3	ST Modes of Operation.....	873
6.17.4	TPH Capability.....	874

6.18	Latency Tolerance Reporting (LTR) Mechanism.....	875
6.19	Optimized Buffer Flush/Fill (OBFF) Mechanism.....	880
6.20	PASID.....	883
6.20.1	Managing PASID Usage	883
6.20.2	PASID Information Layout	884
6.20.2.1	PASID TLP Prefix - Non-Flit Mode	884
6.20.2.2	PASID field (Flit Mode and Non-Flit Mode)	885
6.20.2.3	Execute Requested	886
6.20.2.4	Privileged Mode Requested	887
6.21	Precision Time Measurement (PTM) Mechanism.....	888
6.21.1	Introduction	888
6.21.2	PTM Link Protocol	889
6.21.3	Configuration and Operational Requirements	893
6.21.3.1	PTM Requester Role	894
6.21.3.2	PTM Responder Role	896
6.21.3.3	PTM Time Source Role - Rules Specific to Switches	897
6.22	Readiness Notifications (RN)	898
6.22.1	Device Readiness Status (DRS)	901
6.22.2	Function Readiness Status (FRS)	902
6.22.3	FRS Queuing	902
6.23	Enhanced Allocation	903
6.24	Emergency Power Reduction State	905
6.25	Hierarchy ID Message.....	908
6.26	Flattening Portal Bridge (FPB)	912
6.26.1	Introduction	912
6.26.2	Hardware and Software Requirements.....	916
6.27	Vital Product Data (VPD)	923
6.27.1	VPD Format.....	925
6.27.2	VPD Definitions.....	926
6.27.2.1	VPD Large and Small Resource Data Tags.....	926
6.27.2.2	Read-Only Fields	926
6.27.2.3	Read/Write Fields	928
6.27.2.4	VPD Example.....	928
6.28	Native PCIe Enclosure Management	929
6.29	Conventional PCI Advanced Features Operation.....	934
6.30	Data Object Exchange (DOE).....	936
6.30.1	Data Objects and Features.....	939
6.30.1.1	DOE Discovery Feature.....	941
6.30.1.2	DOE Async Message.....	943
6.30.2	Operation.....	944
6.30.3	Interrupt Generation	946
6.31	Component Measurement and Authentication (CMA-SPDM)	947
6.31.1	Removed.....	953
6.31.2	Removed.....	953
6.31.3	CMA-SPDM Rules	953
6.31.4	Secured CMA-SPDM.....	955
6.32	Deferrable Memory Write	956
6.33	Integrity & Data Encryption (IDE).....	961
6.33.1	IDE Stream and TEE State Machines.....	965
6.33.2	IDE Stream Establishment	967
6.33.3	IDE Key Management (IDE_KM)	968

6.33.4	IDE TLPs	978
6.33.5	IDE TLP Sub-Streams	996
6.33.6	IDE TLP Aggregation	1001
6.33.7	Flow-Through Selective IDE Streams	1002
6.33.8	Other IDE Rules	1003
6.34	Unordered I/O (UIO)	1006
6.34.1	UIO Rules	1006
6.35	MMIO Register Blocks	1007
6.35.1	MMIO Capabilities Register Block (MCAP)	1008
6.35.1.1	MCAP Array Register (Offset 00h)	1009
6.35.1.2	MCAP Header Register Block (Offset Varies)	1011
6.35.1.3	MMIO Mailbox Capability (MMB) (Offset: Varies)	1013
6.35.1.3.1	MMB Operation	1014
6.35.1.3.2	MMB Registers	1015
6.35.1.3.2.1	MMB Capabilities Register (Offset 00h)	1015
6.35.1.3.2.2	MMB Control Register (Offset 04h)	1017
6.35.1.3.2.3	MMB Command Register (Offset 08h)	1018
6.35.1.3.2.4	MMB Status Register (Offset 10h)	1019
6.35.1.3.2.4.1	MMB Command Return Codes	1019
6.35.1.3.2.5	MMB Payload Registers (Offset 20h)	1020
6.35.1.4	Management Message Passthrough (MMPT) Capability (Offset: Varies)	1021
6.35.1.4.1	MMPT Registers	1021
6.35.1.4.1.1	MMPT Capabilities Register (Offset 00h)	1021
6.35.1.4.1.2	MMPT Control Register (Offset 04h)	1022
6.35.1.4.1.3	MMPT Receive Message Notification Register (Offset 08h)	1023
6.35.2	MMIO Designated Vendor-Specific Register Block (MDVS)	1023
6.35.2.1	MDVS Register Block Header Register 1 (Offset 00h)	1024
6.35.3	MDVS Register Block Header Register 2 (Offset 04h)	1025
6.35.4	MDVS Register Block Header Register 3 (Offset 08h)	1025
6.36	MMB Command Interface	1026
6.36.1	Management Message Passthrough (MMPT)	1026
6.36.1.1	MMPT Send Message (Opcode 0100h)	1027
6.36.1.1.1	MMPT Send Message Operation	1028
6.36.1.2	MMPT Receive Message (Opcode 0101h)	1029
6.36.1.2.1	MMPT Receive Message Operation	1030
6.37	Debug Over Link	1030
6.37.1	NOP Flit	1030
7.	Software Initialization and Configuration	1033
7.1	Configuration Topology	1033
7.2	PCI Express Configuration Mechanisms	1034
7.2.1	PCI-compatible Configuration Mechanism	1035
7.2.2	PCI Express Enhanced Configuration Access Mechanism (ECAM)	1036
7.2.2.1	Host Bridge Requirements	1039
7.2.2.2	PCI Express Device Requirements	1039
7.2.3	Root Complex Register Block (RCRB)	1040
7.3	Configuration Transaction Rules	1040
7.3.1	Device Number	1040
7.3.2	Configuration Transaction Addressing	1041
7.3.3	Configuration Request Routing Rules	1041
7.3.4	PCI Special Cycles	1043

7.4	Configuration Register Types.....	1043
7.5	PCI and PCIe Capabilities Required by the Base Spec for all Ports.....	1045
7.5.1	PCI-Compatible Configuration Registers	1045
7.5.1.1	Type 0/1 Common Configuration Space	1045
7.5.1.1.1	Vendor ID Register (Offset 00h)	1046
7.5.1.1.2	Device ID Register (Offset 02h).....	1047
7.5.1.1.3	Command Register (Offset 04h)	1047
7.5.1.1.4	Status Register (Offset 06h)	1050
7.5.1.1.5	Revision ID Register (Offset 08h)	1052
7.5.1.1.6	Class Code Register (Offset 09h).....	1052
7.5.1.1.7	Cache Line Size Register (Offset 0Ch)	1053
7.5.1.1.8	Latency Timer Register (Offset 0Dh).....	1053
7.5.1.1.9	Header Type Register (Offset 0Eh)	1053
7.5.1.1.10	BIST Register (Offset 0Fh)	1054
7.5.1.1.11	Capabilities Pointer (Offset 34h)	1055
7.5.1.1.12	Interrupt Line Register (Offset 3Ch).....	1055
7.5.1.1.13	Interrupt Pin Register (Offset 3Dh)	1056
7.5.1.1.14	Error Registers	1056
7.5.1.2	Type 0 Configuration Space Header	1057
7.5.1.2.1	Base Address Registers (Offset 10h - 24h)	1057
7.5.1.2.2	Cardbus CIS Pointer Register (Offset 28h).....	1061
7.5.1.2.3	Subsystem Vendor ID Register/Subsystem ID Register (Offset 2Ch/2Eh)	1061
7.5.1.2.4	Expansion ROM Base Address Register (Offset 30h)	1062
7.5.1.2.5	Min_Gnt Register/Max_Lat Register (Offset 3Eh/3Fh)	1065
7.5.1.3	Type 1 Configuration Space Header	1065
7.5.1.3.1	Type 1 Base Address Registers (Offset 10h-14h)	1067
7.5.1.3.2	Primary Bus Number Register (Offset 18h)	1067
7.5.1.3.3	Secondary Bus Number Register (Offset 19h).....	1067
7.5.1.3.4	Subordinate Bus Number Register (Offset 1Ah)	1067
7.5.1.3.5	Secondary Latency Timer (Offset 1Bh)	1067
7.5.1.3.6	I/O Base/I/O Limit Registers(Offset 1Ch/1Dh)	1068
7.5.1.3.7	Secondary Status Register (Offset 1Eh).....	1068
7.5.1.3.8	Memory Base Register/Memory Limit Register(Offset 20h/22h).....	1070
7.5.1.3.9	64-bit Memory Base/64-bit Memory Limit Registers (Offset 24h/26h) and 64-bit Base Upper 32 Bits/64-bit Limit Upper 32 Bits Registers (Offset 28h/2Ch).....	1070
7.5.1.3.10	64-bit Base Upper 32 Bits/64-bit Limit Upper 32 Bits Registers (Offset 28h/2Ch).....	1071
7.5.1.3.11	I/O Base Upper 16 Bits/I/O Limit Upper 16 Bits Registers (Offset 30h/32h)	1071
7.5.1.3.12	Expansion ROM Base Address Register (Offset 38h)	1071
7.5.1.3.13	Bridge Control Register (Offset 3Eh).....	1072
7.5.2	PCI Power Management Capability Structure	1074
7.5.2.1	Power Management Capabilities Register (Offset 00h)	1075
7.5.2.2	Power Management Control/Status Register (Offset 04h)	1077
7.5.2.3	Power Management Data Register (Offset 07h)	1078
7.5.3	PCI Express Capability Structure	1080
7.5.3.1	PCI Express Capability List Register (Offset 00h)	1081
7.5.3.2	PCI Express Capabilities Register (Offset 02h)	1082
7.5.3.3	Device Capabilities Register (Offset 04h)	1084
7.5.3.4	Device Control Register (Offset 08h).....	1088
7.5.3.5	Device Status Register (Offset 0Ah)	1095
7.5.3.6	Link Capabilities Register (Offset 0Ch)	1096
7.5.3.7	Link Control Register (Offset 10h).....	1100

7.5.3.8	Link Status Register (Offset 12h)	1107
7.5.3.9	Slot Capabilities Register (Offset 14h)	1110
7.5.3.10	Slot Control Register (Offset 18h)	1112
7.5.3.11	Slot Status Register (Offset 1Ah)	1115
7.5.3.12	Root Control Register (Offset 1Ch)	1117
7.5.3.13	Root Capabilities Register (Offset 1Eh)	1119
7.5.3.14	Root Status Register (Offset 20h)	1119
7.5.3.15	Device Capabilities 2 Register (Offset 24h)	1121
7.5.3.16	Device Control 2 Register (Offset 28h)	1126
7.5.3.17	Device Status 2 Register (Offset 2Ah)	1129
7.5.3.18	Link Capabilities 2 Register (Offset 2Ch)	1130
7.5.3.19	Link Control 2 Register (Offset 30h)	1133
7.5.3.20	Link Status 2 Register (Offset 32h)	1136
7.5.3.21	Slot Capabilities 2 Register (Offset 34h)	1140
7.5.3.22	Slot Control 2 Register (Offset 38h)	1140
7.5.3.23	Slot Status 2 Register (Offset 3Ah)	1141
7.6	PCI Express Extended Capabilities	1141
7.6.1	Extended Capabilities in Configuration Space	1141
7.6.2	Extended Capabilities in the Root Complex Register Block	1141
7.6.3	PCI Express Extended Capability Header	1142
7.7	PCI and PCIe Capabilities Required by the Base Spec in Some Situations	1142
7.7.1	MSI Capability Structures	1142
7.7.1.1	MSI Capability Header (Offset 00h)	1144
7.7.1.2	Message Control Register for MSI (Offset 02h)	1145
7.7.1.3	Message Address Register for MSI (Offset 04h)	1147
7.7.1.4	Message Upper Address Register for MSI (Offset 08h)	1147
7.7.1.5	Message Data Register for MSI (Offset 08h or 0Ch)	1148
7.7.1.6	Extended Message Data Register for MSI (Optional)	1148
7.7.1.7	Mask Bits Register for MSI (Offset 0Ch or 10h)	1149
7.7.1.8	Pending Bits Register for MSI (Offset 10h or 14h)	1149
7.7.2	MSI-X Capability and Table Structure	1150
7.7.2.1	MSI-X Capability Header (Offset 00h)	1153
7.7.2.2	Message Control Register for MSI-X (Offset 02h)	1154
7.7.2.3	Table Offset/Table BIR Register for MSI-X (Offset 04h)	1155
7.7.2.4	PBA Offset/PBA BIR Register for MSI-X (Offset 08h)	1155
7.7.2.5	Message Address Register for MSI-X Table Entries	1156
7.7.2.6	Message Upper Address Register for MSI-X Table Entries	1157
7.7.2.7	Message Data Register for MSI-X Table Entries	1157
7.7.2.8	Vector Control Register for MSI-X Table Entries	1157
7.7.2.9	Pending Bits Register for MSI-X PBA Entries	1158
7.7.3	Secondary PCI Express Extended Capability	1159
7.7.3.1	Secondary PCI Express Extended Capability Header (Offset 00h)	1161
7.7.3.2	Link Control 3 Register (Offset 04h)	1161
7.7.3.3	Lane Error Status Register (Offset 08h)	1162
7.7.3.4	Lane Equalization Control Register (Offset 0Ch)	1163
7.7.4	Data Link Feature Extended Capability	1165
7.7.4.1	Data Link Feature Extended Capability Header (Offset 00h)	1166
7.7.4.2	Data Link Feature Capabilities Register (Offset 04h)	1167
7.7.4.3	Data Link Feature Status Register (Offset 08h)	1168
7.7.5	Physical Layer 16.0 GT/s Extended Capability	1169
7.7.5.1	Physical Layer 16.0 GT/s Extended Capability Header (Offset 00h)	1171

7.7.5.2	16.0 GT/s Capabilities Register (Offset 04h)	1171
7.7.5.3	16.0 GT/s Control Register (Offset 08h)	1172
7.7.5.4	16.0 GT/s Status Register (Offset 0Ch)	1172
7.7.5.5	16.0 GT/s Local Data Parity Mismatch Status Register (Offset 10h).....	1173
7.7.5.6	16.0 GT/s First Retimer Data Parity Mismatch Status Register (Offset 14h)	1174
7.7.5.7	16.0 GT/s Second Retimer Data Parity Mismatch Status Register (Offset 18h)	1174
7.7.5.8	Physical Layer 16.0 GT/s Reserved (Offset 1Ch)	1175
7.7.5.9	16.0 GT/s Lane Equalization Control Register (Offsets 20h to 3Ch).....	1175
7.7.6	Physical Layer 32.0 GT/s Extended Capability.....	1176
7.7.6.1	Physical Layer 32.0 GT/s Extended Capability Header (Offset 00h)	1178
7.7.6.2	32.0 GT/s Capabilities Register (Offset 04h)	1178
7.7.6.3	32.0 GT/s Control Register (Offset 08h)	1179
7.7.6.4	32.0 GT/s Status Register (Offset 0Ch)	1180
7.7.6.5	Received Modified TS Data 1 Register (Offset 10h)	1181
7.7.6.6	Received Modified TS Data 2 Register (Offset 14h)	1182
7.7.6.7	Transmitted Modified TS Data 1 Register (Offset 18h)	1183
7.7.6.8	Transmitted Modified TS Data 2 Register (Offset 1Ch).....	1184
7.7.6.9	32.0 GT/s Lane Equalization Control Register (Offset 20h).....	1185
7.7.7	Physical Layer 64.0 GT/s Extended Capability.....	1186
7.7.7.1	Physical Layer 64.0 GT/s Extended Capability Header (Offset 00h)	1187
7.7.7.2	64.0 GT/s Capabilities Register (Offset 04h)	1188
7.7.7.3	64.0 GT/s Control Register (Offset 08h)	1188
7.7.7.4	64.0 GT/s Status Register (Offset 0Ch)	1189
7.7.7.5	64.0 GT/s Lane Equalization Control Register (Offset 10h).....	1190
7.7.8	Flit Logging Extended Capability.....	1191
7.7.8.1	Flit Logging Extended Capability Header (Offset 00h).....	1192
7.7.8.2	Flit Error Log 1 Register (Offset 04h).....	1193
7.7.8.3	Flit Error Log 2 Register (Offset 08h).....	1195
7.7.8.4	Flit Error Counter Control Register (Offset 0Ch)	1196
7.7.8.5	Flit Error Counter Status Register (Offset 0Eh).....	1197
7.7.8.6	FBER Measurement Control Register (Offset 10h)	1198
7.7.8.7	FBER Measurement Status 1 Register (Offset 14h)	1199
7.7.8.8	FBER Measurement Status 2 Register (Offset 18h)	1200
7.7.8.9	FBER Measurement Status 3 Register (Offset 1Ch)	1200
7.7.8.10	FBER Measurement Status 4 Register (Offset 20h)	1201
7.7.8.11	FBER Measurement Status 5 Register (Offset 24h)	1201
7.7.8.12	FBER Measurement Status 6 Register (Offset 28h)	1201
7.7.8.13	FBER Measurement Status 7 Register (Offset 2Ch)	1202
7.7.8.14	FBER Measurement Status 8 Register (Offset 30h)	1202
7.7.8.15	FBER Measurement Status 9 Register (Offset 34h)	1203
7.7.8.16	FBER Measurement Status 10 Register (Offset 38h)	1203
7.7.9	Device 3 Extended Capability Structure	1203
7.7.9.1	Device 3 Extended Capability Header (Offset 00h)	1204
7.7.9.2	Device Capabilities 3 Register (Offset 04h).....	1204
7.7.9.3	Device Control 3 Register (Offset 08h).....	1207
7.7.9.4	Device Status 3 Register (Offset 0Ch).....	1210
7.7.10	Lane Margining at the Receiver Extended Capability	1211
7.7.10.1	Lane Margining at the Receiver Extended Capability Header (Offset 00h)	1213
7.7.10.2	Margining Port Capabilities Register (Offset 04h)	1213
7.7.10.3	Margining Port Status Register (Offset 06h)	1214
7.7.10.4	Margining Lane Control Register (Offset 08h)	1214

7.7.10.5	Margining Lane Status Register (Offset 0Ah)	1215
7.7.11	ACS Extended Capability	1216
7.7.11.1	ACS Extended Capability Header (Offset 00h)	1217
7.7.11.2	ACS Capability Register (Offset 04h)	1218
7.7.11.3	ACS Control Register (Offset 06h)	1219
7.7.11.4	Egress Control Vector Register (Offset 08h)	1221
7.8	Common PCI and PCIe Capabilities	1223
7.8.1	Power Budgeting Extended Capability	1223
7.8.1.1	Power Budgeting Extended Capability Header (Offset 00h)	1223
7.8.1.2	Power Budgeting Data Select Register (Offset 04h)	1224
7.8.1.3	Power Budgeting Control Register (Offset 06h)	1224
7.8.1.4	Power Budgeting Data Register (Offset 08h)	1226
7.8.1.5	Power Budgeting Capability Register (Offset 0Ch)	1231
7.8.1.6	Power Budgeting Sense Detect Register (Offset 0Dh)	1232
7.8.2	Latency Tolerance Reporting (LTR) Extended Capability	1235
7.8.2.1	LTR Extended Capability Header (Offset 00h)	1236
7.8.2.2	Max Snoop Latency Register (Offset 04h)	1236
7.8.2.3	Max No-Snoop Latency Register (Offset 06h)	1237
7.8.2.4	LTR Capabilities Register (Offset 08h)	1237
7.8.3	L1 PM Substates Extended Capability	1238
7.8.3.1	L1 PM Substates Extended Capability Header (Offset 00h)	1239
7.8.3.2	L1 PM Substates Capabilities Register (Offset 04h)	1240
7.8.3.3	L1 PM Substates Control 1 Register (Offset 08h)	1241
7.8.3.4	L1 PM Substates Control 2 Register (Offset 0Ch)	1243
7.8.3.5	L1 PM Substates Status Register (Offset 10h)	1244
7.8.4	Advanced Error Reporting Extended Capability (AER)	1244
7.8.4.1	Advanced Error Reporting Extended Capability Header (Offset 00h)	1247
7.8.4.2	Uncorrectable Error Status Register (Offset 04h)	1247
7.8.4.3	Uncorrectable Error Mask Register (Offset 08h)	1250
7.8.4.4	Uncorrectable Error Severity Register (Offset 0Ch)	1252
7.8.4.5	Correctable Error Status Register (Offset 10h)	1255
7.8.4.6	Correctable Error Mask Register (Offset 14h)	1256
7.8.4.7	Advanced Error Capabilities and Control Register (Offset 18h)	1257
7.8.4.8	Header Log Register (Offset 1Ch)	1259
7.8.4.9	Root Error Command Register (Offset 2Ch)	1260
7.8.4.10	Root Error Status Register (Offset 30h)	1262
7.8.4.11	Error Source Identification Register (Offset 34h)	1264
7.8.4.12	TLP Prefix Log Register (Offset 38h)	1265
7.8.5	Enhanced Allocation Capability Structure (EA)	1266
7.8.5.1	Enhanced Allocation Capability First DW (Offset 00h)	1266
7.8.5.2	Enhanced Allocation Capability Second DW (Offset 04h) [Type 1 Functions Only]	1267
7.8.5.3	Enhanced Allocation Per-Entry Format (Offset 04h or 08h)	1267
7.8.6	Resizable BAR Extended Capability	1272
7.8.6.1	Resizable BAR Extended Capability Header (Offset 00h)	1275
7.8.6.2	Resizable BAR Capability Register	1275
7.8.6.3	Resizable BAR Control Register	1278
7.8.7	VF Resizable BAR Extended Capability	1280
7.8.7.1	VF Resizable BAR Extended Capability Header (Offset 00h)	1282
7.8.7.2	VF Resizable BAR Capability Register (Offset 04h)	1282
7.8.7.3	VF Resizable BAR Control Register (Offset 08h)	1282

7.8.8	ARI Extended Capability	1284
7.8.8.1	ARI Extended Capability Header (Offset 00h)	1284
7.8.8.2	ARI Capability Register (Offset 04h)	1285
7.8.8.3	ARI Control Register (Offset 06h)	1286
7.8.9	PASID Extended Capability Structure	1286
7.8.9.1	PASID Extended Capability Header (Offset 00h)	1287
7.8.9.2	PASID Capability Register (Offset 04h)	1288
7.8.9.3	PASID Control Register (Offset 06h)	1288
7.8.10	FRS Queueing Extended Capability	1290
7.8.10.1	FRS Queueing Extended Capability Header (Offset 00h)	1290
7.8.10.2	FRS Queueing Capability Register (Offset 04h)	1291
7.8.10.3	FRS Queueing Status Register (Offset 08h)	1291
7.8.10.4	FRS Queueing Control Register (Offset 0Ah)	1292
7.8.10.5	FRS Message Queue Register (Offset 0Ch)	1292
7.8.11	Flattening Portal Bridge (FPB) Capability	1293
7.8.11.1	FPB Capability Header (Offset 00h)	1294
7.8.11.2	FPB Capabilities Register (Offset 04h)	1294
7.8.11.3	FPB RID Vector Control 1 Register (Offset 08h)	1296
7.8.11.4	FPB RID Vector Control 2 Register (Offset 0Ch)	1297
7.8.11.5	FPB MEM Low Vector Control Register (Offset 10h)	1298
7.8.11.6	FPB MEM High Vector Control 1 Register (Offset 14h)	1299
7.8.11.7	FPB MEM High Vector Control 2 Register (Offset 18h)	1301
7.8.11.8	FPB Vector Access Control Register (Offset 1Ch)	1302
7.8.11.9	FPB Vector Access Data Register (Offset 20h)	1303
7.8.12	Flit Performance Measurement Extended Capability	1304
7.8.12.1	Flit Performance Measurement Extended Capability Header (Offset 00h)	1304
7.8.12.2	Flit Performance Measurement Capability Register (Offset 04h)	1305
7.8.12.3	Flit Performance Measurement Control Register (Offset 08h)	1306
7.8.12.4	Flit Performance Measurement Status Register (Offset 0Ch)	1308
7.8.12.5	LTSSM Performance Measurement Status Register (Offsets 10h to 20h)	1309
7.8.13	Flit Error Injection Extended Capability	1310
7.8.13.1	Flit Error Injection Extended Capability Header (Offset 00h)	1311
7.8.13.2	Flit Error Injection Capability Register (Offset 04h)	1312
7.8.13.3	Flit Error Injection Control 1 Register (Offset 08h)	1312
7.8.13.4	Flit Error Injection Control 2 Register (Offset 0Ch)	1314
7.8.13.5	Flit Error Injection Status Register (Offset 10h)	1315
7.8.13.6	Ordered Set Error Injection Control 1 Register (Offset 14h)	1316
7.8.13.7	Ordered Set Error Injection Control 2 Register (Offset 18h)	1317
7.8.13.8	Ordered Set Error Tx Injection Status Register (Offset 1Ch)	1318
7.8.13.9	Ordered Set Error Rx Injection Status Register (Offset 20h)	1319
7.8.14	NOP Flit Extended Capability	1320
7.8.14.1	NOP Flit Extended Capability Header	1321
7.8.14.2	NOP Flit Capabilities Register	1322
7.8.14.3	NOP Flit Control 1 Register	1322
7.8.14.4	NOP Flit Control 2 Register	1324
7.8.14.5	NOP Flit Status Register	1325
7.9	Additional PCI and PCIe Capabilities	1326
7.9.1	Virtual Channel Extended Capability	1326
7.9.1.1	Virtual Channel Extended Capability Header (Offset 00h)	1327
7.9.1.2	Port VC Capability Register 1 (Offset 04h)	1327
7.9.1.3	Port VC Capability Register 2 (Offset 08h)	1329

7.9.1.4	Port VC Control Register (Offset 0Ch)	1329
7.9.1.5	Port VC Status Register (Offset 0Eh)	1330
7.9.1.6	VC Resource Capability Register	1331
7.9.1.7	VC Resource Control Register	1332
7.9.1.8	VC Resource Status Register	1335
7.9.1.9	VC Arbitration Table	1336
7.9.1.10	Port Arbitration Table.....	1336
7.9.2	Multi-Function Virtual Channel Extended Capability	1338
7.9.2.1	MFVC Extended Capability Header (Offset 00h)	1338
7.9.2.2	MFVC Port VC Capability Register 1 (Offset 04h)	1339
7.9.2.3	MFVC Port VC Capability Register 2 (Offset 08h)	1340
7.9.2.4	MFVC Port VC Control Register (Offset 0Ch)	1341
7.9.2.5	MFVC Port VC Status Register (Offset 0Eh)	1342
7.9.2.6	MFVC VC Resource Capability Register	1342
7.9.2.7	MFVC VC Resource Control Register	1343
7.9.2.8	MFVC VC Resource Status Register	1346
7.9.2.9	MFVC VC Arbitration Table	1346
7.9.2.10	Function Arbitration Table	1347
7.9.3	Device Serial Number Extended Capability	1348
7.9.3.1	Device Serial Number Extended Capability Header (Offset 00h)	1349
7.9.3.2	Serial Number Register (Offset 04h)	1350
7.9.4	Vendor-Specific Capability.....	1350
7.9.5	Vendor-Specific Extended Capability	1351
7.9.5.1	Vendor-Specific Extended Capability Header (Offset 00h)	1352
7.9.5.2	Vendor-Specific Header (Offset 04h)	1353
7.9.6	Designated Vendor-Specific Extended Capability (DVSEC)	1353
7.9.6.1	Designated Vendor-Specific Extended Capability Header (Offset 00h)	1354
7.9.6.2	Designated Vendor-Specific Header 1 (Offset 04h).....	1355
7.9.6.3	Designated Vendor-Specific Header 2 (Offset 08h).....	1355
7.9.7	RCRB Header Extended Capability	1356
7.9.7.1	RCRB Header Extended Capability Header (Offset 00h)	1356
7.9.7.2	RCRB Vendor ID and Device ID register (Offset 04h)	1357
7.9.7.3	RCRB Capabilities register (Offset 08h)	1357
7.9.7.4	RCRB Control register (Offset 0Ch)	1358
7.9.8	Root Complex Link Declaration Extended Capability	1358
7.9.8.1	Root Complex Link Declaration Extended Capability Header (Offset 00h).....	1360
7.9.8.2	Element Self Description Register (Offset 04h)	1361
7.9.8.3	Link Entries	1362
7.9.8.3.1	Link Description Register	1362
7.9.8.3.2	Link Address	1363
7.9.8.3.2.1	Link Address for Link Type 0	1363
7.9.8.3.2.2	Link Address for Link Type 1	1363
7.9.9	Root Complex Internal Link Control Extended Capability.....	1364
7.9.9.1	Root Complex Internal Link Control Extended Capability Header (Offset 00h)	1365
7.9.9.2	Root Complex Link Capabilities Register (Offset 04h)	1365
7.9.9.3	Root Complex Link Control Register (Offset 08h)	1368
7.9.9.4	Root Complex Link Status Register (Offset 0Ah)	1369
7.9.10	Root Complex Event Collector Endpoint Association Extended Capability	1370
7.9.10.1	Root Complex Event Collector Endpoint Association Extended Capability Header (Offset 00h) ..	1371
7.9.10.2	Association Bitmap for RCIEPs (Offset 04h)	1372
7.9.10.3	RCEC Associated Bus Numbers Register (Offset 08h)	1372

7.9.11	Multicast Extended Capability	1373
7.9.11.1	Multicast Extended Capability Header (Offset 00h)	1374
7.9.11.2	Multicast Capability Register (Offset 04h)	1375
7.9.11.3	Multicast Control Register (Offset 06h)	1376
7.9.11.4	MC_Base_Address Register (Offset 08h)	1376
7.9.11.5	MC_Receive Register (Offset 10h)	1377
7.9.11.6	MC_Block_All Register (Offset 18h)	1377
7.9.11.7	MC_Block_Untranslated Register (Offset 20h)	1378
7.9.11.8	MC_Overlay_BAR Register (Offset 28h)	1378
7.9.12	Dynamic Power Allocation Extended Capability (DPA Capability)	1379
7.9.12.1	DPA Extended Capability Header (Offset 00h)	1380
7.9.12.2	DPA Capability Register (Offset 04h)	1380
7.9.12.3	DPA Latency Indicator Register (Offset 08h)	1381
7.9.12.4	DPA Status Register (Offset 0Ch)	1382
7.9.12.5	DPA Control Register (Offset 0Eh)	1382
7.9.12.6	DPA Power Allocation Array	1383
7.9.13	TPH Requester Extended Capability	1383
7.9.13.1	TPH Requester Extended Capability Header (Offset 00h)	1384
7.9.13.2	TPH Requester Capability Register (Offset 04h)	1384
7.9.13.3	TPH Requester Control Register (Offset 08h)	1385
7.9.13.4	TPH ST Table (Starting from Offset 0Ch)	1386
7.9.14	DPC Extended Capability	1387
7.9.14.1	DPC Extended Capability Header (Offset 00h)	1390
7.9.14.2	DPC Capability Register (Offset 04h)	1390
7.9.14.3	DPC Control Register (Offset 06h)	1392
7.9.14.4	DPC Status Register (Offset 08h)	1394
7.9.14.5	DPC Error Source ID Register (Offset 0Ah)	1395
7.9.14.6	RP PIO Status Register (Offset 0Ch)	1396
7.9.14.7	RP PIO Mask Register (Offset 10h)	1397
7.9.14.8	RP PIO Severity Register (Offset 14h)	1397
7.9.14.9	RP PIO SysError Register (Offset 18h)	1398
7.9.14.10	RP PIO Exception Register (Offset 1Ch)	1399
7.9.14.11	RP PIO Header Log Register (Offset 20h)	1400
7.9.14.12	RP PIO ImpSpec Log Register (Offset 30h)	1401
7.9.14.13	RP PIO TLP Prefix Log Register (Offset 34h)	1401
7.9.15	Precision Time Measurement Extended Capability (PTM Extended Capability)	1402
7.9.15.1	PTM Extended Capability Header (Offset 00h)	1403
7.9.15.2	PTM Capability Register (Offset 04h)	1403
7.9.15.3	PTM Control Register (Offset 08h)	1405
7.9.16	Readiness Time Reporting Extended Capability	1406
7.9.16.1	Readiness Time Reporting Extended Capability Header (Offset 00h)	1407
7.9.16.2	Readiness Time Reporting 1 Register (Offset 04h)	1408
7.9.16.3	Readiness Time Reporting 2 Register (Offset 08h)	1409
7.9.17	Hierarchy ID Extended Capability	1410
7.9.17.1	Hierarchy ID Extended Capability Header (Offset 00h)	1411
7.9.17.2	Hierarchy ID Status Register (Offset 04h)	1412
7.9.17.3	Hierarchy ID Data Register (Offset 08h)	1413
7.9.17.4	Hierarchy ID GUID 1 Register (Offset 0Ch)	1414
7.9.17.5	Hierarchy ID GUID 2 Register (Offset 10h)	1414
7.9.17.6	Hierarchy ID GUID 3 Register (Offset 14h)	1415
7.9.17.7	Hierarchy ID GUID 4 Register (Offset 18h)	1415

7.9.17.8	Hierarchy ID GUID 5 Register (Offset 1Ch)	1416
7.9.18	Vital Product Data Capability (VPD Capability)	1416
7.9.18.1	VPD Address Register	1417
7.9.18.2	VPD Data Register	1418
7.9.19	Native PCIe Enclosure Management Extended Capability (NPEM Extended Capability)	1418
7.9.19.1	NPEM Extended Capability Header (Offset 00h)	1419
7.9.19.2	NPEM Capability Register (Offset 04h)	1419
7.9.19.3	NPEM Control Register (Offset 08h)	1421
7.9.19.4	NPEM Status Register (Offset 0Ch)	1423
7.9.20	Alternate Protocol Extended Capability	1424
7.9.20.1	Alternate Protocol Extended Capability Header (Offset 00h)	1424
7.9.20.2	Alternate Protocol Capabilities Register (Offset 04h)	1425
7.9.20.3	Alternate Protocol Control Register (Offset 08h)	1425
7.9.20.4	Alternate Protocol Data 1 Register (Offset 0Ch)	1426
7.9.20.5	Alternate Protocol Data 2 Register (Offset 10h)	1427
7.9.20.6	Alternate Protocol Selective Enable Mask Register (Offset 14h)	1427
7.9.21	Conventional PCI Advanced Features Capability (AF)	1428
7.9.21.1	Advanced Features Capability Header (Offset 00h)	1428
7.9.21.2	AF Capabilities Register (Offset 03h)	1429
7.9.21.3	Conventional PCI Advanced Features Control Register (Offset 04h)	1429
7.9.21.4	AF Status Register (Offset 05h)	1430
7.9.22	SFI Extended Capability	1430
7.9.22.1	SFI Extended Capability Header (Offset 00h)	1431
7.9.22.2	SFI Capability Register (Offset 04h)	1431
7.9.22.3	SFI Control Register (Offset 06h)	1432
7.9.22.4	SFI Status Register (Offset 08h)	1435
7.9.22.5	SFI CAM Address Register (Offset 0Ch)	1436
7.9.22.6	SFI CAM Data Register (Offset 10h)	1436
7.9.23	Subsystem ID and Subsystem Vendor ID Capability	1437
7.9.23.1	Subsystem ID and Subsystem Vendor ID Capability Header (Offset 00h)	1438
7.9.23.2	Subsystem ID and Subsystem Vendor ID Capability Data (Offset 04h)	1438
7.9.24	Data Object Exchange Extended Capability	1438
7.9.24.1	DOE Extended Capability Header (Offset 00h)	1439
7.9.24.2	DOE Capabilities Register (Offset 04h)	1440
7.9.24.3	DOE Control Register (Offset 08h)	1441
7.9.24.4	DOE Status Register (Offset 0Ch)	1442
7.9.24.5	DOE Write Data Mailbox Register (Offset 10h)	1443
7.9.24.6	DOE Read Data Mailbox Register (Offset 14h)	1443
7.9.25	Shadow Functions Extended Capability	1444
7.9.25.1	Shadow Functions Extended Capability Header (Offset 00h)	1446
7.9.25.2	Shadow Functions Capability Register (Offset 04h)	1447
7.9.25.3	Shadow Functions Control Register (Offset 08h)	1447
7.9.25.4	Shadow Functions Instance Register Entry	1448
7.9.26	IDE Extended Capability	1448
7.9.26.1	IDE Extended Capability Header (Offset 00h)	1449
7.9.26.2	IDE Capability Register (Offset 04h)	1450
7.9.26.3	IDE Control Register (Offset 08h)	1452
7.9.26.4	Link IDE Register Block	1452
7.9.26.4.1	Link IDE Stream Control Register	1452
7.9.26.4.2	Link IDE Stream Status Register	1454
7.9.26.5	Selective IDE Stream Register Block	1455

7.9.26.5.1	Selective IDE Stream Capability Register	1455
7.9.26.5.2	Selective IDE Stream Control Register	1456
7.9.26.5.3	Selective IDE Stream Status Register	1458
7.9.26.5.4	Selective IDE RID Association Register Block.....	1459
7.9.26.5.4.1	IDE RID Association Register 1	1459
7.9.26.5.4.2	IDE RID Association Register 2	1460
7.9.26.5.5	Selective IDE Address Association Register Block	1460
7.9.26.5.5.1	IDE Address Association Register 1	1460
7.9.26.5.5.2	IDE Address Association Register 2	1461
7.9.26.5.5.3	IDE Address Association Register 3	1461
7.9.27	Null Capability	1462
7.9.28	Null Extended Capability.....	1462
7.9.29	Streamlined Virtual Channel Extended Capability (SVC).....	1463
7.9.29.1	Streamlined Virtual Channel Extended Capability Header (Offset 00h)	1464
7.9.29.2	SVC Port Capability Register 1 (Offset 04h)	1465
7.9.29.3	SVC Port Capability Register 2 (Offset 08h)	1465
7.9.29.4	SVC Port Control Register (Offset 0Ch)	1465
7.9.29.5	SVC Port Status Register (Offset 10h)	1466
7.9.29.6	SVC Resource Capability Register.....	1466
7.9.29.7	SVC Resource Control Register	1467
7.9.29.8	SVC Resource Status Register	1469
7.9.30	MMIO Register Block Locator Extended Capability (MRBL)	1470
7.9.30.1	MRBL Extended Capability Header (Offset 00h).....	1470
7.9.30.2	MRBL Capabilities Register (Offset 04h)	1471
7.9.30.3	MRBL Locator Register (Offset Varies)	1472
8.	Electrical Sub-Block	1473
8.1	Electrical Specification Introduction.....	1473
8.2	Interoperability Criteria	1473
8.2.1	Data Rates	1473
8.2.2	Refclk Architectures	1473
8.3	Transmitter Specification.....	1474
8.3.1	Measurement Setup for Characterizing Transmitters	1474
8.3.1.1	Breakout and Replica Channels	1475
8.3.2	Voltage Level Definitions	1475
8.3.3	Tx Voltage Parameters.....	1476
8.3.3.1	2.5 and 5.0 GT/s Transmitter Equalization	1476
8.3.3.2	8.0, 16.0, 32.0, and 64.0 GT/s Transmitter Equalization.....	1476
8.3.3.3	Tx Equalization Presets for 8.0, 16.0, 32.0, and 64.0 GT/s	1478
8.3.3.4	Measuring Tx Equalization for 2.5 GT/s and 5.0 GT/s	1480
8.3.3.5	Measuring Presets at 8.0, 16.0, 32.0, and 64.0 GT/s	1480
8.3.3.6	Method for Measuring $V_{TX-DIFF-PP}$ at 2.5 GT/s and 5.0 GT/s.....	1481
8.3.3.7	Method for Measuring $V_{TX-DIFF-PP}$ at 8.0, 16.0, 32.0, and 64.0 GT/s	1481
8.3.3.8	Coefficient Range and Tolerance for 8.0, 16.0, 32.0, and 64.0 GT/s.....	1482
8.3.3.9	EIEOS and $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ Limits	1485
8.3.3.10	Reduced Swing Signaling	1486
8.3.3.11	Effective Tx Package Loss at 8.0, 16.0, 32.0, and 64.0 GT/s.....	1486
8.3.3.12	Linear Fit Pulse Response.....	1488
8.3.3.13	Transmitter Signal-to Noise and Distortion Ratio ($SNDR_{TX}$) for 64.0 GT/s	1490
8.3.3.14	Transmitter Ratio of Level Mismatch (R_{LM-TX}) for 64.0 GT/s	1492
8.3.3.14.1	Multi Pulse Response Fit (MPRF) – PAM4 Voltage Variant	1493

8.3.4	Transmitter Margining.....	1497
8.3.5	Tx Jitter Parameters	1498
8.3.5.1	Post Processing Steps to Extract Jitter	1498
8.3.5.2	Applying CTLE or De-embedding	1498
8.3.5.3	Independent Refclk Measurement and Post Processing.....	1499
8.3.5.4	Embedded and Non-Embedded Refclk Measurement and Post Processing	1499
8.3.5.5	Behavioral CDR Characteristics	1500
8.3.5.6	Data Dependent and Uncorrelated Jitter.....	1505
8.3.5.7	Data Dependent Jitter.....	1505
8.3.5.8	Uncorrelated Total Jitter and Deterministic Jitter (Dual Dirac Model) (T_{TX-UTJ} and $T_{TX-UDJDD}$)	1506
8.3.5.9	Random Jitter (T_{TX-RJ}) (informative).....	1506
8.3.5.10	Uncorrelated Total and Deterministic PWJ ($T_{TX-UPW-TJ}$ and $T_{TX-UPW-DJDD}$)	1506
8.3.6	Data Rate Dependent Parameters	1508
8.3.7	Tx and Rx Return Loss for 2.5, 5.0, 8.0, 16.0, and 32.0 GT/s	1511
8.3.8	Tx and Rx Return Loss for 64.0 GT/s	1513
8.3.9	Transmitter PLL Bandwidth and Peaking	1515
8.3.9.1	2.5 GT/s and 5.0 GT/s Tx PLL Bandwidth and Peaking	1515
8.3.9.2	8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s Tx PLL Bandwidth and Peaking	1515
8.3.9.3	Series Capacitors.....	1516
8.3.10	Data Rate Independent Tx Parameters.....	1516
8.4	Receiver Specifications	1517
8.4.1	Receiver Stressed Eye Specification	1517
8.4.1.1	Breakout and Replica Channels	1517
8.4.1.2	Calibration Channel Insertion Loss Characteristics.....	1518
8.4.1.3	Post Processing Procedures	1527
8.4.1.4	Behavioral Rx Package Models	1528
8.4.1.5	Behavioral CDR Model.....	1528
8.4.1.6	No Behavioral Rx Equalization for 2.5 and 5.0 GT/s	1528
8.4.1.7	Behavioral Rx Equalization for 8.0, 16.0, 32.0, and 64.0 GT/s	1528
8.4.1.8	Behavioral CTLE (8.0 and 16.0 GT/s)	1529
8.4.1.9	Behavioral CTLE (32.0 and 64.0 GT/s)	1531
8.4.1.10	Behavioral DFE (8.0, 16.0, 32.0, and 64.0 GT/s Only).....	1534
8.4.2	Stressed Eye Test.....	1536
8.4.2.1	Procedure for Calibrating a Stressed EH/EW Eye.....	1536
8.4.2.1.1	Post Processing Tool Requirements.....	1541
8.4.2.2	Procedure for Testing Rx DUT	1542
8.4.2.2.1	Sj Mask.....	1542
8.4.2.3	Receiver Refclk Modes	1550
8.4.2.3.1	Common Refclk Mode	1550
8.4.2.3.2	Independent Refclk Mode.....	1551
8.4.3	Common Receiver Parameters	1552
8.4.3.1	5.0 GT/s Exit From Idle Detect (EFI).....	1554
8.4.3.2	Receiver Return Loss	1554
8.4.4	Lane Margining at the Receiver - Electrical Requirements.....	1554
8.4.5	Low Frequency and Miscellaneous Signaling Requirements.....	1558
8.4.5.1	ESD Standards	1558
8.4.5.2	Channel AC Coupling Capacitors.....	1558
8.4.5.3	Short Circuit Requirements	1558
8.4.5.4	Transmitter and Receiver Termination.....	1559
8.4.5.5	Electrical Idle	1559
8.4.5.6	DC Common Mode Voltage	1559

8.4.5.7	Receiver Detection	1560
8.5	Channel Tolerancing	1560
8.5.1	Channel Compliance Testing	1560
8.5.1.1	Behavioral Transmitter and Receiver Package Models.....	1562
8.5.1.2	Measuring Package Performance (16.0 GT/s only)	1572
8.5.1.3	Simulation Tool Requirements	1572
8.5.1.3.1	Simulation Tool Chain Inputs	1573
8.5.1.3.2	Processing Steps	1573
8.5.1.3.3	Simulation Tool Outputs.....	1573
8.5.1.3.4	Open Source Simulation Tool.....	1574
8.5.1.4	Behavioral Transmitter Parameters	1574
8.5.1.4.1	Deriving Voltage and Jitter Parameters	1574
8.5.1.4.2	Optimizing Tx/Rx Equalization (8.0, 16.0, 32.0, and 64.0 GT/s only)	1576
8.5.1.4.3	Pass/Fail Eye Characteristics	1576
8.5.1.4.4	Characterizing Channel Common Mode Noise	1579
8.5.1.4.5	Verifying $V_{CH-IDLE-DET-DIFF-pp}$	1580
8.6	Refclk Specifications	1580
8.6.1	Refclk Test Setup	1580
8.6.2	REFCLK AC Specifications	1581
8.6.3	Data Rate Independent Refclk Parameters	1584
8.6.3.1	Low Frequency Refclk Jitter Limits	1585
8.6.4	Refclk Architectures Supported.....	1586
8.6.5	Filtering Functions Applied to Raw Data	1586
8.6.5.1	PLL Filter Transfer Function Example.....	1587
8.6.5.2	CDR Transfer Function Examples.....	1587
8.6.6	Common Refclk Rx Architecture (CC)	1588
8.6.6.1	Determining the Number of PLL BW and peaking Combinations.....	1589
8.6.6.2	CDR and PLL BW and Peaking Limits for Common Refclk.....	1589
8.6.7	Jitter Limits for Refclk Architectures	1590
8.6.8	Form Factor Requirements for RefClock Architectures	1591
9.	I/O Virtualization and Sharing	1593
9.1	IOV Architectural Overview	1593
9.1.1	RID in Endpoints where a PF contains an SIOV Extended Capability Structure	1606
9.1.2	SDI Reset	1607
9.2	SR-IOV Initialization and Resource Allocation	1608
9.2.1	SR-IOV Resource Discovery	1608
9.2.1.1	Configuring SR-IOV Capabilities	1608
9.2.1.1.1	Configuring the VF BAR Mechanisms	1609
9.2.1.2	VF Discovery	1610
9.2.1.3	Function Dependency Lists.....	1612
9.2.1.4	Interrupt Resource Allocation.....	1613
9.2.2	SR-IOV Reset Mechanisms.....	1613
9.2.2.1	SR-IOV Conventional Reset.....	1613
9.2.2.2	FLR That Targets a VF	1613
9.2.2.3	FLR That Targets a PF	1613
9.2.3	IOV Re-initialization and Reallocation.....	1613
9.3	SIOV Initialization and Resource Allocation	1614
9.3.1	SIOV Resource Discovery	1614
9.3.1.1	Configuring SIOV Capabilities.....	1614
9.3.1.1.1	Configuring an SDI	1614

	9.3.1.1.1.1	Behavior of a VI.....	1614
	9.3.1.1.1.2	Requirements of an SI.....	1615
	9.3.1.2	MSI/MSI-X Interrupt Message Resource Allocation.....	1615
9.4		SR-IOV Configuration	1616
	9.4.1	SR-IOV Configuration Overview	1616
	9.4.2	SR-IOV Configuration Space.....	1616
	9.4.3	SR-IOV Extended Capability	1616
	9.4.3.1	SR-IOV Extended Capability Header (Offset 00h).....	1617
	9.4.3.2	SR-IOV Capabilities Register (04h)	1618
	9.4.3.2.1	VF Migration Capable.....	1619
	9.4.3.2.2	ARI Capable Hierarchy Preserved	1619
	9.4.3.2.3	VF Larger-Tag Requester Support.....	1619
	9.4.3.2.4	VF Migration Interrupt Message Number.....	1620
	9.4.3.3	SR-IOV Control Register (Offset 08h)	1620
	9.4.3.3.1	VF Enable.....	1622
	9.4.3.3.2	VF Migration Enable.....	1623
	9.4.3.3.3	VF Migration Interrupt Enable	1623
	9.4.3.3.4	VF MSE (Memory Space Enable)	1623
	9.4.3.3.5	ARI Capable Hierarchy	1624
	9.4.3.4	SR-IOV Status Register (Offset 0Ah)	1624
	9.4.3.4.1	VF Migration Status.....	1625
	9.4.3.5	InitialVFs (Offset 0Ch).....	1625
	9.4.3.6	TotalVFs (Offset 0Eh)	1625
	9.4.3.7	NumVFs (Offset 10h).....	1625
	9.4.3.8	Function Dependency Link (Offset 12h).....	1625
	9.4.3.9	First VF Offset (Offset 14h)	1628
	9.4.3.10	VF Stride (Offset 16h)	1628
	9.4.3.11	VF Device ID (Offset 1Ah).....	1628
	9.4.3.12	Supported Page Sizes (Offset 1Ch)	1629
	9.4.3.13	System Page Size (Offset 20h).....	1629
	9.4.3.14	VF BAR0 (Offset 24h), VF BAR1 (Offset 28h), VF BAR2 (Offset 2Ch), VF BAR3 (Offset 30h), VF BAR4 (Offset 34h), VF BAR5 (Offset 38h).....	1629
	9.4.3.15	VF Migration State Array Offset (Deprecated) (Offset 3Ch)	1630
	9.4.4	PF/VF Configuration Space Header	1630
	9.4.5	PCI Express Capability Changes	1630
	9.4.6	PCI Standard Capabilities	1630
	9.4.7	PCI Express Extended Capabilities Changes	1631
9.5		SIOV Configuration	1634
	9.5.1	SIOV Configuration Overview	1634
	9.5.2	SIOV Configuration Space	1634
	9.5.3	SIOV Extended Capability	1634
	9.5.4	SIOV Extended Capability Header (Offset 00h)	1635
	9.5.5	SIOV Capabilities Register (04h)	1635
	9.5.6	Total SDIs (Offset 08h)	1636
	9.5.7	SIOV Status Register (Offset 0Bh)	1636
	9.5.8	First SDI Offset (Offset 0Ch).....	1637
	9.5.9	SDI Stride (Offset 0Eh).....	1637
10.		Address Translation Services (ATS).....	1639
10.1		ATS Architectural Overview.....	1639
	10.1.1	Address Translation Services (ATS) Overview	1640

10.1.2	Page Request Interface Extension	1644
10.1.3	Process Address Space ID (PASID)	1645
10.1.4	ATS Memory Attributes.....	1646
10.2	ATS Translation Services	1646
10.2.1	Memory Requests with Address Type	1646
10.2.2	Translation Requests.....	1648
10.2.2.1	Attribute Field.....	1650
10.2.2.2	Length Field.....	1650
10.2.2.3	Tag Field.....	1650
10.2.2.4	Untranslated Address Field	1651
10.2.2.5	No Write (NW) Flag	1651
10.2.2.6	PASID on Translation Request	1651
10.2.2.7	CXL Src	1651
10.2.3	Translation Completion	1652
10.2.3.1	Translated Address Field.....	1655
10.2.3.2	Translation Range Size (S) Field	1655
10.2.3.3	Non-snooped (N) Field.....	1656
10.2.3.4	Untranslated Access Only (U) Field	1656
10.2.3.5	Read (R) and Write (W) Fields.....	1657
10.2.3.6	Execute Permitted (Exe)	1658
10.2.3.7	Privileged Mode Access (Priv)	1658
10.2.3.8	Global Mapping (Global)	1659
10.2.3.9	ATS Memory Attributes	1661
10.2.3.10	TEE Exclusive Memory Attribute (TE)	1664
10.2.4	Completions with Multiple Translations	1664
10.3	ATS Invalidation.....	1665
10.3.1	Invalidate Request.....	1665
10.3.2	Invalidate Completion	1667
10.3.3	Invalidate Completion Semantics	1669
10.3.4	Request Acceptance Rules	1670
10.3.5	Invalidate Flow Control.....	1671
10.3.6	Invalidate Ordering Semantics	1671
10.3.7	Implicit Invalidation Events	1672
10.3.8	PASID and Global Invalidate	1673
10.4	Page Request Services	1673
10.4.1	Page Request Message.....	1674
10.4.1.1	PASID Usage.....	1676
10.4.1.2	Managing PASID Usage on PRG Requests	1676
10.4.1.2.1	Stop Marker Messages.....	1677
10.4.2	Page Request Group Response Message.....	1679
10.4.2.1	Response Code Field	1680
10.4.2.2	PASID Usage on PRG Responses.....	1681
10.5	ATS Configuration.....	1681
10.5.1	ATS Extended Capability	1681
10.5.1.1	ATS Extended Capability Header (Offset 00h)	1682
10.5.1.2	ATS Capability Register (Offset 04h)	1682
10.5.1.3	ATS Control Register (Offset 06h).....	1683
10.5.2	Page Request Extended Capability Structure	1684
10.5.2.1	Page Request Extended Capability Header (Offset 00h).....	1685
10.5.2.2	Page Request Control Register (Offset 04h)	1686
10.5.2.3	Page Request Status Register (Offset 06h)	1686

10.5.2.4	Outstanding Page Request Capacity (Offset 08h)	1688
10.5.2.5	Outstanding Page Request Allocation (Offset 0Ch)	1688
11.	TEE Device Interface Security Protocol (TDISP)	1689
11.1	Overview of the TEE-I/O Security Model as it Relates to Devices	1691
11.2	TDISP Rules	1696
11.2.1	TDISP TLP Rules	1700
11.2.2	TDISP Message Transport	1702
11.2.3	Requirements for Requesters (TSM)	1703
11.2.4	Requirements for Responders (DSM)	1704
11.2.5	TDISP Timing Requirements	1704
11.2.6	DSM Tracking and Handling of Locked TDI Configurations (Informative)	1704
11.2.7	TVM Acceptance of a TDI	1707
11.3	TDISP Message Formats and processing	1707
11.3.1	TDISP Request Codes	1707
11.3.2	TDISP Response Codes	1708
11.3.3	TDISP Message Format and Protocol Versioning	1709
11.3.4	GET_TDISP_VERSION	1710
11.3.5	TDISP_VERSION	1710
11.3.6	GET_TDISP_CAPABILITIES	1711
11.3.7	TDISP_CAPABILITIES	1711
11.3.8	LOCK_INTERFACE_REQUEST	1712
11.3.9	LOCK_INTERFACE_RESPONSE	1714
11.3.10	GET_DEVICE_INTERFACE_REPORT	1715
11.3.11	DEVICE_INTERFACE_REPORT	1715
11.3.12	GET_DEVICE_INTERFACE_STATE	1718
11.3.13	DEVICE_INTERFACE_STATE	1718
11.3.14	START_INTERFACE_REQUEST	1719
11.3.15	START_INTERFACE_RESPONSE	1719
11.3.16	STOP_INTERFACE_REQUEST	1719
11.3.17	STOP_INTERFACE_RESPONSE	1720
11.3.18	BIND_P2P_STREAM_REQUEST	1720
11.3.19	BIND_P2P_STREAM_RESPONSE	1721
11.3.20	UNBIND_P2P_STREAM_REQUEST	1722
11.3.21	UNBIND_P2P_STREAM_RESPONSE	1722
11.3.22	SET_MMIO_ATTRIBUTE_REQUEST	1722
11.3.23	SET_MMIO_ATTRIBUTE_RESPONSE	1724
11.3.24	TDISP_ERROR	1724
11.3.25	VDM_REQUEST	1726
11.3.26	VDM_RESPONSE	1726
11.3.27	SET_TDISP_CONFIG_REQUEST	1726
11.3.28	SET_TDISP_CONFIG_RESPONSE	1727
11.4	Device Security Requirements	1727
11.4.1	Device Identity and Authentication	1727
11.4.2	Firmware and Configuration Measurements	1727
11.4.3	Securing Interconnects	1728
11.4.4	Device Attached Memory	1728
11.4.5	TDI Security	1729
11.4.6	Data Integrity Errors	1730
11.4.7	Debug Modes	1730
11.4.8	Conventional Reset	1730

11.4.9	Function Level Reset	1731
11.4.10	Address Translation Services (ATS) and Access Control Services (ACS).....	1731
11.5	Requirements Placed on Host Security due to TDI Requirements	1734
11.5.1	Address Translation.....	1734
11.5.2	MMIO Access Control.....	1734
11.5.3	DMA Access Control.....	1735
11.5.4	Device Binding.....	1736
11.5.5	Securing Interconnects	1736
11.5.6	Data Integrity Errors	1736
11.5.7	TSM Tracking and Handling of Locked Root Port Configurations (Informative)	1736
11.5.8	IDE Extended Capability registers.....	1739
11.6	Overview of Threat Model and Mitigations	1740
11.6.1	Interconnect Security.....	1740
11.6.2	Identity and Measurement Reporting.....	1741
11.6.3	TDI Assignment and Detach.....	1741
12.	Architectural Out-of-Band Management.....	1743
12.1	Introduction.....	1743
12.2	Framework for Sidebands.....	1743
12.3	Sideband Signaling Mechanisms.....	1744
12.3.1	Discrete Sidebands.....	1744
12.3.2	Flex I/O Sidebands	1745
12.3.2.1	Flex I/O Default State Guidelines	1745
12.3.2.2	Flex I/O Discovery Phase Guidelines	1746
12.3.2.3	Flex I/O Compatibility Check Guidelines.....	1746
12.3.2.4	Flex I/O Control Negotiation Guidelines	1747
12.3.2.5	General Flex I/O Control Guidelines	1747
12.3.3	Peripheral Sideband Tunnelling Interface (PESTI) Sidebands	1748
12.3.3.1	PESTI Introduction	1748
12.3.3.2	PESTI Physical Interface.....	1749
12.3.3.3	PESTI Electrical Circuit.....	1749
12.3.3.4	PESTI DC Specifications	1751
12.3.3.5	PESTI Target Detection	1751
12.3.3.6	PESTI Protocol Commands.....	1752
12.3.3.6.1	Discovery Payload Request (DPR)	1752
12.3.3.6.2	PESTI Virtual Wire Exchange (VWE)	1752
12.3.3.6.3	PESTI Fanout MUX Control	1752
12.3.3.7	PESTI Initiator Abort	1753
12.3.3.8	PESTI Broadcast	1753
12.3.3.9	PESTI Initiator Control and Status Registers.....	1754
12.3.3.10	PESTI AC Specifications	1756
12.3.3.11	PESTI Discovery Phase.....	1757
12.3.3.12	PESTI Active Phase.....	1760
12.3.3.13	PESTI Target Reset and Fault Handling.....	1762
12.3.3.14	PESTI Fan-Out.....	1762
12.3.3.15	PESTI Security Considerations	1766
12.4	Managed USB 2.0.....	1766
12.5	2-Wire Interface	1767
12.5.1	2-Wire Interface Use Cases.....	1768
12.5.2	2-Wire Addressing	1768
12.5.3	2-Wire Bus Sharing	1770

12.5.3.1	2-Wire Multi-Drop Topology.....	1770
12.5.3.2	SMBus MUX Use.....	1770
12.5.3.3	2-Wire Hub Use.....	1771
12.5.4	[I3C-Basic] Support on Existing SMBus Signals.....	1772
12.5.4.1	I3C Basic Overview.....	1772
12.5.4.2	I3C Basic Discovery and Mode Changing.....	1773
12.5.4.3	I3C Basic DC and AC Signal Requirements.....	1776
12.6	Field Replacement Unit (FRU) Information.....	1778
12.6.1	FRU Information Device Requirements.....	1778
12.6.1.1	FRU Information Device Requirements Specific to SMBus/I2C Mode.....	1779
12.6.1.2	[SMBus]/[I2C] Access Protocol.....	1780
12.6.2	FRU Information Format.....	1781
12.6.3	Common PCI-SIG MultiRecord Descriptors.....	1783
12.6.3.1	Connector Subdivision (Group ID 0h, Sub-Type 0h).....	1783
12.7	Out-of-Band Management Control Mechanism.....	1785
12.7.1	PCIe-MI Architectural Model.....	1787
12.7.1.1	Operational Times.....	1788
12.7.1.2	PCIe-MI Completer Reset.....	1789
12.7.1.3	Example PCIe-MI Component.....	1791
12.7.1.4	Example PCIe-MI Management Traffic Flows.....	1794
12.7.2	PCIe-MI Message Format.....	1796
12.7.2.1	PCIe-MI Opcodes.....	1799
12.7.2.2	PCIe-MI Status.....	1800
12.7.2.2.1	Invalid Input Error Completion.....	1801
12.7.3	PCIe-MI Commands.....	1802
12.7.3.1	Get Information (Opcode 00h).....	1804
12.7.3.1.1	Supported Information Identifiers (Information Identifier 00h).....	1805
12.7.3.1.2	Supported Opcodes (Information Identifier 01h).....	1805
12.7.3.1.3	Supported Parameters (Information Identifier 02h).....	1806
12.7.3.1.4	Component Information (Information Identifier 03h).....	1807
12.7.3.1.5	Port Information (Information Identifier 04h).....	1808
12.7.3.1.6	Function List (Information Identifier 05h).....	1810
12.7.3.1.7	Function Information (Information Identifier 06h).....	1812
12.7.3.2	Get Parameters (Opcode 01h) and Set Parameters (Opcode 02h).....	1815
12.7.3.2.1	Configuration Space Writability (Parameter Identifier 00h).....	1816
12.7.3.2.1.1	Get Parameters – Configuration Space Writability.....	1817
12.7.3.2.1.2	Set Parameters – Configuration Space Writability.....	1820
12.7.3.2.2	Routing Information (Parameter Identifier 01h).....	1822
12.7.3.2.2.1	Get Parameters – Routing Information.....	1822
12.7.3.2.2.2	Set Parameters – Routing Information.....	1823
12.7.3.2.3	Configuration Space (Parameter Identifier 02h).....	1824
12.7.3.2.3.1	Get Parameters – Configuration Space.....	1824
12.7.3.2.3.2	Set Parameters – Configuration Space.....	1825
12.7.3.2.4	Connector Subdivision (Parameter Identifier 03h).....	1826
12.7.3.2.4.1	Get Parameters – Connector Subdivision.....	1826
12.7.3.2.4.2	Set Parameters – Connector Subdivision.....	1827
12.8	Retimer Management.....	1828
12.9	Internal Cable Management.....	1828
A.	Isochronous Applications.....	1831
A.1	Introduction.....	1831

A.2	Isochronous Contract and Contract Parameters	1832
A.2.1	Isochronous Time Period and Isochronous Virtual Timeslot.....	1833
A.2.2	Isochronous Payload Size	1833
A.2.3	Isochronous Bandwidth Allocation	1833
A.2.4	Isochronous Transaction Latency	1835
A.2.5	An Example Illustrating Isochronous Parameters.....	1836
A.3	Isochronous Transaction Rules	1836
A.4	Transaction Ordering	1836
A.5	Isochronous Data Coherency.....	1836
A.6	Flow Control	1837
A.7	Considerations for Bandwidth Allocation	1837
A.7.1	Isochronous Bandwidth of PCI Express Links.....	1837
A.7.2	Isochronous Bandwidth of Endpoints	1837
A.7.3	Isochronous Bandwidth of Switches.....	1837
A.7.4	Isochronous Bandwidth of Root Complex	1838
A.8	Considerations for PCI Express Components.....	1838
A.8.1	An Endpoint as a Requester	1838
A.8.2	An Endpoint as a Completer	1838
A.8.3	Switches.....	1838
A.8.4	Root Complex	1839
B.	Symbol Encoding	1841
C.	Physical Layer Appendix	1851
C.1	8b/10b Data Scrambling Example	1851
C.2	128b/130b Data Scrambling Example	1856
D.	Request Dependencies	1859
E.	ID-Based Ordering Usage	1863
E.1	Introduction.....	1863
E.2	Potential Benefits with IDO Use.....	1864
E.2.1	Benefits for MFD/RP Direct Connect.....	1864
E.2.2	Benefits for Switched Environments.....	1864
E.2.3	Benefits for Integrated Endpoints	1864
E.2.4	IDO Use in Conjunction with RO	1865
E.3	When to Use IDO	1865
E.4	When Not to Use IDO	1865
E.4.1	When Not to Use IDO with Endpoints.....	1865
E.4.2	When Not to Use IDO with Root Ports	1866
E.5	Software Control of IDO Use	1866
E.5.1	Software Control of Endpoint IDO Use	1866
E.5.2	Software Control of Root Port IDO Use	1867
F.	Message Code Usage	1869
G.	Protocol Multiplexing.....	1871
G.1	Protocol Multiplexing Interactions with PCI Express.....	1872
G.2	PMUX Packets	1876
G.3	PMUX Packet Layout	1877
G.3.1	PMUX Packet Layout for 8b/10b Encoding.....	1877
G.3.2	PMUX Packet Layout at 128b/130b Encoding	1878

G.4	PMUX Control	1881
G.5	PMUX Extended Capability.....	1881
G.5.1	PMUX Extended Capability Header (Offset 00h)	1882
G.5.2	PMUX Capability Register (Offset 04h)	1882
G.5.3	PMUX Control Register (Offset 08h)	1883
G.5.4	PMUX Status Register (Offset 0Ch).....	1885
G.5.5	PMUX Protocol Array (Offsets 10h through 48h)	1886
H.	Flow Control Update Latency and ACK Update Latency Calculations.....	1889
H.1	Flow Control Update Latency	1889
H.2	Ack Latency.....	1891
I.	Async Hot-Plug Reference Model.....	1895
I.1	Async Hot-Plug Initial Configuration	1897
I.2	Async Removal Configuration and Interrupt Handling	1898
I.3	Async Hot-Add Configuration and Interrupt Handling	1900
J.	Alpha Power and Reverse lookup assignment.....	1903
J.1	Alpha Powers	1904
J.2	84 Byte to 86 Byte Encoder	1914
J.3	250 Byte to 256 Byte Encoder example	1915
J.4	86 Byte to 84 Byte Decoder	1918
J.5	256 Byte to 250 Byte decoder	1922
K.	MATLAB created generator matrix for CRC code	1927
K.1	Generator Matrix output	1928
K.2	Flit 8 byte LCRC.....	1960
L.	Example IDE TLPs and Test Keys	2201
L.1	Example NFM IDE TLP Without Partial Header Encryption	2201
L.2	Example FM IDE TLP Without Partial Header Encryption.....	2202
L.3	Example NFM IDE TLP With Partial Header Encryption.....	2204
L.4	Example FM IDE TLP With Partial Header Encryption.....	2206
L.5	IDE Test Keys.....	2208

List of Figures

Figure 1	Old Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Transmit side	85
Figure 2	New Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Transmit side	86
Figure 3	Old Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Receive side	86
Figure 4	New Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Receive side	87
Figure 5	Old Figure: Powers of alpha for the check bits for Bytes 0 to 84	88
Figure 6	New Figure: Powers of alpha for the check bits for Bytes 0 to 84	88
Figure 1-1	PCI Express Link	138
Figure 1-2	Example PCI Express Topology	140
Figure 1-3	Logical Block Diagram of a Switch	143
Figure 1-4	High-Level Layering Diagram	145
Figure 1-5	Packet Flow Through the Layers	145
Figure 2-1	Layering Diagram Highlighting the Transaction Layer	151
Figure 2-2	Serial View of a Non-Flit Mode TLP	153
Figure 2-3	Generic TLP Format - Non-Flit Mode	154
Figure 2-4	Fields Present in All Non-Flit Mode TLPs	155
Figure 2-5	Fields Present in All Non-Flit Mode TLP Headers	156
Figure 2-6	First DW of Header Base	158
Figure 2-7	OHC-A1	171
Figure 2-8	OHC-A2	171
Figure 2-9	OHC-A3	172
Figure 2-10	OHC-A4	172
Figure 2-11	OHC-A5	172
Figure 2-12	OHC-B	173
Figure 2-13	OHC-C	173
Figure 2-14	Example Topology Illustrating Multiple Segments and NFM Subtrees	176
Figure 2-15	Examples of Completer Target Memory Access for FetchAdd	183
Figure 2-16	32-bit Address Routing - Non-Flit Mode	185
Figure 2-17	64-bit Address Routing - Non-Flit Mode	185
Figure 2-18	32-bit Address Routing - Flit Mode	186
Figure 2-19	64-bit Address Routing - Flit Mode - 4 DW	186
Figure 2-20	64-bit Address Routing - Flit Mode - 5 DW	186
Figure 2-21	64-bit Address Routing - Flit Mode - 6 DW	187
Figure 2-22	64-bit Address Routing - Flit Mode - 7 DW	187
Figure 2-23	Non-ARI ID Routing with 4 DW Header - Non-Flit Mode	189
Figure 2-24	ARI ID Routing with 4 DW Header - Non-Flit Mode	190
Figure 2-25	Non-ARI ID Routing with 3 DW Header - Non-Flit Mode	190
Figure 2-26	ARI ID Routing with 3 DW Header - Non-Flit Mode	190
Figure 2-27	ID Routing with 3 DW Header - Flit Mode	190
Figure 2-28	ID Routing with 4 DW Header - Flit Mode	191
Figure 2-29	ID Routing with 5 DW Header - Flit Mode	191
Figure 2-30	ID Routing with 6 DW Header - Flit Mode	191
Figure 2-31	ID Routing with 7 DW Header - Flit Mode	192
Figure 2-32	Location of Byte Enables in TLP Header - Non-Flit Mode	193
Figure 2-33	Transaction Descriptor	196

Figure 2-34	Transaction ID	196
Figure 2-35	Attributes Field of Transaction Descriptor	204
Figure 2-36	Memory Request for 64-bit Addressing Header - Non-Flit Mode	207
Figure 2-37	Memory Request for 32-bit Addressing Header - Non-Flit Mode	207
Figure 2-38	I/O Request Header - Non-Flit Mode	208
Figure 2-39	Configuration Request Header - Non-Flit Mode.....	209
Figure 2-40	TPH TLP Prefix - Non-Flit Mode	209
Figure 2-41	Location of PH[1:0] in a 4 DW Request Header - Non-Flit Mode.....	210
Figure 2-42	Location of PH[1:0] in a 3 DW Request Header - Non-Flit Mode.....	211
Figure 2-43	Location of ST[7:0] in the Memory Write Request Header - Non-Flit Mode	211
Figure 2-44	Location of ST[7:0] in Memory Read, DMWr, and AtomicOp Request Headers - Non-Flit Mode.....	212
Figure 2-45	Memory Request with 64-bit Addressing Header - Flit Mode.....	213
Figure 2-46	Memory Request with 32-bit Addressing Header - Flit Mode.....	213
Figure 2-47	I/O Request Header - Flit Mode.....	213
Figure 2-48	Configuration Request Header - Flit Mode.....	214
Figure 2-49	Message Request Header - Non-Flit Mode	215
Figure 2-50	Message Request Header - Flit Mode.....	215
Figure 2-51	ERR_COR Message - Non-Flit Mode	222
Figure 2-52	ERR_COR Message - Flit Mode	222
Figure 2-53	Header for Vendor-Defined Messages - Non-Flit Mode.....	225
Figure 2-54	Header for Vendor-Defined Messages - Flit Mode.....	225
Figure 2-55	Header for PCI-SIG-Defined VDMs - Non-Flit Mode.....	226
Figure 2-56	Header for PCI-SIG-Defined VDMs - Flit Mode.....	226
Figure 2-57	DRS Message - Non-Flit Mode	227
Figure 2-58	DRS Message - Flit Mode	228
Figure 2-59	FRS Message - Non-Flit Mode.....	229
Figure 2-60	FRS Message - Flit Mode.....	229
Figure 2-61	Hierarchy ID Message - Non-Flit Mode	231
Figure 2-62	Hierarchy ID Message - Flit Mode.....	231
Figure 2-63	LTR Message - Non-Flit Mode	233
Figure 2-64	LTR Message - Flit Mode	233
Figure 2-65	OBFF Message - Non-Flit Mode.....	234
Figure 2-66	OBFF Message - Flit Mode	234
Figure 2-67	PTM Request/Response Message - Non-Flit Mode.....	235
Figure 2-68	PTM ResponseD Message - Non-Flit Mode	236
Figure 2-69	PTM Request/Response Message - Flit Mode.....	236
Figure 2-70	PTM ResponseD Message - Flit Mode	236
Figure 2-71	IDE Sync Message for Link IDE Stream - Non-Flit Mode.....	238
Figure 2-72	IDE Sync Message for Link IDE Stream - Flit Mode.....	238
Figure 2-73	IDE Sync Message for Selective IDE Stream - Non-Flit Mode.....	239
Figure 2-74	IDE Sync Message for Selective IDE Stream - Flit Mode.....	239
Figure 2-75	IDE Fail Message for Link IDE Stream - Non-Flit Mode.....	239
Figure 2-76	IDE Fail Message for Link IDE Stream - Flit Mode	240
Figure 2-77	IDE Fail Message for Selective IDE Stream - Non-Flit Mode.....	240
Figure 2-78	IDE Fail Message for Selective IDE Stream - Flit Mode	240
Figure 2-79	Completion Header Format - Non-Flit Mode	241
Figure 2-80	(Non-ARI) Completer ID.....	242
Figure 2-81	ARI Completer ID	242
Figure 2-82	Completion Header Base Format - Non-UIO Flit Mode	244
Figure 2-83	Completion Header Base Format – UIOWrCpl and UIORdCpl - Flit Mode	244
Figure 2-84	Completion Header Base Format - UIORdCplD - Flit Mode	244
Figure 2-85	Flit Mode Local TLP Prefix.....	248

Figure 2-86	OHC-E1.....	250
Figure 2-87	OHC-E2.....	251
Figure 2-88	OHC-E4.....	251
Figure 2-89	Flowchart for Handling of Received TLPs	253
Figure 2-90	Flowchart for Switch Handling of TLPs.....	255
Figure 2-91	Flowchart for Handling of Received Requests	260
Figure 2-92	Example Completion Data when some Byte Enables are 0b.....	263
Figure 2-93	Deadlock Examples with Intersystem Interconnects	274
Figure 2-94	Virtual Channel Concept - An Illustration.....	281
Figure 2-95	Virtual Channel Concept - Switch Internals (Upstream Flow).....	281
Figure 2-96	An Example of TC/VC Configurations	284
Figure 2-97	Relationship Between Requester and Ultimate Completer	285
Figure 2-98	Credit Block Example	294
Figure 2-99	Calculation of 32-bit ECRC for TLP End to End Data Integrity Protection	311
Figure 3-1	Layering Diagram Highlighting the Data Link Layer	319
Figure 3-2	Data Link Control and Management State Machine	321
Figure 3-3	VC0 Flow Control Initialization Example with 8b/10b Encoding-based Framing - Non-Flit Mode	333
Figure 3-4	DLLP Type and CRC Fields (Non-Flit Mode)	335
Figure 3-5	DLLP Type Field (Flit Mode)	335
Figure 3-6	Data Link Layer Packet Format for Ack and Nak (Non-Flit Mode)	339
Figure 3-7	Data Link Layer Packet Format for NOP	339
Figure 3-8	Data Link Layer Packet Format for NOP2 (Flit Mode)	339
Figure 3-9	Data Link Layer Packet Format for InitFC1.....	340
Figure 3-10	Data Link Layer Packet Format for InitFC2.....	340
Figure 3-11	Data Link Layer Packet Format for UpdateFC	341
Figure 3-12	Data Link Layer Packet Format for Power Management	341
Figure 3-13	Data Link Layer Packet Format for Vendor-Specific	341
Figure 3-14	Data Link Layer Packet Format for Data Link Feature DLLP	342
Figure 3-15	Data Link Packet Layer Format for Link Management (Flit Mode)	342
Figure 3-16	Diagram of CRC Calculation for DLLPs	344
Figure 3-17	TLP with LCRC and TLP Sequence Number Applied - Non-Flit Mode	345
Figure 3-18	TLP Following Application of TLP Sequence Number and 4 Bits.....	347
Figure 3-19	Calculation of LCRC.....	349
Figure 3-20	Received DLLP Error Check Flowchart	353
Figure 3-21	Ack/Nak DLLP Processing Flowchart	354
Figure 3-22	Receive Data Link Layer Handling of TLPs Flowchart	357
Figure 4-1	Layering Diagram Highlighting Physical Layer	361
Figure 4-2	Character to Symbol Mapping	364
Figure 4-3	Bit Transmission Order on Physical Lanes - x1 Example	364
Figure 4-4	Bit Transmission Order on Physical Lanes - x4 Example	365
Figure 4-5	TLP with Framing Symbols Applied	368
Figure 4-6	DLLP with Framing Symbols Applied	369
Figure 4-7	Framed TLP on a x1 Link	369
Figure 4-8	Framed TLP on a x2 Link	370
Figure 4-9	Framed TLP on a x4 Link	370
Figure 4-10	LFSR with 8b/10b Scrambling Polynomial.....	371
Figure 4-11	Example of Bit Transmission Order in a x1 Link Showing 130 Bits of a Block	372
Figure 4-12	Example of Bit Placement in a x4 Link with One Block per Lane	373
Figure 4-13	Layout of Framing Tokens.....	376
Figure 4-14	TLP and DLLP Layout	378
Figure 4-15	Packet Transmission in a x8 Link	378
Figure 4-16	Nullified TLP Layout in a x8 Link with Other Packets	378

Figure 4-17	SKP Ordered Set of Length 66-bit in a x8 Link	379
Figure 4-18	LFSR with Scrambling Polynomial in 8.0 GT/s and Above Data Rate	386
Figure 4-19	Alternate Implementation of the LFSR for Descrambling	388
Figure 4-20	Precoding working the scrambler/ de-scrambler	390
Figure 4-21	Example of Symbol placement in a x4 Link with 1b/1b encoding	393
Figure 4-22	Transmit side at 64.0 GT/s	394
Figure 4-23	Receive side at 64.0 GT/s	394
Figure 4-24	PAM4 Signaling at UI level: Voltage levels, 2-bit encoding, and their corresponding DC balance values	395
Figure 4-25	The Sequence of Gray Coding, Precoding, and PAM4 voltage translation on an aligned 2-bit boundary on a per Lane	396
Figure 4-26	Processing of Ordered Sets during or at the end of a Data Stream in Flit Mode at 64.0 GT/s Data Rate	404
Figure 4-27	Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Transmit side	409
Figure 4-28	Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Receive side	410
Figure 4-29	DLP Byte to Bit Number Assignment	414
Figure 4-30	DLP Bit usage	414
Figure 4-31	Optimized_Update_FC	415
Figure 4-32	Flit_Marker	416
Figure 4-33	Flit Ack, Nak, and Discard Rules Flow Chart (Zoom-In to View)	435
Figure 4-34	Flit Ack/Nak/Replay Example	440
Figure 4-35	NOP Flit Common Header	442
Figure 4-36	NOP.Empty Flit Payload	443
Figure 4-37	NOP.Debug Flit Debug Chunk	444
Figure 4-38	Example Debug Chunk with one DW Debug Chunk Header and one DW of Debug Chunk Payload ..	445
Figure 4-39	Example Debug Chunk with two DW Debug Chunk Header and one DW of Debug Chunk Payload ..	445
Figure 4-40	Example Debug Chunk with four DW Debug Chunk Header and one DW of Debug Chunk Payload	445
Figure 4-41	Example NOP.Debug Flit Payload with a single Debug Chunk with a one DW Debug Chunk Header ..	446
Figure 4-42	Example NOP.Debug Flit Payload with multiple Debug Chunks with one DW Debug Chunk Headers	446
Figure 4-43	Empty Debug Chunk	448
Figure 4-44	Start Capture Trigger Debug Chunk	448
Figure 4-45	Stop Capture Trigger Debug Chunk	449
Figure 4-46	FC Information Tracked by Transmitter Debug Chunk	449
Figure 4-47	FC Information Tracked by Receiver Debug Chunk	451
Figure 4-48	Flit Mode Transmitter Retry Flags and Counters Debug Chunk	452
Figure 4-49	Flit Mode Receiver Retry Flags and Counters Debug Chunk	453
Figure 4-50	Buffer Occupancy Debug Chunk	454
Figure 4-51	Link Debug Request Debug Chunk	455
Figure 4-52	NOP.Vendor Flit Payload	455
Figure 4-53	CRC generation/ checking in Flit	456
Figure 4-54	FEC Table: i to α^i	457
Figure 4-55	FEC Log Table: α^i to i	458
Figure 4-56	H-matrix of the FEC	458
Figure 4-57	Weight of check bits for different Bytes/bits	459
Figure 4-58	ECC Decoder function	461
Figure 4-59	3-way ECC decode followed by CRC check of flit on the Receive side	462
Figure 4-60	8.0 GT/s Equalization Flow	473
Figure 4-61	16.0 GT/s Equalization Flow	473

Figure 4-62	64.0 GT/s Equalization Flow	474
Figure 4-63	Equalization Bypass Example	475
Figure 4-64	Alternate Protocol Negotiation and Equalization Bypass LTSSM States	500
Figure 4-65	Electrical Idle Exit Ordered Set for 8.0 GT/s to 32.0 GT/s Data Rates (EIEOS)	505
Figure 4-66	Example of L0p flow in a x16 Link	525
Figure 4-67	Main State Diagram for Link Training and Status State Machine	529
Figure 4-68	Detect Substate Machine	531
Figure 4-69	Polling Substate Machine	539
Figure 4-70	Configuration Substate Machine	558
Figure 4-71	Recovery Substate Machine	590
Figure 4-72	L0s Substate Machine	595
Figure 4-73	L1 Substate Machine	597
Figure 4-74	L2 Substate Machine	598
Figure 4-75	Loopback Substate Machine	606
Figure 4-76	Margin PHY Payload for Control SKP Ordered Set with 1b/1b Encoding	614
Figure 4-77	LFSR PHY Payload for Control SKP Ordered Set with 1b/1b Encoding	615
Figure 4-78	Polling.Compliance PHY Payload for Control SKP Ordered Set with 1b/1b Encoding	615
Figure 4-79	Receiver Number Assignment	633
Figure 4-80	Supported Retimer Topologies	649
Figure 4-81	Supported Retimer Topology Using an Optical Retimer Solution	650
Figure 4-82	Optical Aware Retimer Example – Flow Diagram of Steps 1 to 6	654
Figure 4-83	Sample Retimer Topology Using PCIe Optical Aware Retimers	655
Figure 4-84	Flit Interleaving in a x8 Link	657
Figure 4-85	Flit Interleaving of a x8 Link in a 1:2 Electrical Lane to Optical Channel Mapping	658
Figure 4-86	Flit Interleaving of a x8 Link in a 2:1 Electrical Lane to Optical Channel MappingLink	658
Figure 4-87	Retimer CLKREQ# Connection Topology	687
Figure 5-1	Link Power Management State Flow Diagram	694
Figure 5-2	Entry into the L1 Link State	702
Figure 5-3	Exit from L1 Link State Initiated by Upstream Component	704
Figure 5-4	Conceptual Diagrams Showing Two Example Cases of WAKE# Routing	707
Figure 5-5	A Conceptual PME Control State Machine	710
Figure 5-6	L1 Transition Sequence Ending with a Rejection (L0s Enabled)	721
Figure 5-7	L1 Successful Transition Sequence	722
Figure 5-8	Example of L1 Exit Latency Computation	723
Figure 5-9	State Diagram for L1 PM Substates	729
Figure 5-10	Downstream Port with a Single PLL	730
Figure 5-11	Multiple Downstream Ports with a shared PLL	731
Figure 5-12	Example: L1.1 Waveforms Illustrating Upstream Port Initiated Exit	733
Figure 5-13	Example: L1.1 Waveforms Illustrating Downstream Port Initiated Exit	734
Figure 5-14	L1.2 Substates	735
Figure 5-15	Example: Illustration of Boundary Condition due to Different Sampling of CLKREQ#	736
Figure 5-16	Example: L1.2 Waveforms Illustrating Upstream Port Initiated Exit	737
Figure 5-17	Example: L1.2 Waveforms Illustrating Downstream Port Initiated Exit	738
Figure 5-18	Function Power Management State Transitions	742
Figure 5-19	PCI Express Bridge Power Management Diagram	745
Figure 6-1	Error Classification	758
Figure 6-2	Flowchart Showing Sequence of Device Error Signaling and Logging Operations	773
Figure 6-3	Pseudo Logic Diagram for Selected Error Message Control and Status Bits	774
Figure 6-4	TC Filtering Example	792
Figure 6-5	TC to VC Mapping Example	793
Figure 6-6	An Example of Traffic Flow Illustrating Ingress and Egress	794
Figure 6-7	An Example of Differentiated Traffic Flow Through a Switch	794

Figure 6-8	Switch Arbitration Structure.....	795
Figure 6-9	VC ID and Priority Order - An Example.....	796
Figure 6-10	Multi-Function Arbitration Model.....	798
Figure 6-11	Root Complex Represented as a Single Component.....	844
Figure 6-12	Root Complex Represented as Multiple Components.....	844
Figure 6-13	Example System Topology with ARI Devices.....	859
Figure 6-14	Segmentation of the Multicast Address Range.....	860
Figure 6-15	Latency Fields Format for LTR Messages.....	875
Figure 6-16	CLKREQ# and Clock Power Management.....	879
Figure 6-17	Use of LTR and Clock Power Management.....	880
Figure 6-18	Codes and Equivalent WAKE# Patterns.....	881
Figure 6-19	Example Platform Topology Showing a Link Where OBFF is Carried by Messages.....	882
Figure 6-20	PASID TLP Prefix.....	885
Figure 6-21	Example System Topologies using PTM.....	889
Figure 6-22	Precision Time Measurement Link Protocol.....	890
Figure 6-23	Precision Time Measurement Example.....	892
Figure 6-24	PTM Requester Operation.....	895
Figure 6-25	PTM Timestamp Capture Example.....	898
Figure 6-26	Example Illustrating Application of Enhanced Allocation.....	904
Figure 6-27	Emergency Power Reduction State: Example Add-in Card.....	908
Figure 6-28	FPB High Level Diagram and Example Topology.....	913
Figure 6-29	Example Illustrating “Flattening” of a Switch.....	914
Figure 6-30	Vector Mechanism for Address Range Decoding.....	915
Figure 6-31	Relationship between FPB and non-FPB Decode Mechanisms.....	916
Figure 6-32	Routing IDs (RIDs) and Supported Granularities.....	918
Figure 6-33	Addresses in Memory Below 4 GB and Effect of Granularity.....	920
Figure 6-34	VPD Format.....	925
Figure 6-35	Example NPEM Configuration using a Downstream Port.....	930
Figure 6-36	Example NPEM Configuration using an Upstream Port.....	931
Figure 6-37	NPEM Command Flow.....	932
Figure 6-38	Stack Diagram Illustration of Multiple Sessions and Connections.....	938
Figure 6-39	Example Showing Relationships of Software and Hardware Elements.....	939
Figure 6-40	DOE Data Object Format.....	939
Figure 6-41	DOE Data Object Header 1.....	940
Figure 6-42	DOE Data Object Header 2.....	940
Figure 6-43	DOE Discovery Request Data Object Contents (3rd DW).....	941
Figure 6-44	DOE Discovery Response Data Object Contents (3rd DW).....	941
Figure 6-45	DOE Discovery Response Data Object Contents (4th DW).....	942
Figure 6-46	CMA-SPDM as Part of a Layered Architecture.....	948
Figure 6-47	Example System Showing Multiple Access Mechanisms.....	950
Figure 6-48	Example Add-In-Card Supporting CMA-SPDM.....	951
Figure 6-49	Byte Mapping of SPDM Messages Including Example Payload.....	954
Figure 6-50	Byte Mapping of Secured CMA-SPDM Messages Including Example Payload.....	955
Figure 6-51	Example DMW _r Data Payload Template.....	960
Figure 6-52	IDE Secures TLPs Between Ports.....	962
Figure 6-53	IDE Stream State Machine.....	966
Figure 6-54	IDE Stream State Machine.....	967
Figure 6-55	IDE Key Management (IDE_KM) and Related Specifications & Capabilities.....	969
Figure 6-56	Query (QUERY) Data Object.....	974
Figure 6-57	Query Response (QUERY_RESP) Data Object.....	975
Figure 6-58	Key Programming (KEY_PROG) Data Object with Example 256b Key.....	975
Figure 6-59	Key Programming Acknowledgement (KP_ACK) Data Object.....	976

Figure 6-60	Key Set Go (K_SET_GO) Data Object	976
Figure 6-61	Key Set Stop (K_SET_STOP) Data Object	976
Figure 6-62	Key Set Go/Stop Acknowledgement (K_GOSTOP_ACK) Data Object	976
Figure 6-63	IDE_KM Example	978
Figure 6-64	IDE TLP Prefix (NFM).....	979
Figure 6-65	MAC Layout.....	980
Figure 6-66	Example of IDE TLP for a Link IDE Stream without Aggregation (Non-Flit Mode)	980
Figure 6-67	IDE TLP – Example Showing Aggregation of Two TLPs for a Link IDE Stream (Non-Flit Mode)	981
Figure 6-68	IDE TLP – Example of IDE TLP for a Selective IDE Stream without Aggregation (Non-Flit Mode).....	981
Figure 6-69	IDE TLP – Example Showing Aggregation of Two TLPs for a Selective IDE Stream (Non-Flit Mode) ...	981
Figure 6-70	Example of IDE TLP for a Link IDE Stream without Aggregation (Flit Mode)	982
Figure 6-71	IDE TLP – Example Showing Aggregation of Two TLPs for a Link IDE Stream (Flit Mode)	982
Figure 6-72	IDE TLP – Example of IDE TLP for a Selective IDE Stream without Aggregation (Flit Mode).....	983
Figure 6-73	IDE TLP – Example Showing Aggregation of Two TLPs for a Selective IDE Stream (Flit Mode)	983
Figure 6-74	High Level Flow For Partial Header Encryption	984
Figure 6-75	Partial Header Encryption in NFM with Byte Enables.....	986
Figure 6-76	Partial Header Encryption in NFM without Byte Enables.....	987
Figure 6-77	Partial Header Encryption in FM with OHC-A1.....	988
Figure 6-78	Partial Header Encryption in FM without OHC-A1	989
Figure 6-79	Example Illustrating PCRC Application to Two Aggregated IDE TLPs for a Link IDE Stream (NFM)	990
Figure 6-80	Example – Posted Requests Allowed to Bypass Non-Posted Requests	999
Figure 6-81	Example – Non-Posted Requests Never Allowed to Bypass Posted Requests.....	1000
Figure 6-82	Example – Secure Non-Posted Request Reordering Not Allowed Over PCIe Fabric	1000
Figure 6-83	MMIO Register Blocks.....	1008
Figure 6-84	MCAP Register Block	1009
Figure 6-85	MCAP Array Register Block.....	1010
Figure 6-86	MCAP Array Register 1	1010
Figure 6-87	MCAP Array Register 2	1011
Figure 6-88	MCAP Header Register Block	1011
Figure 6-89	MCAP Header Register 1.....	1011
Figure 6-90	MCAP Header Register 2.....	1012
Figure 6-91	MCAP Header Register 3.....	1012
Figure 6-92	MCAP Header Register 4.....	1013
Figure 6-93	MMB Registers	1015
Figure 6-94	MMB Capabilities Register	1016
Figure 6-95	MMB Control Register.....	1017
Figure 6-96	MMB Command Register.....	1018
Figure 6-97	MMB Status Register.....	1019
Figure 6-98	MMB Payload Registers	1021
Figure 6-99	MMPT Registers	1021
Figure 6-100	MMPT Capabilities Register	1022
Figure 6-101	MMPT Control Register.....	1022
Figure 6-102	MMPT Receive Message Notification Register	1023
Figure 6-103	MDVS Register Block	1024
Figure 6-104	MDVS Register Block Header Register 1	1024
Figure 6-105	MDVS Register Block Header Register 2	1025
Figure 6-106	MDVS Register Block Header Register 3	1025
Figure 7-1	PCI Express Root Complex Device Mapping.....	1033
Figure 7-2	PCI Express Switch Device Mapping	1034
Figure 7-3	PCI Express Configuration Space Layout	1035
Figure 7-4	Common Configuration Space Header	1046
Figure 7-5	Command Register.....	1047

Figure 7-6	Status Register.....	1050
Figure 7-7	Class Code Register	1053
Figure 7-8	Header Type Register	1054
Figure 7-9	BIST Register.....	1055
Figure 7-10	Type 0 Configuration Space Header.....	1057
Figure 7-11	Base Address Register for Memory.....	1058
Figure 7-12	Base Address Register for I/O.....	1058
Figure 7-13	Expansion ROM Base Address Register	1063
Figure 7-14	Type 1 Configuration Space Header.....	1066
Figure 7-15	Secondary Status Register	1069
Figure 7-16	Bridge Control Register	1072
Figure 7-17	PCI Power Management Capability Structure	1074
Figure 7-18	Power Management Capabilities Register	1075
Figure 7-19	Power Management Control/Status Register.....	1077
Figure 7-20	Power Management Data Register	1079
Figure 7-21	PCI Express Capability Structure	1081
Figure 7-22	PCI Express Capability List Register.....	1082
Figure 7-23	PCI Express Capabilities Register.....	1082
Figure 7-24	Device Capabilities Register.....	1084
Figure 7-25	Device Control Register.....	1088
Figure 7-26	Device Status Register.....	1095
Figure 7-27	Link Capabilities Register.....	1097
Figure 7-28	Link Control Register.....	1101
Figure 7-29	Link Status Register	1108
Figure 7-30	Slot Capabilities Register	1110
Figure 7-31	Slot Control Register	1113
Figure 7-32	Slot Status Register	1116
Figure 7-33	Root Control Register	1118
Figure 7-34	Root Capabilities Register	1119
Figure 7-35	Root Status Register	1120
Figure 7-36	Device Capabilities 2 Register.....	1121
Figure 7-37	Device Control 2 Register.....	1126
Figure 7-38	Link Capabilities 2 Register.....	1130
Figure 7-39	Link Control 2 Register	1133
Figure 7-40	Link Status 2 Register	1136
Figure 7-41	Slot Capabilities 2 Register	1140
Figure 7-42	PCI Express Extended Configuration Space Layout.....	1141
Figure 7-43	PCI Express Extended Capability Header	1142
Figure 7-44	MSI Capability Structure for 32-bit Message Address.....	1143
Figure 7-45	MSI Capability Structure for 64-bit Message Address.....	1143
Figure 7-46	MSI Capability Structure for 32-bit Message Address and PVM.....	1143
Figure 7-47	MSI Capability Structure for 64-bit Message Address and PVM.....	1144
Figure 7-48	MSI Capability Header.....	1144
Figure 7-49	Message Control Register for MSI	1145
Figure 7-50	Message Address Register for MSI	1147
Figure 7-51	Message Upper Address Register for MSI	1147
Figure 7-52	Message Data Register for MSI	1148
Figure 7-53	Extended Message Data Register for MSI.....	1148
Figure 7-54	Mask Bits Register for MSI	1149
Figure 7-55	Pending Bits Register for MSI	1149
Figure 7-56	MSI-X Capability Structure	1150
Figure 7-57	MSI-X Table Structure.....	1151

Figure 7-58	MSI-X PBA Structure	1151
Figure 7-59	MSI-X Capability Header	1153
Figure 7-60	Message Control Register for MSI-X	1154
Figure 7-61	Table Offset/Table BIR Register for MSI-X	1155
Figure 7-62	PBA Offset/PBA BIR Register for MSI-X	1155
Figure 7-63	Message Address Register for MSI-X Table Entries	1156
Figure 7-64	Message Upper Address Register for MSI-X Table Entries	1157
Figure 7-65	Message Data Register for MSI-X Table Entries	1157
Figure 7-66	Vector Control Register for MSI-X Table Entries	1158
Figure 7-67	Pending Bits Register for MSI-X PBA Entries	1158
Figure 7-68	Secondary PCI Express Extended Capability Structure	1160
Figure 7-69	Secondary PCI Express Extended Capability Header	1161
Figure 7-70	Link Control 3 Register	1161
Figure 7-71	Lane Error Status Register	1162
Figure 7-72	Lane Equalization Control Register	1163
Figure 7-73	Lane Equalization Control Register Entry	1163
Figure 7-74	Data Link Feature Extended Capability	1166
Figure 7-75	Data Link Feature Extended Capability Header	1166
Figure 7-76	Data Link Feature Capabilities Register	1167
Figure 7-77	Data Link Feature Status Register	1168
Figure 7-78	Physical Layer 16.0 GT/s Extended Capability	1170
Figure 7-79	Physical Layer 16.0 GT/s Extended Capability Header	1171
Figure 7-80	16.0 GT/s Capabilities Register	1171
Figure 7-81	16.0 GT/s Control Register	1172
Figure 7-82	16.0 GT/s Status Register	1172
Figure 7-83	16.0 GT/s Local Data Parity Mismatch Status Register	1173
Figure 7-84	16.0 GT/s First Retimer Data Parity Mismatch Status Register	1174
Figure 7-85	16.0 GT/s Second Retimer Data Parity Mismatch Status Register	1174
Figure 7-86	16.0 GT/s Lane Equalization Control Register Entry	1175
Figure 7-87	Physical Layer 32.0 GT/s Extended Capability	1177
Figure 7-88	Physical Layer 32.0 GT/s Extended Capability Header	1178
Figure 7-89	32.0 GT/s Capabilities Register	1178
Figure 7-90	32.0 GT/s Control Register	1179
Figure 7-91	32.0 GT/s Status Register	1180
Figure 7-92	Received Modified TS Data 1 Register	1181
Figure 7-93	Received Modified TS Data 2 Register	1182
Figure 7-94	Transmitted Modified TS Data 1 Register	1183
Figure 7-95	Transmitted Modified TS Data 2 Register	1184
Figure 7-96	32.0 GT/s Lane Equalization Control Register Entry	1185
Figure 7-97	Physical Layer 64.0 GT/s Extended Capability	1187
Figure 7-98	Physical Layer 64.0 GT/s Extended Capability Header	1187
Figure 7-99	64.0 GT/s Capabilities Register	1188
Figure 7-100	64.0 GT/s Control Register	1188
Figure 7-101	64.0 GT/s Status Register	1189
Figure 7-102	64.0 GT/s Lane Equalization Control Register Entry	1190
Figure 7-103	Flit Logging Extended Capability Structure	1192
Figure 7-104	Flit Logging Extended Capability Header	1192
Figure 7-105	Flit Error Log 1 Register	1194
Figure 7-106	Flit Error Log 2 Register	1196
Figure 7-107	Flit Error Counter Control Register	1196
Figure 7-108	Flit Error Counter Status Register	1197
Figure 7-109	FBER Measurement Control Register	1198

Figure 7-110	FBER Measurement Status 1 Register.....	1199
Figure 7-111	FBER Measurement Status 2 Register.....	1200
Figure 7-112	FBER Measurement Status 3 Register.....	1200
Figure 7-113	FBER Measurement Status 4 Register.....	1201
Figure 7-114	FBER Measurement Status 5 Register.....	1201
Figure 7-115	FBER Measurement Status 6 Register.....	1201
Figure 7-116	FBER Measurement Status 7 Register.....	1202
Figure 7-117	FBER Measurement Status 8 Register.....	1202
Figure 7-118	FBER Measurement Status 9 Register.....	1203
Figure 7-119	FBER Measurement Status 10 Register.....	1203
Figure 7-120	Device 3 Extended Capability Structure	1204
Figure 7-121	Device 3 Extended Capability Header	1204
Figure 7-122	Device Capabilities 3 Register.....	1205
Figure 7-123	Device Control 3 Register	1208
Figure 7-124	Device Status 3 Register	1210
Figure 7-125	Lane Margining at the Receiver Extended Capability	1212
Figure 7-126	Lane Margining at the Receiver Extended Capability Header	1213
Figure 7-127	Margining Port Capabilities Register	1213
Figure 7-128	Margining Port Status Register	1214
Figure 7-129	Lane N: Margining Control Register Entry	1215
Figure 7-130	Lane N: Margining Lane Status Register Entry	1216
Figure 7-131	ACS Extended Capability	1217
Figure 7-132	ACS Extended Capability Header.....	1217
Figure 7-133	ACS Capability Register	1218
Figure 7-134	ACS Control Register	1219
Figure 7-135	Egress Control Vector Register.....	1222
Figure 7-136	Power Budgeting Extended Capability.....	1223
Figure 7-137	Power Budgeting Extended Capability Header.....	1224
Figure 7-138	Power Budgeting Control Register	1225
Figure 7-139	Power Budgeting Data Register	1227
Figure 7-140	Power Budgeting Capability Register.....	1231
Figure 7-141	Power Budgeting Sense Detect Register	1233
Figure 7-142	LTR Extended Capability Structure	1236
Figure 7-143	LTR Extended Capability Header	1236
Figure 7-144	Max Snoop Latency Register.....	1236
Figure 7-145	Max No-Snoop Latency Register	1237
Figure 7-146	LTR Capabilities Register.....	1238
Figure 7-147	L1 PM Substates Extended Capability	1238
Figure 7-148	L1 PM Substates Extended Capability Header	1239
Figure 7-149	L1 PM Substates Capabilities Register.....	1240
Figure 7-150	L1 PM Substates Control 1 Register	1241
Figure 7-151	L1 PM Substates Control 2 Register	1243
Figure 7-152	L1 PM Substates Status Register	1244
Figure 7-153	Advanced Error Reporting Extended Capability - Functions that do not support Flit Mode Structure	1245
Figure 7-154	Advanced Error Reporting Extended Capability - Functions that support Flit Mode Structure	1246
Figure 7-155	Advanced Error Reporting Extended Capability Header	1247
Figure 7-156	Uncorrectable Error Status Register	1248
Figure 7-157	Uncorrectable Error Mask Register.....	1250
Figure 7-158	Uncorrectable Error Severity Register	1253
Figure 7-159	Correctable Error Status Register	1255
Figure 7-160	Correctable Error Mask Register	1256

Figure 7-161	Advanced Error Capabilities and Control Register	1257
Figure 7-162	Header Log Register	1260
Figure 7-163	Root Error Command Register	1260
Figure 7-164	Root Error Status Register	1263
Figure 7-165	Error Source Identification Register	1265
Figure 7-166	TLP Prefix Log Register	1266
Figure 7-167	First DW of Enhanced Allocation Capability	1266
Figure 7-168	Second DW of Enhanced Allocation Capability	1267
Figure 7-169	First DW of Each Entry for Enhanced Allocation Capability	1268
Figure 7-170	Format of Entry for Enhanced Allocation Capability	1270
Figure 7-171	Example Entry with 64b Base and 64b MaxOffset	1271
Figure 7-172	Example Entry with 64b Base and 32b MaxOffset	1272
Figure 7-173	Example Entry with 32b Base and 64b MaxOffset	1272
Figure 7-174	Example Entry with 32b Base and 32b MaxOffset	1272
Figure 7-175	Resizable BAR Extended Capability	1274
Figure 7-176	Resizable BAR Extended Capability Header	1275
Figure 7-177	Resizable BAR Capability Register	1276
Figure 7-178	Resizable BAR Control Register	1278
Figure 7-179	VF Resizable BAR Extended Capability	1281
Figure 7-180	VF Resizable BAR Extended Capability Header	1282
Figure 7-181	VF Resizable BAR Control Register	1283
Figure 7-182	ARI Extended Capability	1284
Figure 7-183	ARI Extended Capability Header	1284
Figure 7-184	ARI Capability Register	1285
Figure 7-185	ARI Control Register	1286
Figure 7-186	PASID Extended Capability Structure	1287
Figure 7-187	PASID Extended Capability Header	1287
Figure 7-188	PASID Capability Register	1288
Figure 7-189	PASID Control Register	1289
Figure 7-190	FRS Queueing Extended Capability	1290
Figure 7-191	FRS Queueing Extended Capability Header	1290
Figure 7-192	FRS Queueing Capability Register	1291
Figure 7-193	FRS Queueing Status Register	1291
Figure 7-194	FRS Queueing Control Register	1292
Figure 7-195	FRS Message Queue Register	1292
Figure 7-196	FPB Capability Structure	1293
Figure 7-197	FPB Capability Header	1294
Figure 7-198	FPB Capabilities Register	1294
Figure 7-199	FPB RID Vector Control 1 Register	1296
Figure 7-200	FPB RID Vector Control 2 Register	1298
Figure 7-201	FPB MEM Low Vector Control Register	1298
Figure 7-202	FPB MEM High Vector Control 1 Register	1300
Figure 7-203	FPB MEM High Vector Control 2 Register	1301
Figure 7-204	FPB Vector Access Control Register	1302
Figure 7-205	FPB Vector Access Data Register	1303
Figure 7-206	Flit Performance Measurement Extended Capability Structure	1304
Figure 7-207	Flit Performance Measurement Extended Capability Header	1305
Figure 7-208	Flit Performance Measurement Capability Register	1305
Figure 7-209	Flit Performance Measurement Control Register	1306
Figure 7-210	Flit Performance Measurement Status Register	1308
Figure 7-211	LTSSM Performance Measurement Status Register	1309
Figure 7-212	Flit Error Injection Extended Capability Structure	1311

Figure 7-213	Flit Error Injection Extended Capability Header	1311
Figure 7-214	Flit Error Injection Capability Register	1312
Figure 7-215	Flit Error Injection Control 1 Register	1312
Figure 7-216	Flit Error Injection Control 2 Register	1314
Figure 7-217	Flit Error Injection Status Register	1315
Figure 7-218	Ordered Set Error Injection Control 1 Register	1316
Figure 7-219	Ordered Set Error Injection Control 2 Register	1317
Figure 7-220	Ordered Set Tx Error Injection Status Register	1318
Figure 7-221	Ordered Set Rx Error Injection Status Register	1319
Figure 7-222	NOP Flit Extended Capability	1321
Figure 7-223	NOP Flit Extended Capability Header	1321
Figure 7-224	NOP Flit Capabilities Register	1322
Figure 7-225	NOP Flit Control 1 Register	1322
Figure 7-226	NOP Flit Control 2 Register	1324
Figure 7-227	NOP Flit Status Register	1325
Figure 7-228	Virtual Channel Extended Capability Structure	1326
Figure 7-229	Virtual Channel Extended Capability Header	1327
Figure 7-230	Port VC Capability Register 1	1328
Figure 7-231	Port VC Capability Register 2	1329
Figure 7-232	Port VC Control Register	1330
Figure 7-233	Port VC Status Register	1331
Figure 7-234	VC Resource Capability Register	1331
Figure 7-235	VC Resource Control Register	1333
Figure 7-236	VC Resource Status Register	1335
Figure 7-237	Example VC Arbitration Table with 32 Phases	1336
Figure 7-238	Example Port Arbitration Table with 128 Phases and 2-bit Table Entries	1337
Figure 7-239	MFVC Capability Structure	1338
Figure 7-240	MFVC Extended Capability Header	1339
Figure 7-241	MFVC Port VC Capability Register 1	1339
Figure 7-242	MFVC Port VC Capability Register 2	1340
Figure 7-243	MFVC Port VC Control Register	1341
Figure 7-244	MFVC Port VC Status Register	1342
Figure 7-245	MFVC VC Resource Capability Register	1343
Figure 7-246	MFVC VC Resource Control Register	1344
Figure 7-247	MFVC VC Resource Status Register	1346
Figure 7-248	Device Serial Number Extended Capability Structure	1349
Figure 7-249	Device Serial Number Extended Capability Header	1349
Figure 7-250	Serial Number Register	1350
Figure 7-251	Vendor-Specific Capability	1351
Figure 7-252	VSEC Capability Structure	1352
Figure 7-253	Vendor-Specific Extended Capability Header	1352
Figure 7-254	Vendor-Specific Header	1353
Figure 7-255	Designated Vendor-Specific Extended Capability	1354
Figure 7-256	Designated Vendor-Specific Extended Capability Header	1354
Figure 7-257	Designated Vendor-Specific Header 1	1355
Figure 7-258	Designated Vendor-Specific Header 2	1355
Figure 7-259	RCRB Header Extended Capability Structure	1356
Figure 7-260	RCRB Header Extended Capability Header	1356
Figure 7-261	RCRB Vendor ID and Device ID register	1357
Figure 7-262	RCRB Capabilities register	1358
Figure 7-263	RCRB Control register	1358
Figure 7-264	Root Complex Link Declaration Extended Capability	1360

Figure 7-265	Root Complex Link Declaration Extended Capability Header.....	1360
Figure 7-266	Element Self Description Register	1361
Figure 7-267	Link Entry.....	1362
Figure 7-268	Link Description Register	1362
Figure 7-269	Link Address for Link Type 0	1363
Figure 7-270	Link Address for Link Type 1	1364
Figure 7-271	Root Complex Internal Link Control Extended Capability.....	1365
Figure 7-272	Root Complex Internal Link Control Extended Capability Header.....	1365
Figure 7-273	Root Complex Link Capabilities Register	1366
Figure 7-274	Root Complex Link Control Register	1368
Figure 7-275	Root Complex Link Status Register	1370
Figure 7-276	Root Complex Event Collector Endpoint Association Extended Capability	1371
Figure 7-277	Root Complex Event Collector Endpoint Association Extended Capability Header	1371
Figure 7-278	RCEC Associated Bus Numbers Register	1372
Figure 7-279	Multicast Extended Capability Structure.....	1374
Figure 7-280	Multicast Extended Capability Header	1374
Figure 7-281	Multicast Capability Register	1375
Figure 7-282	Multicast Control Register.....	1376
Figure 7-283	MC_Base_Address Register.....	1376
Figure 7-284	MC_Receive Register	1377
Figure 7-285	MC_Block_All Register	1377
Figure 7-286	MC_Block_Untranslated Register	1378
Figure 7-287	MC_Overlay_BAR Register	1379
Figure 7-288	Dynamic Power Allocation Extended Capability Structure	1379
Figure 7-289	DPA Extended Capability Header.....	1380
Figure 7-290	DPA Capability Register.....	1380
Figure 7-291	DPA Latency Indicator Register.....	1381
Figure 7-292	DPA Status Register	1382
Figure 7-293	DPA Control Register	1382
Figure 7-294	DPA Power Allocation Array	1383
Figure 7-295	Substate Power Allocation Register (0 to Substate_Max).....	1383
Figure 7-296	TPH Extended Capability Structure.....	1384
Figure 7-297	TPH Requester Extended Capability Header.....	1384
Figure 7-298	TPH Requester Capability Register.....	1384
Figure 7-299	TPH Requester Control Register	1385
Figure 7-300	TPH ST Table.....	1386
Figure 7-301	TPH ST Table Entry.....	1387
Figure 7-302	DPC Extended Capability – Non-Flit Mode	1388
Figure 7-303	DPC Extended Capability – Flit Mode	1389
Figure 7-304	DPC Extended Capability Header	1390
Figure 7-305	DPC Capability Register	1390
Figure 7-306	DPC Control Register.....	1392
Figure 7-307	DPC Status Register	1394
Figure 7-308	DPC Error Source ID Register	1395
Figure 7-309	RP PIO Status Register.....	1396
Figure 7-310	RP PIO Mask Register	1397
Figure 7-311	RP PIO Severity Register	1398
Figure 7-312	RP PIO SysError Register	1399
Figure 7-313	RP PIO Exception Register.....	1400
Figure 7-314	RP PIO Header Log Register	1401
Figure 7-315	RP PIO ImpSpec Log Register	1401
Figure 7-316	RP PIO TLP Prefix Log Register	1402

Figure 7-317	PTM Extended Capability Structure.....	1403
Figure 7-318	PTM Extended Capability Header	1403
Figure 7-319	PTM Capability Register	1404
Figure 7-320	PTM Control Register.....	1405
Figure 7-321	Readiness Time Reporting Extended Capability.....	1407
Figure 7-322	Readiness Time Encoding.....	1407
Figure 7-323	Readiness Time Reporting Extended Capability Header	1408
Figure 7-324	Readiness Time Reporting 1 Register.....	1408
Figure 7-325	Readiness Time Reporting 2 Register.....	1409
Figure 7-326	Hierarchy ID Extended Capability	1411
Figure 7-327	Hierarchy ID Extended Capability Header	1411
Figure 7-328	Hierarchy ID Status Register	1412
Figure 7-329	Hierarchy ID Data Register	1413
Figure 7-330	Hierarchy ID GUID 1 Register.....	1414
Figure 7-331	Hierarchy ID GUID 2 Register.....	1414
Figure 7-332	Hierarchy ID GUID 3 Register.....	1415
Figure 7-333	Hierarchy ID GUID 4 Register.....	1415
Figure 7-334	Hierarchy ID GUID 5 Register.....	1416
Figure 7-335	VPD Capability Structure	1417
Figure 7-336	VPD Address Register	1418
Figure 7-337	VPD Data Register.....	1418
Figure 7-338	NPEM Extended Capability	1419
Figure 7-339	NPEM Extended Capability Header	1419
Figure 7-340	NPEM Capability Register.....	1420
Figure 7-341	NPEM Control Register	1421
Figure 7-342	NPEM Status Register	1423
Figure 7-343	Alternate Protocol Extended Capability	1424
Figure 7-344	Alternate Protocol Extended Capability Header	1424
Figure 7-345	Alternate Protocol Capabilities Register	1425
Figure 7-346	Alternate Protocol Control Register.....	1425
Figure 7-347	Alternate Protocol Data 1 Register.....	1426
Figure 7-348	Alternate Protocol Data 2 Register.....	1427
Figure 7-349	Alternate Protocol Selective Enable Mask Register	1427
Figure 7-350	Conventional PCI Advanced Features Capability (AF)	1428
Figure 7-351	Advanced Features Capability Header	1428
Figure 7-352	AF Capabilities Register	1429
Figure 7-353	Conventional PCI Advanced Features Control Register.....	1429
Figure 7-354	AF Status Register.....	1430
Figure 7-355	SFI Extended Capability	1430
Figure 7-356	SFI Extended Capability Header	1431
Figure 7-357	SFI Capability Register	1431
Figure 7-358	SFI Control Register.....	1432
Figure 7-359	SFI Status Register.....	1435
Figure 7-360	SFI CAM Address Register.....	1436
Figure 7-361	SFI CAM Data Register	1436
Figure 7-362	Subsystem ID and Subsystem Vendor ID Capability.....	1437
Figure 7-363	Subsystem ID and Subsystem Vendor ID Capability Header.....	1438
Figure 7-364	Subsystem ID and Subsystem Vendor ID Capability Data	1438
Figure 7-365	Data Object Exchange Extended Capability	1439
Figure 7-366	DOE Extended Capability Header	1439
Figure 7-367	DOE Capabilities Register	1440
Figure 7-368	DOE Control Register.....	1441

Figure 7-369	DOE Status Register	1442
Figure 7-370	DOE Write Data Mailbox Register	1443
Figure 7-371	DOE Read Data Mailbox Register	1443
Figure 7-372	Shadow Functions Extended Capability Structure	1446
Figure 7-373	Shadow Functions Extended Capability Header	1446
Figure 7-374	Shadow Functions Capability Register	1447
Figure 7-375	Shadow Functions Control Register	1447
Figure 7-376	Shadow Functions Instance Register Entry	1448
Figure 7-377	IDE Extended Capability Structure	1449
Figure 7-378	IDE Extended Capability Header	1449
Figure 7-379	IDE Capability Register	1450
Figure 7-380	IDE Control Register	1452
Figure 7-381	Link IDE Stream Control Register	1452
Figure 7-382	Link IDE Stream Status Register	1454
Figure 7-383	Selective IDE Stream Capability Register	1455
Figure 7-384	Selective IDE Stream Control Register	1456
Figure 7-385	Selective IDE Stream Status Register	1458
Figure 7-386	IDE RID Association Register 1 (Offset +00h)	1459
Figure 7-387	IDE RID Association Register 2 (Offset +04h)	1460
Figure 7-388	IDE Address Association Register 1 (Offset +00h)	1460
Figure 7-389	IDE Address Association Register 2 (Offset +04h)	1461
Figure 7-390	IDE Address Association Register 3 (Offset +04h)	1461
Figure 7-391	Null Capability	1462
Figure 7-392	Null Extended Capability	1462
Figure 7-393	Streamlined Virtual Channel Extended Capability Structure	1464
Figure 7-394	Streamlined Virtual Channel Extended Capability Header	1464
Figure 7-395	SVC Port Capability Register 1	1465
Figure 7-396	SVC Port Control Register	1465
Figure 7-397	SVC Port Status Register	1466
Figure 7-398	SVC Resource Capability Register	1467
Figure 7-399	SVC Resource Control Register	1467
Figure 7-400	SVC Resource Status Register	1469
Figure 7-401	MRBL Extended Capability	1470
Figure 7-402	MRBL Extended Capability Header	1471
Figure 7-403	MRBL Capabilities Register	1471
Figure 7-404	MRBL Locator Register	1472
Figure 8-1	Tx Test Board for Non-Embedded Refclk	1474
Figure 8-2	Tx Test board for Embedded Refclk	1474
Figure 8-3	Single-ended and Differential Levels	1476
Figure 8-4	Tx Equalization FIR Representation for 8.0, 16.0, and 32.0 GT/s	1477
Figure 8-5	Tx Equalization FIR Representation for 64.0 GT/s	1478
Figure 8-6	Definition of Tx Voltage Levels and Equalization Ratios	1479
Figure 8-7	Methodology for measuring Tx equalization coefficients and presets	1481
Figure 8-8	$V_{TX-DIFF-PP}$ and $V_{TX-DIFF-PP-LOW}$ Measurement	1482
Figure 8-9	Transmit Equalization Coefficient Space Triangular Matrix Example for 8.0, 16.0, and 32.0 GT/s	1483
Figure 8-10	Transmit Equalization Coefficient Space Triangular Matrix Example for 64.0 GT/s	1484
Figure 8-11	Measuring $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ at 8.0 GT/s	1486
Figure 8-12	Compliance Pattern and Resulting Package Loss Test Waveform	1487
Figure 8-13	Example of Normalized Four Symbol Linear Pulse Responses	1495
Figure 8-14	Example of Un-normalized Four Symbol Linear Pulse Responses	1496
Figure 8-15	2.5 and 5.0 GT/s Transmitter Margining Voltage Levels and Codes	1497
Figure 8-16	First Order CC Behavioral CDR Transfer Functions	1501

Figure 8-17	2 nd Order Behavioral SRIS CDR Transfer Functions for 2.5 GT/s and 5.0 GT/s	1502
Figure 8-18	Behavioral SRIS CDR Function for 8.0 GT/s, and SRIS and CC CDR for 16.0 and 32.0 GT/s.....	1503
Figure 8-19	Behavioral SRIS and CC CDR for 64.0 GT/s	1504
Figure 8-20	Relation Between Data Edge PDFs and Recovered Data Clock	1506
Figure 8-21	Derivation of T_{TX-UTJ} and $T_{TX-UDJDD}$	1506
Figure 8-22	PWJ Relative to Consecutive Edges 1 UI Apart.....	1507
Figure 8-23	Definition of $T_{TX-UPW-DJDD}$ and $T_{TX-UPW-TJ}$ Data Rate Dependent Transmitter Parameters.....	1507
Figure 8-24	Tx, Rx Differential Return Loss Mask with 50 Ohm Reference	1512
Figure 8-25	Tx, Rx Common Mode Return Loss Mask with 50 Ohm Reference	1513
Figure 8-26	64.0 GT/s Tx, Rx Differential Return Loss Mask with 50 Ohm Reference	1514
Figure 8-27	64.0 GT/s Tx, Rx Common Mode Return Loss Mask with 50 Ohm Reference	1515
Figure 8-28	Rx Test board Topology for 16.0 and 32.0 GT/s	1518
Figure 8-29	Example Calibration Channel IL Mask Excluding Rx Package for 8.0 GT/s	1519
Figure 8-30	Example 16.0 GT/s Calibration Channel	1523
Figure 8-31	Stackup for Example 16.0 GT/s Calibration Channel	1523
Figure 8-32	CEM Connector Drill Hole Pad Stack	1524
Figure 8-33	Pad Stack for SMA Drill Holes.....	1525
Figure 8-34	Example 32.0 GT/s Calibration Channel	1527
Figure 8-35	Stack-up for Example 32.0 GT/s Calibration Channel	1527
Figure 8-36	Transfer Function for 8.0 GT/s Behavioral CTLE	1529
Figure 8-37	Loss Curves for 8.0 GT/s Behavioral CTLE	1530
Figure 8-38	Loss Curves for 16.0 GT/s Behavioral CTLE	1530
Figure 8-39	Loss Curves for 32.0 GT/s Behavioral CTLE	1532
Figure 8-40	Loss Curves for 64.0 GT/s Behavioral CTLE	1534
Figure 8-41	Variables Definition and Diagram for 1-tap DFE	1535
Figure 8-42	Diagram for 2-tap DFE	1535
Figure 8-43	Layout for Calibrating the Stressed Jitter Eye at 8.0 GT/s	1539
Figure 8-44	Layout for Calibrating the Stressed Jitter Eye at 16.0, 32.0, and 64.0 GT/s.....	1540
Figure 8-45	Sj Mask for Receivers Operating in IR mode at 8.0 GT/s.....	1543
Figure 8-46	Sj Mask for Receivers Operating in SRIS mode at 16.0 GT/s	1544
Figure 8-47	Sj Mask for Receivers Operating in CC mode at 16.0 GT/s	1545
Figure 8-48	Sj Mask for Receivers Operating in SRIS mode at 32.0 GT/s	1546
Figure 8-49	Sj Mask for Receivers Operating in CC mode at 32.0 GT/s	1547
Figure 8-50	Sj Mask for Receivers Operating in SRIS mode at 64.0 GT/s	1548
Figure 8-51	Sj Mask for Receivers Operating in CC mode at 64.0 GT/s	1549
Figure 8-52	Sj Masks for Receivers Operating in CC Mode at 8.0 GT/s	1550
Figure 8-53	Layout for Jitter Testing Common Refclk Rx at 16.0 GT/s.....	1551
Figure 8-54	Layout for Jitter Testing for Independent Refclk Rx at 16.0 GT/s	1551
Figure 8-55	Exit from Idle Voltage and Time Margins.....	1554
Figure 8-56	Allowed Ranges for Maximum NRZ Timing and Voltage Margin	1555
Figure 8-57	Allowed Ranges for Maximum PAM4 Timing and Voltage Margins	1556
Figure 8-58	Flow Diagram for Channel Tolerancing at 2.5 and 5.0 GT/s.....	1561
Figure 8-59	Flow Diagram for Channel Tolerancing at 8.0 and 16.0 GT/s.....	1561
Figure 8-60	Tx/Rx Behavioral Package Models	1562
Figure 8-61	Behavioral Tx and Rx S-Port Designation for 8.0 and 16.0 GT/s Packages	1563
Figure 8-62	SDD21 Plots for Root and Non-Root Packages for 16.0 GT/s	1563
Figure 8-63	Insertion Loss for Root Reference Package for 32.0 GT/s.....	1564
Figure 8-64	Return Loss for Root Reference Package for 32.0 GT/s	1564
Figure 8-65	NEXT for Root Reference Package (Worst-Case) for 32.0 GT/s.....	1565
Figure 8-66	FEXT for Root Reference Package (Worst-Case) for 32.0 GT/s.....	1565
Figure 8-67	Insertion Loss for Non-Root Reference Package for 32.0 GT/s	1566
Figure 8-68	Return Loss for Non-Root Reference Package for 32.0 GT/s	1566

Figure 8-69	NEXT for Non-Root Reference Package (Worst-Case) for 32.0 GT/s.....	1567
Figure 8-70	FEXT for Non-Root Reference Package (Worst-Case) for 32.0 GT/s	1567
Figure 8-71	Insertion Loss for Root Reference Package for 64.0 GT/s.....	1568
Figure 8-72	Return Loss for Root Reference Package for 64.0 GT/s	1568
Figure 8-73	NEXT for Root Reference Package (Worst Case) for 64.0 GT/s	1569
Figure 8-74	FEXT for Root Reference Package (Worst Case) for 64.0 GT/s.....	1569
Figure 8-75	Insertion Loss for Non-Root Reference Package for 64.0 GT/s	1570
Figure 8-76	Return Loss for Non-Root Reference Package for 64.0 GT/s	1570
Figure 8-77	NEXT for Non-Root Reference Package (Worst Case) for 64.0 GT/s	1571
Figure 8-78	FEXT for Non-Root Reference Package (Worst Case) for 64.0 GT/s.....	1571
Figure 8-79	32.0 and 64.0 GT/s Reference Package Port Connections for Pin to Pin Channel Evaluation	1572
Figure 8-80	Example Derivation of 8.0 GT/s Jitter Parameters for	1574
Figure 8-81	EH, EW Mask	1577
Figure 8-82	Oscilloscope Refclk Test Setup for All Cases Except Jitter at 32.0 and 64.0 GT/s	1581
Figure 8-83	Single-Ended Measurement Points for Absolute Cross Point and Swing.....	1583
Figure 8-84	Single-Ended Measurement Points for Delta Cross Point	1583
Figure 8-85	Single-Ended Measurement Points for Rise and Fall Time Matching	1583
Figure 8-86	Differential Measurement Points for Duty Cycle and Period	1584
Figure 8-87	Differential Measurement Points for Rise and Fall Time	1584
Figure 8-88	Differential Measurement Points for Ringback.....	1584
Figure 8-89	Limits for phase jitter from the Reference with 5000 ppm SSC.....	1586
Figure 8-90	5 MHz PLL Transfer Function Example	1587
Figure 8-91	Common Refclk Rx Architecture for all Data Rates Except 32.0 and 64.0 GT/s	1588
Figure 8-92	Common Refclk PLL and CDR Characteristics for 2.5 GT/s	1589
Figure 8-93	Common Refclk PLL and CDR Characteristics for 5.0 GT/s	1590
Figure 8-94	Common Refclk PLL and CDR Characteristics for 8.0 and 16.0 GT/s	1590
Figure 8-95	Common Refclk PLL and CDR Characteristics for 32.0 GT/s	1590
Figure 8-96	Common Refclk PLL and CDR Characteristics for 64.0 GT/s	1590
Figure 9-1	Generic Platform Configuration	1593
Figure 9-2	Generic Platform Configuration with a VI and Multiple SI.....	1594
Figure 9-3	Generic Platform Configuration with SR-IOV and IOV Enablers.....	1596
Figure 9-4	Example Multi-Function Device.....	1598
Figure 9-5	Example SR-IOV Single PF Capable Device	1599
Figure 9-6	Example SR-IOV Multi-PF Capable Device	1601
Figure 9-7	Example SR-IOV Device with Multiple Bus Numbers	1603
Figure 9-8	Example SR-IOV Device with a Mixture of Function Types	1604
Figure 9-9	Example SIOV Capable Device	1605
Figure 9-10	BAR Space Example for Single BAR Device	1610
Figure 9-11	SR-IOV Extended Capability	1617
Figure 9-12	SR-IOV Extended Capability Header	1617
Figure 9-13	SR-IOV Capabilities Register	1618
Figure 9-14	SR-IOV Control Register.....	1621
Figure 9-15	SR-IOV Status.....	1625
Figure 9-16	SIOV Extended Capability	1635
Figure 9-17	SIOV Extended Capability Header	1635
Figure 9-18	SIOV Capabilities Register	1636
Figure 9-19	SIOV Status Register	1636
Figure 10-1	Example Illustrating a Platform with TA, ATPT, and ATC Elements.....	1640
Figure 10-2	Example ATS Translation Request/Completion Exchange	1640
Figure 10-3	Example Multi-Function Device with ATC per Function.....	1643
Figure 10-4	Invalidation Protocol with a Single Invalidate Request and Completion.....	1643
Figure 10-5	Single Invalidate Request with Multiple Invalidate Completions	1644

Figure 10-6	Memory Request Header with 64-bit Address Highlighting AT field	1647
Figure 10-7	Memory Request Header with 32-bit Address Highlighting AT field	1647
Figure 10-8	Memory Request Header with 64-bit Address Highlighting AT field - Flit Mode	1647
Figure 10-9	Memory Request Header with 32-bit Address Highlighting AT field - Flit Mode	1648
Figure 10-10	Translation Request with 64-bit Address - Non-Flit Mode.....	1649
Figure 10-11	Translation Request with 32-bit Address - Non-Flit Mode.....	1649
Figure 10-12	Translation Request with 64-bit Address - Flit Mode	1649
Figure 10-13	Translation Request with 32-bit Address - Flit Mode	1650
Figure 10-14	Translation Completion Data Entry.....	1653
Figure 10-15	Example Translation Completion with 1 TLP	1660
Figure 10-16	Example Translation Completion with 2 TLPs	1661
Figure 10-17	ATS Memory Attributes Example	1663
Figure 10-18	Invalidate Request Message - Non-Flit Mode	1665
Figure 10-19	Invalidate Request Message - Flit Mode	1666
Figure 10-20	Invalidate Request Message Body	1667
Figure 10-21	Invalidate Completion Message Format - Non-Flit Mode.....	1668
Figure 10-22	Invalidate Completion Message - Flit Mode	1668
Figure 10-23	Page Request Message - Non-Flit Mode	1675
Figure 10-24	Page Request Message - Flit Mode.....	1675
Figure 10-25	Stop Marker Message - Non-Flit Mode.....	1678
Figure 10-26	Stop Marker Message - Flit Mode	1678
Figure 10-27	PRG Response Message - Non-Flit Mode.....	1680
Figure 10-28	PRG Response Message - Flit Mode	1680
Figure 10-29	ATS Extended Capability Structure.....	1681
Figure 10-30	ATS Extended Capability Header	1682
Figure 10-31	ATS Capability Register (Offset 04h)	1682
Figure 10-32	ATS Control Register	1684
Figure 10-33	Page Request Extended Capability Structure	1685
Figure 10-34	Page Request Extended Capability Header	1685
Figure 10-35	Page Request Control Register	1686
Figure 10-36	Page Request Status Register	1687
Figure 11-1	Conceptual View with Example Host and Device and Logical Communication Paths.....	1690
Figure 11-2	TDISP Host/Device Reference Architecture.....	1693
Figure 11-3	Identification of Requests	1696
Figure 11-4	TDI Identifier – INTERFACE_ID	1697
Figure 11-5	TDISP State Machine	1697
Figure 11-6	TDISP Request/Response Encapsulation.....	1703
Figure 11-7	Example Flow Where DSM is Unable to Return Full Length Report	1718
Figure 12-1	Example PESTI Application.....	1749
Figure 12-2	UART Data Framing	1749
Figure 12-3	PESTI Circuit Diagram	1750
Figure 12-4	PESTI Broadcast Command.....	1754
Figure 12-5	PESTI Protocol Phases	1756
Figure 12-6	PESTI Discovery Command and Response Format	1758
Figure 12-7	Single Byte PESTI Virtual Wire Exchange.....	1760
Figure 12-8	Multi-byte PESTI Virtual Wire Exchange	1760
Figure 12-9	PESTI Fan-out Methods.....	1764
Figure 12-10	PESTI Mux Switch Control Format.....	1765
Figure 12-11	[CEM] form factor example circuit for repurposing legacy JTAG to USB 2.0 mode.....	1767
Figure 12-12	Example of 2-wire, 8-bit addressing for a card carrier with N end form factors in SMBus mode	1770
Figure 12-13	Example of 2-Wire Hub Use.....	1772
Figure 12-14	SMBus to I3C Transition Flow	1775

Figure 12-15	Component Timing Diagram for Transition to I3C Signaling Voltage.....	1777
Figure 12-16	SMBus/I2C-based FRU Information Device Writes with Two-Byte Addressing	1780
Figure 12-17	FRU Information Device Reads with Two-Byte Addressing.....	1780
Figure 12-18	PCIe-MI Component Protocol Layers	1787
Figure 12-19	Example PCIe-MI Component.....	1793
Figure 12-20	Example PCIe-MI Message Flows.....	1796
Figure 12-21	PCIe-MI Message Format.....	1797
Figure 12-22	Example Tiers Involving Sidebands.....	1829
Figure A-1	An Example Showing Endpoint-to-Root-Complex and Peer-to-Peer Communication Models	1831
Figure A-2	Two Basic Bandwidth Resourcing Problems: Over-Subscription and Congestion	1832
Figure A-3	A Simplified Example Illustrating PCI Express Isochronous Parameters	1836
Figure C-1	Scrambling Spectrum at 2.5 GT/s for Data Value of 0	1856
Figure E-1	Reference Topology for IDO Use.....	1863
Figure G-1	Device and Processor Connected Using a PMUX Link	1871
Figure G-2	PMUX Link.....	1871
Figure G-3	PMUX Packet Flow Through the Layers.....	1872
Figure G-4	PMUX Packet.....	1876
Figure G-5	TLP and PMUX Packet Framing (8b/10b Encoding)	1877
Figure G-6	TLP and PMUX Packet Framing (128b/130b Encoding)	1879
Figure G-7	PMUX Extended Capability.....	1882
Figure G-8	PMUX Extended Capability Header.....	1882
Figure G-9	PMUX Capability Register.....	1883
Figure G-10	PMUX Control Register	1884
Figure G-11	PMUX Status Register	1885
Figure G-12	PMUX Protocol Array Entry	1886
Figure L-1	Example Memory Write TLP (NFM)	2201
Figure L-2	Example NFM Memory Write IDE TLP	2202
Figure L-3	Example Memory Write TLP (FM).....	2203
Figure L-4	Example Memory Write IDE TLP (FM)	2204
Figure L-5	Example Memory Write TLP with Partial Header Encryption (NFM)	2205
Figure L-6	Example NFM Memory Write IDE TLP with Partial Header Encryption	2206
Figure L-7	Example Memory Write TLP with Partial Header Encryption (FM)	2207
Figure L-8	Example Memory Write IDE TLP with Partial Header Encryption (FM)	2208

List of Equations

Equation 2-1	CREDITS_CONSUMED	295
Equation 2-2	SHARED_CREDITS_CONSUMED.....	295
Equation 2-3	SUM_SHARED_CREDITS_CONSUMED	296
Equation 2-4	TLP_SHARED_CREDITS_CONSUMED_CURRENTLY.....	296
Equation 2-5	FC_SHARED_CREDITS_CONSUMED_CURRENTLY.....	296
Equation 2-6	SUM_SHARED_CREDIT_LIMIT.....	297
Equation 2-7	SHARED_CUMULATIVE_CREDITS_REQUIRED	298
Equation 2-8	Shared Transmitter Gate non-[Merged]	298
Equation 2-9	Shared Transmitter Gate [Merged]	299
Equation 2-10	Shared Transmitter Usage Limit Gate non-[Merged].....	299
Equation 2-11	Shared Transmitter Usage Limit Gate [Merged].....	299
Equation 2-12	CUMULATIVE_CREDITS_REQUIRED	299
Equation 2-13	Transmitter Gate.....	300
Equation 2-14	CREDITS_ALLOCATED.....	301
Equation 2-15	CREDITS_RECEIVED.....	301
Equation 2-16	Receiver Overflow Error Check Non-Flit / Dedicated.....	302
Equation 2-17	Receiver Overflow Error Check Non-Posted / Not [Merged].....	302
Equation 2-18	Receiver Overflow Error Check [Merged]	303
Equation 3-1	Tx SEQ Stall.....	346
Equation 3-2	Tx SEQ Update.....	347
Equation 4-1	Parity bytes.....	457
Equation 4-2	Check bytes	457
Equation 4-3	Retimer Latency with SRIS.....	685
Equation 6-1	MC_Overlay Transform rules	864
Equation 6-2	PTM Master Time	890
Equation 7-1	MSI-X Starting Address	1153
Equation 7-2	MSI-X PBA QWORD Access.....	1153
Equation 7-3	MSI-X PBA DWORD Access	1153
Equation 7-4	Egress Control Vector Access	1221
Equation 8-1	$V_{DIFFp-p}$	1475
Equation 8-2	$V_{TX-AC-CM-PP}$	1475
Equation 8-3	Y	1488
Equation 8-4	x_r	1489
Equation 8-5	X	1489
Equation 8-6	P	1489
Equation 8-7	E	1489
Equation 8-8	y	1490
Equation 8-9	X_{NP}	1490
Equation 8-10	P_{EO}	1490
Equation 8-11	e	1491
Equation 8-12	$\sigma_{L,j}$	1491
Equation 8-13	$\mu_{L,j}$	1491
Equation 8-14	σ_L	1492
Equation 8-15	σ_n	1492
Equation 8-16	SNDR	1492
Equation 8-17	R_{LM}	1493

Equation 8-18	$W_{\text{sym0_3}}$	1494
Equation 8-19	X_{sym}	1494
Equation 8-20	P_{sym}	1494
Equation 8-21	$P_{\text{sym_un_normalized}}$	1496
Equation 8-22	Behavioral SRIS CDR at 8.0 GT/s and SRIS and CC Behavioral CDR at 16.0 GT/s	1503
Equation 8-23	SRIS Behavioral CDR Parameters at 8.0 GT/s	1503
Equation 8-24	SRIS and CC Behavioral CDR Parameters at 16.0 GT/s.....	1504
Equation 8-25	SRIS and CC Behavioral CDR Parameters at 32.0 and 64.0 GT/s.....	1505
Equation 8-26	Behavioral CTLE at 32.0 GT/s	1531
Equation 8-27	Behavioral CTLE at 64.0 GT/s	1533
Equation 8-28	Relationship between 2 nd order PLL natural frequency and 3 dB point	1587
Equation A-1	Isochronous Bandwidth.....	1832
Equation A-2	Isochronous Payload Size	1833
Equation A-3	N_{max}	1833
Equation A-4	BW_{max}	1834
Equation A-5	$BW_{\text{granularity}}$	1834
Equation A-6	N_{link}	1834
Equation A-7	Max Isochronous Transaction Latency	1835
Equation H-1	Max UpdateFC Latency	1889
Equation H-2	Max Ack Latency	1892

List of Tables

Table 1-1	PCIe Signaling Characteristics	139
Table 2-1	Transaction Types for Different Address Spaces.....	152
Table 2-2	Fmt[2:0] Field Values.....	156
Table 2-3	Non-Flit Mode Fmt[2:0] and Type[4:0] Field Encodings	156
Table 2-4	Length[9:0] Field Encoding	158
Table 2-5	Flit Mode TLP Header Type Encodings.....	160
Table 2-6	OHC-Ax Included Fields for OHC-A1 through OHC-A5 (see through)	170
Table 2-7	Address Field Mapping.....	187
Table 2-8	Header Field Locations for non-ARI ID Routing - Non-Flit Mode.....	189
Table 2-9	Header Field Locations for ARI ID Routing - Non-Flit Mode.....	189
Table 2-10	Byte Enables Location and Correspondence	194
Table 2-11	Tag Enables, Sizes, and Permitted Ranges for non-UIO Transactions.....	199
Table 2-12	Ordering Attributes	204
Table 2-13	Cache Coherency Management Attribute	205
Table 2-14	Definition of TC Field Encodings.....	205
Table 2-15	Length Field Values for AtomicOp Requests	206
Table 2-16	TPH TLP Prefix Bit Mapping - Non-Flit Mode	210
Table 2-17	Location of PH[1:0] in TLP Header - Non-Flit Mode	211
Table 2-18	Processing Hint Encoding	211
Table 2-19	Location of ST[7:0] in TLP Headers – Non-Flit Mode	212
Table 2-20	Message Routing	216
Table 2-21	INTx Mechanism Messages	217
Table 2-22	Bridge Mapping for INTx Virtual Wires.....	219
Table 2-23	Power Management Messages	220
Table 2-24	Error Signaling Messages.....	221
Table 2-25	ERR_COR Subclass (ECS) Field Encodings	222
Table 2-26	Unlock Message.....	223
Table 2-27	Set_Slot_Power_Limit Message	223
Table 2-28	Vendor-Defined Messages.....	224
Table 2-29	DRS Message.....	227
Table 2-30	FRS Message	229
Table 2-31	Hierarchy ID Message	230
Table 2-32	Ignored Messages.....	232
Table 2-33	LTR Message.....	232
Table 2-34	OBFF Message	234
Table 2-35	Precision Time Measurement Messages	235
Table 2-36	IDE Messages	238
Table 2-37	Completion Status Field Values.....	242
Table 2-38	Local TLP Prefix Types	247
Table 2-39	End-End TLP Prefix Types	248
Table 2-40	Calculating Byte Count from Length and First/Last Byte Enables	264
Table 2-41	Calculating Lower Address from First DW Byte Enables.....	265
Table 2-42	Non-UIO Ordering Rules	270
Table 2-43	UIO TLP Ordering Rules	275
Table 2-44	UIO Acceptance Dependency Rules – Downstream Ports	275
Table 2-45	UIO Acceptance Dependency Rules – Upstream Ports	276
Table 2-46	Streamlined Virtual Channel (SVC) TC/VC Default Assignments.....	279

Table 2-47	TC to VC Mapping Example	283
Table 2-48	Flow Control Credit Types	287
Table 2-49	TLP Flow Control Credit Consumption	287
Table 2-50	Minimum Initial Flow Control Advertisements	289
Table 2-51	[Field Size] Values.....	292
Table 2-52	Maximum UpdateFC Transmission Latency Guidelines for 2.5 GT/s (Symbol Times).....	306
Table 2-53	Maximum UpdateFC Transmission Latency Guidelines for 5.0 GT/s (Symbol Times).....	306
Table 2-54	Maximum UpdateFC Transmission Latency Guidelines for 8.0 GT/s and Higher Data Rates (Symbol Times)	307
Table 2-55	Mapping of Bits into ECRC Field	308
Table 3-1	Data Link Feature Supported Bit Definition.....	325
Table 3-2	InitFC1 / InitFC2 Options – Non-Flit Mode.....	328
Table 3-3	InitFC1 / InitFC2 Options – Flit Mode.....	328
Table 3-4	Scaled Flow Control Scaling Factors	334
Table 3-5	DLLP Type Encodings.....	336
Table 3-6	HdrScale and DataScale Encodings	338
Table 3-7	Mapping of Bits into CRC Field	342
Table 3-8	Mapping of Bits into LCRC Field.....	347
Table 3-10	Maximum Ack Latency Limits for 2.5 GT/s (Symbol Times) (-0%/+0%)	358
Table 3-11	Maximum Ack Latency Limits for 5.0 GT/s (Symbol Times) (-0%/+0%)	358
Table 3-12	Maximum Ack Latency Limits for 8.0 GT/s and higher data rates (Symbol Times).....	358
Table 4-1	Valid Encoding and Data Stream Mode Combinations.....	362
Table 4-2	Valid Encoding for Ordered Sets.....	362
Table 4-3	Special Symbols in 8b/10b Encoding.....	365
Table 4-4	Framing Token Encoding	375
Table 4-6	Effect of +/-1 voltage level error on the wire for various PAM4 voltage levels – at most one bit flips with an error on a UI.....	397
Table 4-7	Truth Table for Precoding on the Transmit side	397
Table 4-8	Truth Table for Precoding on the Receive side	398
Table 4-9	Example of precoding with an error in the channel and DFE error propagation at the Receiver	399
Table 4-10	Flit Layout in a x16 Link	405
Table 4-11	Flit interleaving in a x8 Link	406
Table 4-12	Flit interleaving in a x4 Link	407
Table 4-13	Flit interleaving in a x2 Link	407
Table 4-14	Flit arrangement in a x1 Link	408
Table 4-15	Example TLP Placement in Flit Mode on a x16 Link	412
Table 4-16	Flit Types.....	414
Table 4-17	DLP Bytes in the Flit	414
Table 4-18	Optimized_Update_FC.....	415
Table 4-19	Flit_Marker.....	416
Table 4-20	Nak Ignore Window	421
Table 4-21	Ack/Nak Latency	421
Table 4-22	NOP Flit Types	441
Table 4-23	NOP Flit Common Header Fields.....	442
Table 4-24	NOP.Debug Flit Debug Chunk Fields	444
Table 4-25	PCI-SIG Defined Debug Chunk Opcode Values	446
Table 4-26	FC Information Tracked by Transmitter Encodings	449
Table 4-27	FC Information Tracked by Receiver Encodings	451
Table 4-28	Flit Mode Transmitter Retry Flags and Counters Fields.....	452
Table 4-29	Flit Mode Receiver Retry Flags and Counters Fields.....	453
Table 4-30	Buffer Occupancy Encodings.....	454
Table 4-32	Ordered Set insertion interval once Data Stream starts in terms of number of Flits	463

Table 4-33	Equalization Requirements Under Different Conditions.....	468
Table 4-34	Transmitter Preset Encoding	476
Table 4-35	Receiver Preset Hint Encoding for 8.0 GT/s	477
Table 4-36	TS1 Ordered Set in 8b/10b and 128b/130b Encoding.....	481
Table 4-37	TS2 Ordered Set in 8b/10b and 128b/130b Encoding.....	488
Table 4-38	Modified TS1/TS2 Ordered Set (8b/10b encoding)	491
Table 4-39	TS1/TS2 Ordered Set with 1b/1b Encoding.....	494
Table 4-40	TS0 Ordered Set	496
Table 4-41	Modified TS Information 1 field in Modified TS1/TS2 Ordered Sets if Modified TS Usage = 010b (Alternate Protocol)	501
Table 4-42	Electrical Idle Ordered Set (EIOS) for 2.5 GT/s and 5.0 GT/s Data Rates.....	503
Table 4-43	Electrical Idle Ordered Set (EIOS) for 128b/130b Encoding	503
Table 4-44	Electrical Idle Ordered Set (EIOS) for 1b/1b Encoding	503
Table 4-45	Electrical Idle Exit Ordered Set (EIEOS) for 5.0 GT/s Data Rate.....	503
Table 4-46	Electrical Idle Exit Ordered Set (EIEOS) for 8.0 GT/s Data Rate.....	504
Table 4-47	Electrical Idle Exit Ordered Set (EIEOS) for 16.0 GT/s Data Rate.....	504
Table 4-48	Electrical Idle Exit Ordered Set (EIEOS) for 32.0 GT/s Data Rate.....	504
Table 4-49	Electrical Idle Exit Ordered Set (EIEOS) for 64.0 GT/s Data Rate.....	504
Table 4-50	Electrical Idle Inference Conditions	508
Table 4-51	FTS for 8.0 GT/s and Above Data Rates	510
Table 4-52	SDS Ordered Set (for 8.0 GT/s and 16.0 GT/s Data Rate)	511
Table 4-53	SDS Ordered Set (for 32.0 GT/s)	511
Table 4-54	SDS Ordered Set (for 64.0 GT/s)	511
Table 4-55	Summary of L0p Transmitter/Receiver Behavior	522
Table 4-56	Link Management DLLP	522
Table 4-60	Link Status Mapped to the LTSSM.....	527
Table 4-61	Compliance Pattern Settings	534
Table 4-63	Use of TS0 or TS1 Ordered Sets in different phases.....	566
Table 4-64	Standard SKP Ordered Set with 128b/130b Encoding	610
Table 4-65	Control SKP Ordered Set with 128b/130b Encoding	611
Table 4-66	Control SKP Ordered Set with 1b/1b Encoding	613
Table 4-67	PHY Payload for Control SKP Ordered Set with 1b/1b Encoding.....	616
Table 4-71	Illustration of Modified Compliance Pattern	623
Table 4-74	Margin Command Related Fields in the Control SKP Ordered Set	631
Table 4-75	Margin Commands and Corresponding Responses	634
Table 4-76	Maximum Retimer Exit Latency	662
Table 4-77	Inferring Electrical Idle.....	664
Table 4-78	Retimer Latency Limit not SRIS (Symbol times)	684
Table 4-79	Retimer Latency Limit SRIS (Symbol times)	684
Table 5-1	Summary of PCI Express Link Power Management States	695
Table 5-2	Relation Between Power Management States of Link and Components	701
Table 5-3	Encoding of the ASPM Support Field	724
Table 5-4	Description of the Slot Clock Configuration Bit	724
Table 5-5	Description of the Common Clock Configuration Bit	724
Table 5-6	Encoding of the L0s Exit Latency Field	725
Table 5-7	Encoding of the L1 Exit Latency Field.....	725
Table 5-8	Encoding of the Endpoint L0s Acceptable Latency Field	726
Table 5-9	Encoding of the Endpoint L1 Acceptable Latency Field	726
Table 5-10	Encoding of the ASPM Control Field.....	726
Table 5-11	L1.2 Timing Parameters	739
Table 5-12	Aux Power Source and Availability	740
Table 5-13	Power Management System Messages and DLLPs	741

Table 5-14	PCI Function State Transition Delays	742
Table 6-1	Error Messages	760
Table 6-2	General PCI Express Error List.....	775
Table 6-3	Physical Layer Error List.....	775
Table 6-4	Data Link Layer Error List.....	775
Table 6-5	Transaction Layer Error List.....	776
Table 6-6	Multi-Function Arbitration Error Model Example.....	799
Table 6-7	Elements of Hot-Plug	813
Table 6-8	Attention Indicator States	814
Table 6-9	Power Indicator States	814
Table 6-10	Power Budgeting Deployments.....	837
Table 6-11	ACS P2P Request Redirect and ACS P2P Egress Control Interactions.....	852
Table 6-12	ECRC Rules for MC_Overlay	864
Table 6-13	Processing Hint Mapping.....	873
Table 6-14	ST Modes of Operation.....	873
Table 6-15	PASID TLP Prefix.....	885
Table 6-16	Emergency Power Reduction Supported Values	905
Table 6-17	System GUID Authority ID Encoding.....	909
Table 6-19	Small Resource Data Type Tag Bit Definitions	924
Table 6-20	Large Resource Data Type Tag Bit Definitions	924
Table 6-21	Resource Data Type Flags for a Typical VPD	924
Table 6-22	Example of Add-in Serial Card Number.....	925
Table 6-23	VPD Large and Small Resource Data Tags.....	926
Table 6-24	VPD Read-Only Fields.....	926
Table 6-25	VPD Read/Write Fields.....	928
Table 6-26	VPD Example.....	928
Table 6-27	NPEM States.....	933
Table 6-28	DOE Data Object Header 1	940
Table 6-29	DOE Data Object Header 2	940
Table 6-30	DOE Discovery Request Data Object Contents (3rd DW)	941
Table 6-31	DOE Discovery Response Data Object Contents (3rd DW).....	941
Table 6-32	DOE Discovery Response Data Object Contents (4th DW).....	942
Table 6-33	PCI-SIG Defined Data Object Types (Vendor ID = 0001h)	942
Table 6-34	DOE Async Message Data Object Contents (1 DW).....	944
Table 6-35	TLP Types for Selective IDE Streams	991
Table 6-36	IDE Revised Ordering Rules for Flow-Through non-UIO IDE Streams - Per Stream.....	1003
Table 6-37	IDE Revised Ordering Rules for Flow-Through UIO IDE Streams - Per Stream	1003
Table 6-38	MCAP Array Register 1	1010
Table 6-39	MCAP Array Register 2	1011
Table 6-40	MCAP Header Register 1.....	1012
Table 6-41	MCAP Header Register 2.....	1012
Table 6-42	MCAP Header Register 3.....	1012
Table 6-43	MCAP Header Register 4.....	1013
Table 6-44	PCI-SIG Defined MCAP Identifiers (MCAP Vendor ID = 0001h)	1013
Table 6-45	MMB Capabilities Register	1016
Table 6-46	MMB Control Register.....	1017
Table 6-47	MMB Command Register.....	1018
Table 6-48	MMB Status Register.....	1019
Table 6-49	MMB PCI-SIG Defined Command Return Codes (Vendor ID = 0001h)	1020
Table 6-50	MMPT Capabilities Register	1022
Table 6-51	MMPT Control Register.....	1023
Table 6-52	MMPT Receive Message Notification Register.....	1023

Table 6-53	MDVS Register Block Header Register 1	1024
Table 6-54	MDVS Register Block Header Register 2	1025
Table 6-55	MDVS Register Block Header Register 3	1025
Table 6-56	PCI-SIG Defined MMB Command Opcodes (Vendor ID = 0001h)	1026
Table 6-57	MMPT Send Message Input Payload	1027
Table 6-58	MMPT Send Message Output Payload	1028
Table 6-59	MMPT Receive Message Input Payload	1029
Table 6-60	MMPT Receive Message Output Payload	1029
Table 7-1	Enhanced Configuration Address Mapping	1037
Table 7-2	Register and Register Bit-Field Types	1043
Table 7-3	Special Field Types to Indicate VF Attributes	1045
Table 7-4	Command Register	1048
Table 7-5	Status Register	1050
Table 7-6	Class Code Register	1053
Table 7-7	Header Type Register	1054
Table 7-8	BIST Register	1055
Table 7-9	Memory Base Address Register Bits 2:1 Encoding	1059
Table 7-10	Expansion ROM Base Address Register	1063
Table 7-11	I/O Addressing Capability	1068
Table 7-12	Secondary Status Register	1069
Table 7-13	Bridge Control Register	1072
Table 7-14	Power Management Capabilities Register	1075
Table 7-15	Power Management Control/Status Register	1077
Table 7-16	Power Management Data Register	1079
Table 7-17	Power Consumption/Dissipation Reporting	1079
Table 7-18	PCI Express Capability List Register	1082
Table 7-19	PCI Express Capabilities Register	1082
Table 7-20	Device Capabilities Register	1084
Table 7-21	Device Control Register	1088
Table 7-22	Device Status Register	1095
Table 7-23	Link Capabilities Register	1097
Table 7-24	Link Control Register	1101
Table 7-26	Link Status Register	1108
Table 7-27	Slot Capabilities Register	1110
Table 7-28	Slot Control Register	1113
Table 7-29	Slot Status Register	1116
Table 7-30	Root Control Register	1118
Table 7-31	Root Capabilities Register	1119
Table 7-32	Root Status Register	1120
Table 7-33	Device Capabilities 2 Register	1121
Table 7-34	Device Control 2 Register	1126
Table 7-35	Link Capabilities 2 Register	1130
Table 7-36	Link Control 2 Register	1133
Table 7-37	Link Status 2 Register	1137
Table 7-38	Slot Capabilities 2 Register	1140
Table 7-39	PCI Express Extended Capability Header	1142
Table 7-40	MSI Capability Header	1145
Table 7-41	Message Control Register for MSI	1145
Table 7-42	Message Address Register for MSI	1147
Table 7-43	Message Upper Address Register for MSI	1147
Table 7-44	Message Data Register for MSI	1148
Table 7-45	Extended Message Data Register for MSI	1148

Table 7-46	Mask Bits Register for MSI	1149
Table 7-47	Pending Bits Register for MSI	1150
Table 7-48	MSI-X Capability Header	1154
Table 7-49	Message Control Register for MSI-X	1154
Table 7-50	Table Offset/Table BIR Register for MSI-X	1155
Table 7-51	PBA Offset/PBA BIR Register for MSI-X	1156
Table 7-52	Message Address Register for MSI-X Table Entries	1156
Table 7-53	Message Upper Address Register for MSI-X Table Entries	1157
Table 7-54	Message Data Register for MSI-X Table Entries	1157
Table 7-55	Vector Control Register for MSI-X Table Entries	1158
Table 7-56	Pending Bits Register for MSI-X PBA Entries	1159
Table 7-57	Secondary PCI Express Extended Capability Header	1161
Table 7-58	Link Control 3 Register	1161
Table 7-59	Lane Error Status Register	1163
Table 7-60	Lane Equalization Control Register Entry	1163
Table 7-63	Data Link Feature Extended Capability Header	1166
Table 7-64	Data Link Feature Capabilities Register	1167
Table 7-65	Data Link Feature Status Register	1168
Table 7-66	Physical Layer 16.0 GT/s Extended Capability Header	1171
Table 7-67	16.0 GT/s Capabilities Register	1171
Table 7-68	16.0 GT/s Control Register	1172
Table 7-69	16.0 GT/s Status Register	1172
Table 7-70	16.0 GT/s Local Data Parity Mismatch Status Register	1173
Table 7-71	16.0 GT/s First Retimer Data Parity Mismatch Status Register	1174
Table 7-72	16.0 GT/s Second Retimer Data Parity Mismatch Status Register	1175
Table 7-73	16.0 GT/s Lane Equalization Control Register Entry	1175
Table 7-75	Physical Layer 32.0 GT/s Extended Capability Header	1178
Table 7-76	32.0 GT/s Capabilities Register	1178
Table 7-77	32.0 GT/s Control Register	1179
Table 7-78	32.0 GT/s Status Register	1180
Table 7-79	Received Modified TS Data 1 Register	1182
Table 7-80	Received Modified TS Data 2 Register	1182
Table 7-81	Transmitted Modified TS Data 1 Register	1183
Table 7-82	Transmitted Modified TS Data 2 Register	1184
Table 7-83	32.0 GT/s Lane Equalization Control Register Entry	1186
Table 7-85	Physical Layer 64.0 GT/s Extended Capability Header	1188
Table 7-86	64.0 GT/s Capabilities Register	1188
Table 7-87	64.0 GT/s Control Register	1188
Table 7-88	64.0 GT/s Status Register	1189
Table 7-89	64.0 GT/s Lane Equalization Control Register Entry	1190
Table 7-91	Flit Logging Extended Capability Header	1193
Table 7-92	Flit Error Log Interpretation	1193
Table 7-93	Flit Error Log 1 Register	1194
Table 7-94	Flit Error Log 2 Register	1196
Table 7-95	Flit Error Counter Control Register	1197
Table 7-96	Flit Error Counter Status Register	1197
Table 7-97	FBER Measurement Control Register	1199
Table 7-98	FBER Measurement Status 1 Register	1199
Table 7-99	FBER Measurement Status 2 Register	1200
Table 7-100	FBER Measurement Status 3 Register	1200
Table 7-101	FBER Measurement Status 4 Register	1201
Table 7-102	FBER Measurement Status 5 Register	1201

Table 7-103	FBER Measurement Status 6 Register.....	1202
Table 7-104	FBER Measurement Status 7 Register.....	1202
Table 7-105	FBER Measurement Status 8 Register.....	1202
Table 7-106	FBER Measurement Status 9 Register.....	1203
Table 7-107	FBER Measurement Status 10 Register.....	1203
Table 7-108	Device 3 Extended Capability Header	1204
Table 7-109	Device Capabilities 3 Register.....	1205
Table 7-110	Device Control 3 Register.....	1208
Table 7-111	Device Status 3 Register	1210
Table 7-112	Lane Margining at the Receiver Extended Capability Header	1213
Table 7-113	Margining Port Capabilities Register	1213
Table 7-114	Margining Port Status Register	1214
Table 7-115	Lane N: Margining Control Register Entry	1215
Table 7-116	Lane N: Margining Lane Status Register Entry	1216
Table 7-117	ACS Extended Capability Header.....	1217
Table 7-118	ACS Capability Register	1218
Table 7-119	ACS Control Register	1219
Table 7-120	Egress Control Vector Register.....	1222
Table 7-121	Power Budgeting Extended Capability Header.....	1224
Table 7-122	Power Budgeting Control Register	1225
Table 7-123	Power Budgeting Data Register	1227
Table 7-124	Power Budgeting Capability Register.....	1231
Table 7-125	Power Budgeting Sense Detect Register	1233
Table 7-126	Power Budgeting Sense Detect Encodings	1233
Table 7-127	LTR Extended Capability Header	1236
Table 7-128	Max Snoop Latency Register.....	1237
Table 7-129	Max No-Snoop Latency Register.....	1237
Table 7-130	LTR Capabilities Register.....	1238
Table 7-131	L1 PM Substates Extended Capability Header	1239
Table 7-132	L1 PM Substates Capabilities Register.....	1240
Table 7-133	L1 PM Substates Control 1 Register	1241
Table 7-134	L1 PM Substates Control 2 Register	1243
Table 7-135	L1 PM Substates Status Register	1244
Table 7-136	Advanced Error Reporting Extended Capability Header	1247
Table 7-137	Uncorrectable Error Status Register	1248
Table 7-138	Uncorrectable Error Mask Register.....	1250
Table 7-139	Uncorrectable Error Severity Register.....	1253
Table 7-140	Correctable Error Status Register	1255
Table 7-141	Correctable Error Mask Register	1256
Table 7-142	Advanced Error Capabilities and Control Register	1258
Table 7-143	Header Log Register	1260
Table 7-144	Root Error Command Register.....	1261
Table 7-145	Root Error Status Register	1263
Table 7-146	Error Source Identification Register	1265
Table 7-147	TLP Prefix Log Register.....	1266
Table 7-148	First DW of Enhanced Allocation Capability.....	1266
Table 7-149	Second DW of Enhanced Allocation Capability.....	1267
Table 7-150	First DW of Each Entry for Enhanced Allocation Capability	1268
Table 7-152	Enhanced Allocation Entry Field Value Definitions for both the Primary Properties and Secondary Properties Fields	1270
Table 7-153	Resizable BAR Extended Capability Header.....	1275
Table 7-154	Resizable BAR Capability Register	1276

Table 7-155	Resizable BAR Control Register	1278
Table 7-156	VF Resizable BAR Extended Capability Header	1282
Table 7-157	VF Resizable BAR Control Register	1283
Table 7-158	ARI Extended Capability Header	1284
Table 7-159	ARI Capability Register	1285
Table 7-160	ARI Control Register	1286
Table 7-161	PASID Extended Capability Header	1287
Table 7-162	PASID Capability Register	1288
Table 7-163	PASID Control Register	1289
Table 7-164	FRS Queueing Extended Capability Header	1290
Table 7-165	FRS Queueing Capability Register	1291
Table 7-166	FRS Queueing Status Register	1292
Table 7-167	FRS Queueing Control Register	1292
Table 7-168	FRS Message Queue Register	1293
Table 7-169	FPB Capability Header	1294
Table 7-170	FPB Capabilities Register	1294
Table 7-174	FPB RID Vector Control 1 Register	1296
Table 7-177	FPB RID Vector Control 2 Register	1298
Table 7-178	FPB MEM Low Vector Control Register	1298
Table 7-181	FPB MEM High Vector Control 1 Register	1300
Table 7-184	FPB MEM High Vector Control 2 Register	1301
Table 7-186	FPB Vector Access Control Register	1302
Table 7-188	FPB Vector Access Data Register	1303
Table 7-189	Flit Performance Measurement Extended Capability Header	1305
Table 7-190	Flit Performance Measurement Capability Register	1305
Table 7-191	Flit Performance Measurement Control Register	1306
Table 7-192	Flit Performance Measurement Status Register	1308
Table 7-193	LTSSM Performance Measurement Status Register	1309
Table 7-194	Flit Error Injection Extended Capability Header	1311
Table 7-195	Flit Error Injection Capability Register	1312
Table 7-196	Flit Error Injection Control 1 Register	1312
Table 7-197	Flit Error Injection Control 2 Register	1314
Table 7-198	Flit Error Injection Status Register	1315
Table 7-199	Ordered Set Error Injection Control 1 Register	1316
Table 7-200	Ordered Set Error Injection Control 2 Register	1318
Table 7-201	Ordered Set Tx Error Injection Status Register	1318
Table 7-202	Ordered Set Rx Error Injection Status Register	1319
Table 7-203	NOP Flit Extended Capability Header	1321
Table 7-204	NOP Flit Capabilities Register	1322
Table 7-205	NOP Flit Control 1 Register	1322
Table 7-206	NOP Flit Control 2 Register	1324
Table 7-207	NOP Flit Status Register	1325
Table 7-208	Virtual Channel Extended Capability Header	1327
Table 7-209	Port VC Capability Register 1	1328
Table 7-210	Port VC Capability Register 2	1329
Table 7-211	Port VC Control Register	1330
Table 7-212	Port VC Status Register	1331
Table 7-213	VC Resource Capability Register	1331
Table 7-214	VC Resource Control Register	1333
Table 7-215	VC Resource Status Register	1335
Table 7-216	Definition of the 4-bit Entries in the VC Arbitration Table	1336
Table 7-217	Length of the VC Arbitration Table	1336

Table 7-218	Length of Port Arbitration Table	1337
Table 7-219	MFVC Extended Capability Header	1339
Table 7-220	MFVC Port VC Capability Register 1	1340
Table 7-221	MFVC Port VC Capability Register 2	1341
Table 7-222	MFVC Port VC Control Register.....	1341
Table 7-223	MFVC Port VC Status Register.....	1342
Table 7-224	MFVC VC Resource Capability Register	1343
Table 7-225	MFVC VC Resource Control Register	1344
Table 7-226	MFVC VC Resource Status Register	1346
Table 7-227	Length of Function Arbitration Table	1348
Table 7-228	Device Serial Number Extended Capability Header	1349
Table 7-229	Serial Number Register	1350
Table 7-230	Vendor-Specific Capability	1351
Table 7-231	Vendor-Specific Extended Capability Header	1352
Table 7-232	Vendor-Specific Header	1353
Table 7-233	Designated Vendor-Specific Extended Capability Header	1354
Table 7-234	Designated Vendor-Specific Header 1	1355
Table 7-235	Designated Vendor-Specific Header 2	1356
Table 7-236	RCRB Header Extended Capability Header	1357
Table 7-237	RCRB Vendor ID and Device ID register	1357
Table 7-238	RCRB Capabilities register	1358
Table 7-239	RCRB Control register	1358
Table 7-240	Root Complex Link Declaration Extended Capability Header	1360
Table 7-241	Element Self Description Register	1361
Table 7-242	Link Description Register	1362
Table 7-243	Link Address for Link Type 1	1364
Table 7-244	Root Complex Internal Link Control Extended Capability Header.....	1365
Table 7-245	Root Complex Link Capabilities Register	1366
Table 7-246	Root Complex Link Control Register	1369
Table 7-247	Root Complex Link Status Register	1370
Table 7-248	Root Complex Event Collector Endpoint Association Extended Capability Header	1371
Table 7-249	RCEC Associated Bus Numbers Register	1372
Table 7-250	Multicast Extended Capability Header	1374
Table 7-251	Multicast Capability Register	1375
Table 7-252	Multicast Control Register.....	1376
Table 7-253	MC_Base_Address Register.....	1376
Table 7-254	MC_Receive Register	1377
Table 7-255	MC_Block_All Register	1378
Table 7-256	MC_Block_Untranslated Register	1378
Table 7-257	MC_Overlay_BAR Register	1379
Table 7-258	DPA Extended Capability Header.....	1380
Table 7-259	DPA Capability Register.....	1380
Table 7-260	DPA Latency Indicator Register.....	1381
Table 7-261	DPA Status Register	1382
Table 7-262	DPA Control Register	1382
Table 7-263	Substate Power Allocation Register (0 to Substate_Max)	1383
Table 7-264	TPH Requester Extended Capability Header.....	1384
Table 7-265	TPH Requester Capability Register.....	1385
Table 7-266	TPH Requester Control Register	1386
Table 7-267	TPH ST Table Entry.....	1387
Table 7-268	DPC Extended Capability Header	1390
Table 7-269	DPC Capability Register	1391

Table 7-270	DPC Control Register	1392
Table 7-271	DPC Status Register	1394
Table 7-272	DPC Error Source ID Register	1396
Table 7-273	RP PIO Status Register	1396
Table 7-274	RP PIO Mask Register	1397
Table 7-275	RP PIO Severity Register	1398
Table 7-276	RP PIO SysError Register	1399
Table 7-277	RP PIO Exception Register	1400
Table 7-278	RP PIO Header Log Register	1401
Table 7-279	RP PIO ImpSpec Log Register	1401
Table 7-280	RP PIO TLP Prefix Log Register	1402
Table 7-281	PTM Extended Capability Header	1403
Table 7-282	PTM Capability Register	1404
Table 7-283	PTM Control Register	1405
Table 7-285	Readiness Time Reporting Extended Capability Header	1408
Table 7-286	Readiness Time Reporting 1 Register	1408
Table 7-287	Readiness Time Reporting 2 Register	1409
Table 7-288	Hierarchy ID Extended Capability Header	1411
Table 7-289	Hierarchy ID Status Register	1412
Table 7-290	Hierarchy ID Data Register	1413
Table 7-291	Hierarchy ID GUID 1 Register	1414
Table 7-292	Hierarchy ID GUID 2 Register	1414
Table 7-293	Hierarchy ID GUID 3 Register	1415
Table 7-294	Hierarchy ID GUID 4 Register	1415
Table 7-295	Hierarchy ID GUID 5 Register	1416
Table 7-296	VPD Address Register	1418
Table 7-297	VPD Data Register	1418
Table 7-298	NPEM Extended Capability Header	1419
Table 7-299	NPEM Capability Register	1420
Table 7-300	NPEM Control Register	1421
Table 7-301	NPEM Status Register	1423
Table 7-302	Alternate Protocol Extended Capability Header	1424
Table 7-303	Alternate Protocol Capabilities Register	1425
Table 7-304	Alternate Protocol Control Register	1425
Table 7-305	Alternate Protocol Data 1 Register	1426
Table 7-306	Alternate Protocol Data 2 Register	1427
Table 7-307	Alternate Protocol Selective Enable Mask Register	1427
Table 7-308	Advanced Features Capability Header	1428
Table 7-309	AF Capabilities Register	1429
Table 7-310	Conventional PCI Advanced Features Control Register	1429
Table 7-311	AF Status Register	1430
Table 7-312	SFI Extended Capability Header	1431
Table 7-313	SFI Capability Register	1432
Table 7-314	SFI Control Register	1433
Table 7-315	SFI Status Register	1435
Table 7-316	SFI CAM Address Register	1436
Table 7-317	SFI CAM Data Register	1436
Table 7-318	Subsystem ID and Subsystem Vendor ID Capability Header	1438
Table 7-319	Subsystem ID and Subsystem Vendor ID Capability Data	1438
Table 7-320	DOE Extended Capability Header	1439
Table 7-321	DOE Capabilities Register	1440
Table 7-322	DOE Control Register	1441

Table 7-323	DOE Status Register	1442
Table 7-324	DOE Write Data Mailbox Register	1443
Table 7-325	DOE Read Data Mailbox Register	1444
Table 7-326	Shadow Functions Extended Capability Header	1446
Table 7-327	Shadow Functions Capability Register	1447
Table 7-328	Shadow Functions Control Register	1447
Table 7-329	Shadow Functions Instance Register Entry	1448
Table 7-330	IDE Extended Capability Header	1449
Table 7-331	IDE Capability Register	1450
Table 7-332	IDE Control Register	1452
Table 7-333	Link IDE Stream Control Register	1453
Table 7-334	Link IDE Stream Status Register	1454
Table 7-335	Selective IDE Stream Capability Register	1455
Table 7-336	Selective IDE Stream Control Register	1456
Table 7-337	Selective IDE Stream Status Register	1459
Table 7-338	IDE RID Association Register 1 (Offset +00h)	1459
Table 7-339	IDE RID Association Register 2 (Offset +04h)	1460
Table 7-340	IDE Address Association Register 1 (Offset +00h)	1461
Table 7-341	IDE Address Association Register 2 (Offset +04h)	1461
Table 7-342	IDE Address Association Register 3 (Offset +04h)	1461
Table 7-343	Null Capability	1462
Table 7-344	Null Extended Capability	1463
Table 7-345	Streamlined Virtual Channel Extended Capability Header	1464
Table 7-346	SVC Port Capability Register 1	1465
Table 7-347	SVC Port Control Register	1466
Table 7-348	SVC Port Status Register	1466
Table 7-349	SVC Resource Capability Register	1467
Table 7-350	SVC Resource Control Register	1468
Table 7-351	SVC Resource Status Register	1469
Table 7-352	MRBL Extended Capability Header	1471
Table 7-353	MRBL Capabilities Register	1471
Table 7-354	MRBL Locator Register	1472
Table 8-1	Tx Preset Ratios and Corresponding Coefficient Values for 8.0, 16.0, and 32.0 GT/s	1479
Table 8-2	Tx Preset Ratios and Corresponding Coefficient Values for 64.0 GT/s	1480
Table 8-3	Cases that the Reference Packages and ps21 _{TX} Parameter are Normative	1487
Table 8-4	Recommended De-embedding Cutoff Frequency	1498
Table 8-5	Tx Measurement and Post Processing For Different Refclks	1500
Table 8-6	Data Rate Dependent Transmitter Parameters	1508
Table 8-7	Data Rate Independent Tx Parameters	1516
Table 8-8	Calibration Channel IL Limits	1519
Table 8-11	Stressed Jitter Eye Parameters	1540
Table 8-12	Common Receiver Parameters	1552
Table 8-13	Lane Margining	1556
Table 8-14	Package Model Capacitance Values	1562
Table 8-15	Jitter/Voltage Parameters for Channel Tolerancing	1574
Table 8-16	Channel Tolerancing Eye Mask Values	1577
Table 8-17	EIEOS Signaling Parameters	1580
Table 8-18	REFCLK DC Specifications and AC Timing Requirements	1581
Table 8-19	Data Rate Independent Refclk Parameters	1585
Table 8-20	Jitter Limits for CC Architecture	1591
Table 8-21	Form Factor Clocking Architecture Requirements	1591
Table 8-22	Form Factor Common Clock Architecture Details	1592

Table 8-23	Form Factor Clocking Architecture Requirements Example	1592
Table 8-24	Form Factor Common Clock Architecture Details Example	1592
Table 9-1	VF Routing ID Algorithm	1611
Table 9-2	SR-IOV Extended Capability Header	1618
Table 9-3	SR-IOV Capabilities Register	1618
Table 9-4	SR-IOV Control Register	1621
Table 9-5	SR-IOV Status	1625
Table 9-8	BAR Offsets	1630
Table 9-9	SR-IOV Usage of PCI Standard Capabilities	1631
Table 9-10	SR-IOV Usage of PCI Express Extended Capabilities	1631
Table 9-11	SIOV Extended Capability Header	1635
Table 9-12	SIOV Capabilities Register	1636
Table 9-13	SIOV Status Register	1636
Table 10-1	Address Type (AT) Field Encodings	1648
Table 10-2	Translation Completion Status Codes	1652
Table 10-3	Translation Completion Data Fields	1653
Table 10-5	Examples of Translation Size Using S Field	1656
Table 10-6	Page Request Message Data Fields	1675
Table 10-7	PRG Response Message Data Fields	1680
Table 10-8	Response Codes	1680
Table 10-9	ATS Extended Capability Header	1682
Table 10-10	ATS Capability Register (Offset 04h)	1683
Table 10-11	ATS Control Register	1684
Table 10-13	Page Request Extended Capability Header	1685
Table 10-14	Page Request Control Register	1686
Table 10-15	Page Request Status Register	1687
Table 11-1	XT Bit and T Bit Encodings	1694
Table 11-2	INTERFACE_ID Definition	1697
Table 11-3	Example DSM Tracking and Handling for Architected Registers	1704
Table 11-4	TDISP Request Codes	1707
Table 11-5	TDISP Response Codes	1708
Table 11-6	TDISP Message Format	1709
Table 11-7	Generic Error Response Codes	1710
Table 11-8	TDISP_VERSION	1710
Table 11-9	GET_TDISP_CAPABILITIES	1711
Table 11-10	TDISP_CAPABILITIES	1711
Table 11-11	LOCK_INTERFACE_REQUEST	1713
Table 11-12	LOCK_INTERFACE_RESPONSE	1714
Table 11-13	LOCK_INTERFACE_REQUEST Error Codes	1714
Table 11-14	GET_DEVICE_INTERFACE_REPORT	1715
Table 11-15	DEVICE_INTERFACE_REPORT	1715
Table 11-16	TDI Report Structure	1716
Table 11-17	GET_DEVICE_INTERFACE_REPORT Error Response Codes	1718
Table 11-18	DEVICE_INTERFACE_STATE	1718
Table 11-19	START_INTERFACE_REQUEST	1719
Table 11-20	START_INTERFACE_REQUEST Error Response Codes	1719
Table 11-21	BIND_P2P_STREAM_REQUEST	1721
Table 11-22	BIND_P2P_STREAM_REQUEST Error Response Codes	1721
Table 11-23	UNBIND_P2P_STREAM_REQUEST	1722
Table 11-24	UNBIND_P2P_STREAM_REQUEST Error Response Codes	1722
Table 11-25	SET_MMIO_ATTRIBUTE_REQUEST	1723
Table 11-26	SET_MMIO_ATTRIBUTE_REQUEST Error Response Codes	1724

Table 11-27	TDISP_ERROR.....	1724
Table 11-28	Error Code and Error Data.....	1724
Table 11-29	EXTENDED_ERROR_DATA	1725
Table 11-30	VDM_REQUEST	1726
Table 11-31	VDM_RESPONSE.....	1726
Table 11-32	SET_TDISP_CONFIG_REQUEST	1727
Table 11-33	SET_TDISP_CONFIG_RESPONSE.....	1727
Table 11-34	Example TSM Tracking and Handling for Root Port Configurations	1737
Table 12-1	Relative Comparisons of Typical Architectural Out-of-Band Interfaces	1743
Table 12-2	PESTI DC Specifications	1751
Table 12-3	PESTI Initiator Control and Status Registers.....	1754
Table 12-4	PESTI Discovery Status State Transitions.....	1754
Table 12-5	PESTI AC Specifications	1756
Table 12-6	PESTI Discovery Payload.....	1758
Table 12-7	Example VWIRE_OUT_0 (Initiator to Target)	1761
Table 12-8	Example VWIRE_IN_0 (Target to Initiator).....	1761
Table 12-9	MSC_CTRL_VAL (Initiator to PESTI Snooper Target) Data Byte Value = 02h	1764
Table 12-10	MSC_STAT_VAL (PESTI Snooper Target to Initiator)	1765
Table 12-11	2-Wire Interface Example Usages	1768
Table 12-12	Baseline SMBus Recommended Default Target Addresses.....	1769
Table 12-13	I3C Basic Logic Signaling DC Specification	1776
Table 12-14	I3C Timing Requirements	1776
Table 12-15	PCI-SIG MultiRecord	1781
Table 12-17	PCI-SIG MultiRecord Descriptor.....	1782
Table 12-18	Descriptor Sub-Types for Group ID 0h.....	1783
Table 12-19	Connector Subdivision Combinations Descriptor	1784
Table 12-20	Connector Subdivision Descriptor	1784
Table 12-21	PCIe-MI Command	1797
Table 12-24	PCIe-MI Completion	1798
Table 12-27	PCIe-MI Opcodes	1799
Table 12-28	PCIe-MI Status Codes	1800
Table 12-29	Invalid Input Error Completion.....	1801
Table 12-31	Information Identifiers.....	1804
Table 12-32	Get Information Command.....	1804
Table 12-33	Get Information – Supported Information Identifiers Completion	1805
Table 12-34	Get Information – Supported Opcodes Completion	1806
Table 12-35	Get Information – Supported Parameters Completion	1806
Table 12-39	Get Information – Component Information Completion	1807
Table 12-40	Get Information – Port Information Command	1808
Table 12-41	Get Information – Port Information Completion	1808
Table 12-44	Get Information – PCIe Port-Specific Information Completion	1809
Table 12-47	Get Information – Function List Command.....	1811
Table 12-49	Get Information – Function List Completion	1812
Table 12-50	Function List Data Structure	1812
Table 12-51	Get Information – Function Information Command	1812
Table 12-52	Get Information – Function Information Completion	1813
Table 12-55	Capability Structure Data Structure	1815
Table 12-56	Parameter Identifiers	1815
Table 12-57	Get Parameters Command	1816
Table 12-58	Set Parameters Command.....	1816
Table 12-59	Get Parameters – Configuration Space Writability Command	1818
Table 12-60	Get Parameters – Configuration Space Writability Completion.....	1818

Table 12-61	Configuration Space Writability Data Structure.....	1819
Table 12-66	Set Parameters – Configuration Space Writability Command	1820
Table 12-71	Get Parameters – Routing Information Command	1823
Table 12-72	Get Parameters – Routing Information Completion	1823
Table 12-73	Set Parameters – Routing Information Command	1824
Table 12-74	Get Parameters – Configuration Space Command	1825
Table 12-75	Get Parameters – Configuration Space Completion	1825
Table 12-76	Set Parameters – Configuration Space Command	1826
Table 12-77	Get Parameters – Connector Subdivision Command	1826
Table 12-78	Get Parameters – Connector Subdivision Completion.....	1826
Table 12-79	Set Parameters – Connector Subdivision Command	1827
Table 12-80	Set Parameters – Connector Subdivision Completion	1828
Table A-1	Isochronous Bandwidth Ranges and Granularities.....	1834
Table B-1	8b/10b Data Symbol Codes	1841
Table B-2	8b/10b Special Character Symbol Codes.....	1849
Table F-1	Message Code Usage.....	1869
Table F-2	PCI-SIG-Defined VDM Subtype Usage.....	1870
Table G-1	PCI Express Attribute Impact on Protocol Multiplexing.....	1873
Table G-2	PMUX Attribute Impact on PCI Express	1875
Table G-3	PMUX Packet Layout (8b/10b Encoding).....	1877
Table G-4	PMUX Packet Layout (128b/130b Encoding).....	1879
Table G-5	Symbol 1 Bits [6:3].....	1880
Table G-6	PMUX Extended Capability Header.....	1882
Table G-7	PMUX Capability Register.....	1883
Table G-8	PMUX Control Register	1884
Table G-9	PMUX Status Register	1885
Table G-10	PMUX Protocol Array Entry	1887
Table H-1	Maximum UpdateFC Transmission Latency Guidelines for 2.5 GT/s Mode Operation by Link Width and Max Payload (Symbol Times).....	1890
Table H-2	Maximum UpdateFC Transmission Latency Guidelines for 5.0 GT/s Mode Operation by Link Width and Max Payload (Symbol Times).....	1890
Table H-3	Maximum UpdateFC Transmission Latency Guidelines for 8.0 GT/s Operation by Link Width and Max Payload (Symbol Times)	1891
Table H-5	Maximum Ack Latency Limit and AckFactor for 2.5 GT/s (Symbol Times)	1892
Table H-6	Maximum Ack Transmission Latency Limit and AckFactor for 5.0 GT/s (Symbol Times)	1893
Table H-7	Maximum Ack Transmission Latency Limit and AckFactor for 8.0 GT/s (Symbol Times)	1893
Table L-1	Inputs and Outputs for Example IDE TLP	2201
Table L-2	Inputs and Outputs for Example IDE TLP (FM)	2203
Table L-3	Inputs and Outputs for Example IDE TLP with Partial Header Encryption (mode 0100b - Address[41:2] Encrypted) (NFM)	2205
Table L-4	Inputs and Outputs for Example IDE TLP with Partial Header Encryption (mode 0100b - Address[41:2] Encrypted) (FM)	2207
Table L-5	IDE Test Keys.....	2208

Status of this Document

This section describes the status of this document at the time of its publication. Other documents may supersede this document. A list of current PCISIG publications and the latest revision of this specification can be found at pcisig.com

This is the *PCI Express Base 6.4 Specification*. This consists of Base 6.3 plus approved ECNs, errata, and editorial corrections. See [Critical Errata](#) and [Important Errata and Approved ECNs](#).

- The **NCB-PCI_Express_Base_6.4.pdf** is normative (i.e., the official specification). It contains no changebars.
- The **CB-PCI_Express_Base_6.4-vs-6.3.pdf** is informative. It contains changebars relative to the *PCI Express Base 6.3 Specification* [PCIe-6.3].

NOTE: Background on the new Document Process §

The new PCISIG document system is a variant of the w3c Respec tool (see <https://github.com/w3c/respec/wiki>). Respec is a widely used tool written to support the World Wide Web specifications. The PCISIG variant is <https://github.com/sglaser/respec>. Both Respec and the PCISIG variant are open source (MIT License) Javascript libraries. They operate in the author's browser and provide a rapid edit / review cycle without requiring any special tools be installed.

Respec is built on top of HTML5, the document format for the World Wide Web <http://www.w3.org/TR/html5/>. HTML is a text-based document format that allows us to deploy tools commonly used for software development (git, continuous integration, build scripts, etc.) to better manage and control the spec development process.

PCISIG enhancements to Respec support document formatting closer to existing PCISIG practice as well as automatic creation of register figures (eliminating about half of the manually drawn figures).

NOTE: Navigating in changebar documents §

The Base 6.0.1 version introduces a new errata delivery process. Instead of having a separate “change this to that” document, a new version of the specification is produced with the changes integrated into the document. This makes it easier to consume and less likely that errata will be overlooked.

All changes are annotated with the  character. Searching for this character in a PDF reader will step through all changes. There is a  character in the upper right of page 1 that can be copied into the search string.

Inserted text is yellow and underlined. It contains the  character.

Deleted text is red and struck through. It contains the  character.

All errata are identified by a “red box” near the first change in a chunk of changes. This box contains one or more triangle characters that can navigate through the changes for the associated errata.



The upward pointing triangle is a link to the associated errata table entry.



The right pointing triangle is a link to the next change associated with this errata. If this is the last change, this triangle is not present.



The left pointing triangle is a link to the previous change associated with this errata. If this is the first change, this triangle is not present.

Changes that are not marked with a red box are considered editorial in nature and are not associated with an errata.

Where the automated process can't correctly identify changes (e.g., figures and equations), the revision history contains the change description. When fields are added or removed from registers, the automatically drawn figure before the register table may not show these fields. The table will correctly show such fields as will the drawings in the non-changebar version of the document. Since the table accurately shows the change, the revision history doesn't call out the “missing” figure change.

Publication does not imply endorsement by the PCI-SIG Membership. This document may be updated, replaced or obsoleted by other documents at any time.

This document is governed by the PCI-SIG Specification Development Procedures.

Revision History

Revision	Revision History	Date
1.0	Initial release.	07/22/2002
1.0a	Incorporated Errata C1-C66 and E1-E4.17.	04/15/2003
1.1	Incorporated approved Errata and ECNs.	03/28/2005
2.0	Added 5.0 GT/s data rate and incorporated approved Errata and ECNs.	12/20/2006
2.1	Incorporated <i>Errata for the PCI Express Base Specification, Rev. 2.0</i> (February 27, 2009), and added the following ECNs: • Internal Error Reporting ECN (April 24, 2008) • Multicast ECN (December 14, 2007, approved by PWG May 8, 2008) • Atomic Operations ECN (January 15, 2008, approved by PWG April 17, 2008) • Resizable BAR Capability ECN (January 22, 2008, updated and approved by PWG April 24, 2008) • Dynamic Power Allocation ECN (May 24, 2008) • ID-Based Ordering ECN (January 16, 2008, updated 29 May 2008) • Latency Tolerance Reporting ECN (22 January 2008, updated 14 August 2008) • Alternative Routing-ID Interpretation (ARI) ECN (August 7, 2006, last updated June 4, 2007) • Extended Tag Enable Default ECN (September 5, 2008) • TLP Processing Hints ECN (September 11, 2008) • TLP Prefix ECN (December 15, 2008)	03/04/2009
3.0	Added 8.0 GT/s data rate, incorporated approved Errata, and the following ECNs: • Optimized Buffer Flush/Fill ECN (8 February 2008, updated 30 April 2009) • ASPM Optionality ECN (June 19, 2009, approved by the PWG August 20, 2009) • Incorporated End-End TLP Changes for RCs ECN (26 May 2010) and Protocol Multiplexing ECN (17 June 2010)	11/10/2010
3.1	Incorporated feedback from Member Review Incorporated Errata for the PCI Express® Base Specification Revision 3.0 Incorporated M-PCIe Errata (3p1_active_errata_list_mpcie_28Aug2014.doc and 3p1_active_errata_list_mpcie_part2_11Sept2014.doc) Incorporated the following ECNs: • ECN: Downstream Port Containment (DPC) • ECN: Separate Refclk Independent SSC Architecture (SRIS) • ECN: Process Address Space ID (PASID) • ECN: Lightweight Notification (LN) Protocol	10/8/2014

Revision	Revision History	Date
	• ECN: Precision Time Measurement • ECN: Enhanced DPC (eDPC) • ECN: 8.0 GT/s Receiver Impedance • ECN: L1 PM Substates with CLKREQ • ECN: Change Root Complex Event Collector Class Code • ECN: M-PCIe • ECN: Readiness Notifications (RN) • ECN: Separate Refclk Independent SSC Architecture (SRIS) JTOL and SSC Profile Requirements	
3.1a	Minor update: Corrected: Equation 4.3.9 in Section 4.3.8.5., Separate Refclk With Independent SSC (SRIS) Architecture. Added missing square (exponent=2) in the definition of B. $B = 2.2 \times 10^{12} \times (2\pi)^2$ where ^= exponent.	12/5/2015
	Version 0.3: Based on PCI Express® Base Specification Revision 3.1 (October 8, 2014) with some editorial feedback received in December 2013. • Added § Chapter 9. , Electrical Sub-block: Added § Chapter 9. (Rev0.3-11-30-13_final.docx) • Changes related to Revision 0.3 release • Incorporated PCIe-relevant material from PCI Bus Power Management Interface Specification (Revision 1.2, dated March 3, 2004). This initial integration of the material will be updated as necessary and will supersede the standalone Power Management Interface specification.	
	Version 0.5 (12/22/14, minor revisions on 1/26/15, minor corrections 2/6/15) • Added front matter with notes on expected discussions and changes. • Added ECN:Retimer (dated October 6, 2014) • Corrected § Chapter 4. title to, “Physical Layer Logical Block”. • Added Encoding subteam feedback on § Chapter 4.	2/6/2015
4.0	• Added Electrical work group changes from PCIe Electrical Specification Rev 0.5 RC1 into § Chapter 9.	
	Version 0.7: Based on PCI Express® Base Specification Version 4.0 Revision 0.5 (11/23/2015) • Added ECN_DVSEC-2015-08-04 • Applied ECN PASID-ATS dated 2011-03-31 • Applied PCIe Base Spec Errata: PCIe_Base_r3 1_Errata_2015-09-18 except: ◦ B216; RCIE ◦ B256; grammar is not clear • Changes to Chapter 7. Software Initialization and Configuration per PCIe_4.0_regs_0-3F_gord_7.docx • Added Chapter SR-IOV Spec Rev 1.2 (Rev 1.1 dated September 8, 2009 plus: ◦ SR-IOV_11_errata_table.doc	11/24/2015

Revision	Revision History	Date
	◦ DVSEC ◦ 3.1 Base Spec errata • Added Chapter ATS Spec Rev 1.2 (Rev 1.1 dated January 26, 2009 plus: ◦ ECN-PASID-ATS ◦ 3.1 Base Spec errata	
	2/18/2016 Changes from the Protocol Working Group • Applied changes from the following documents: ◦ FC Init/Revision scaled-flow-control-pcie-base40-2016-01-07.pdf (Steve.G) ◦ Register updates for integrated legacy specs PCIe_4.0_regs_0-3F_gord_8.docx (GordC) ◦ Tag Scaling PCIe 4_0 Tag Field scaling 2015-11-23 clean.docx (JoeC) ◦ MSI/MSI-X PCIe 4_0 MSI & MSI-X 2015-12-18 clean.docx (JoeC); register diagrams TBD on next draft. ◦ REPLAY_TIMER/Ack/FC Limits Ack_FC_Replay_Timers_ver8 (PeterJ)	2/18/16
	Chapter 10. SR-IOV related changes: • Incorporated “SR-IOV and Sharing Specification” Revision 1.1 dated January 20, 2010 (sr-iov1_1_20Jan10.pdf) as § Chapter 10. , with changes from the following documents ◦ Errata for the PCI Express® Base Specification Revision 3.1, Single Root I/O Virtualization and Sharing Revision 1.1, Address Translation and Sharing Revision 1.1, and M.2 Specification Revision 1.0: PCIe_Base_r3 1_Errata_2015-09-18_clean.pdf ◦ ECN__Integrated_Endpoints_and_IOV_updates__19 Nov 2015_Final.pdf ◦ Changes marked “editorial” only in marked PDF: sr-iov1_1_20Jan10-steve-manning-comments.pdf	4/26/16 [snapshot]
	Chapter 9. Electrical Sub-Block related changes: Source: WG approved word document from Dan Froelich (FileName: Electrical-PCI_Express_Base_4.0r0.7_April_7_wg_approved_redo_for_figure_corruption.docx)	5/23/16[snapshot]
	Version 0.7 continued... Chapter 4. PHY Logical Changes based on: • Chapter4-PCI_Express_Base_4 0r0 7_May3_2016_draft.docx Chapter 7. . PHY Logical Changes based on: • PCI_Express_Base_4 0r0 7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx	
	----- Changes incorporated into the August 2016 4.0 r0.7 Draft PDF ----- June 16 Feedback from PWG on the May 2016 snapshot PWG Feedback on 4.0 r0.7 Feb-Apr-May-2016 Drafts	8/30/16

Revision	Revision History	Date
	*EWG Feedback: -CB-PCI_Express_Base_4.0r0.7_May-2016 (Final).fdf -EWG f/b: Electrical-PCI_Express_Base_4.0r0.7_April_7_wg_approved_redo_for_figure_corruption_Broadco.docx *PWG Feedback: -PWG 0.7 fix list part1 and part 2.docx -PWG 0.7 fix list part3a.docx -PCI_Express_Base_4.0r0.7_pref_April-2016_chp5_PM_stuff_only_ver3.docx -PCI_Express_Base_4.0r0.7_pref_April-2016_chp5_PM_stuff_only_ver3.docx -scaled-flow-control-pcie-base40-2016-07-07.pdf -ECN_NOP_DLLP-2014-06-11_clean.pdf -ECN_RN_29_Aug_2013.pdf -3p1_active_errata_list_mpcie_28Aug2014.doc -3p1_active_errata_list_mpcie_part2_11Sept2014.doc -lane-margining-capability-snapshot-2016-06-16.pdf -Emergency Power Reduction Mechanism with PWRBRK Signal ECN -PWG 0.7 fix list part4.docx -ECN_Conventional_Adv_Caps_27Jul06.pdf -10-bit Tag related SR-IOV Updates *Other: -Merged Acknowledgements back pages from SR-IOV and ATS specifications into the main base spec. Acknowledgements page.	
	----- Changes since August 2016 for the September 2016 4.0 r0.7 Draft PDF----- Applied: PWG Feedback/Corrections on August draft ECN_SR-IOV_Table_Updates_16-June-2016.doc	9/28/16
	----- Changes since September 28 2016 for the October 2016 4.0 r0.7 Draft PDF---- EWG: Updates to <u>§ Chapter 9</u>. - Electrical Sub-block (Sections: 9.4.1.4, 9.6.5.1, 9.6.5.2, 9.6.7) PWG: Updates to Sections: 3.2.1, 3.3, 3.5.1, 7.13, 7.13.3 (Figure: Data Link Status Register)	10/7/16
	----- Changes to the October 13 2016 4.0 r0.7 Draft PDF---- EWG:	10/21/16

Revision	Revision History	Date
	Updates to § Chapter 9. - Electrical Sub-block (§ Section 9.4.3.9 and Figure 9-9 caption)	
	----- Changes to the November 3 2016 4.0 r0.7 Draft PDF----- § Section 2.6.1 Flow Control Rules: Updated Scaled Flow Control sub-bullet under FC initialization bullet (before Table 2-43)	11/3/16
	----- Changes to the November 11 2016 4.0 r0.7 Draft PDF----- Added M-PCIe statement to the Open Issues page Updated date to November 11, 2016	11/11/16
	----- Version 0.9: Based on PCI Express® Base Specification Version 4.0 Revision 0.7 (11/11/2016) Incorporated the following ECNs: -ECN-Hierarchy_ID-2017-02-23 -ECN_FPB_9_Feb_2017 -ECN Expanded Resizable BARs 2016-04-18 -ECN-VF-Resizable-BARs_6-July-2016 - § Chapter 7. reorganized: • New section 7.6 created per a PWG-approved reorganization to move sections 7.5, 7.6., and 7.10 to subsections 7.6.1 through 7.6.3 resp. • New section 7.7 created per a PWG-approved reorganization to move sections 7.7, 7.8., 7.12, 7.13, 7.40, 7.41 and 7.20 to subsections 7.7.1 through 7.7.7 resp. • New section 7.9 created per a PWG-approved reorganization to move sections 7.15, 7.22, 7.16, 7.23, 7.39, 7.24, 7.17, 7.18, 7.21, 7.25, 7.28, 7.30, 7.33, 7.34, 7.35, 7.38, and 7.42 to subsections 7.9.1 through 7.9.17 resp. -Removed § Chapter 8. : M-PCIe Logical Sub-Block -Updated § Chapter 9. (8 now), EWG Updates to § Chapter 9. - Electrical Sub-block per: Chapter9-PCI_Express_Base_4 0r09_March_30-2017_approved.docx -Updated § Chapter 4. : Physical Layer Logical Block per PCI_Express_Base_4 0_r0 9_Chapter4_Final_Draft.docx -Updated Figures in § Chapter 10. : ATS Specification -Removed § Appendix H. : M-PCIe timing Diagrams -Removed Appendix I: M-PCIe Compliance Patterns, pursuant to removing the M-PCIe Chapter this 0.9 version of the 4.0 Base Spec. -Added § Appendix H. : Flow Control Update Latency and ACK Update Latency Calculations -Added Appendix I: Vital Product Data (VPD)	April 28 2017
	-Updated editorial feedback on the Appendix section per: PCI_Express_Base_4.0r0.7_appendixes_November-11-2016_combined-editorial.docx	May 26, 2017

Revision	Revision History	Date
	-Deleted references to M-PCIe throughout the document. -Updated § Chapter 9. (8 now), EWG Updates to § Chapter 9. - Electrical Sub-block per: Chapter9-PCI_Express_Base_4 0r09_March_30-2017_approved.docx -Updated § Chapter 4. : Physical Layer Logical Block per PCI_Express_Base_4 0_r0 9_Chapter4_Final_Draft.docx -Updated Figures in § Chapter 10. : ATS Specification -Added § Appendix H. : Flow Control Update Latency and ACK Update Latency Calculations -Following items that were marked deleted in the Change Bar version of the April 28th snapshot have been “accepted” to no longer show up: pp 1070: Lane Equalization Control 2 Register (Offset TBD) Comment: Deleted per: PCI_Express_Base_4 0r0 7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx pp 1074: Physical Layer 16.0 GT/s Margining Extended Capability section Comment: Deleted per: PCI_Express_Base_4 0r0 7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx Comment: Replaced by Section Lane Margining at the Receiver Extended Capability per Fix3a #83 lane-margining-capability-snapshot-2016-06-16.pdf -Incorporated: PCIe 4_0 Tag Field scaling 2017-03-31.docx -Vital Product Data (VPD) -Added § Section 6.27 -Added § Section 7.9.4 -Incorporated feedback from April 28th snapshot.[source: 3 fdf files] -Completed editorial feedback on the Appendix section per: PCI_Express_Base_4.0r0.7_appendixes_November-11-2016_combined-editorial.docx -Incorporated ECN EMD for MSI 2016-05-10 -Updated per: PWG F2F changes from: PCI_Express_Base_4.0r0.7_pref_November-11-2016-F2F-2017-03-16-2017-03-30-sdg.docx -Updated figures per following lists (Gord Caruk): PCIe_4 0_fix_drawing_items.doc PCIe_4 0_fix_drawing_items_part2.doc	
	Version 0.91 <i>***Note this version will be used as the base for the PCI Express® Base Specification Revision 5.0***</i> <i>Item numbers are with reference to PWG CheckList (https://members.pcisig.com/wg/PCIe-Protocol/document/10642)</i> -Moved Flattening Portal Bridge Section 7.10 to Section 7.8.10. PWG Checklist Items #12.1 -Fixed misc. feedback that needed clarification from the 0.9 version. Issues fall under the categories of figure updates, broken cross references. Also incorporated feedback received from member review of the 4.0 version rev. 0.9 Base Spec. -Updated to reconcile issues related to incorporating the Extended Message Data for MSI ECN. PWG Checklist Items #22 -Completed incorporating all resolved editorial items from PWG Checklist Items #14, 14.1,15.1, 36, 42. TBD: Some minor editorial items from #13, #14 and #15 have been deferred to post 0.91 by reviewers. TBD: Errata and NPEM ECN	August 17, 2017

Revision	Revision History	Date
	ECN: ECN_Native_PCl_e_Enclosure_Management_v10August2017.docx Deleted Section 5.11.1 through Section 5.14 Changes tracked by items 34.01 34.02 34.04 34.05 34.11 in the PWG checklist Errata: B265, C266, 267, 268, B269, A270, A271, B274, C275, B276, B277, B278, B279, B280, B281, B283, B284, B285, B286, B288, B289, B292, B293, B294, B295, B297, B299, B300, B301 Other minor edits per: NCB-PCI_Express_Base_4.0r0.91_August-17-2017__dh_sdg_Annot_2.fdf	August 28, 2017
	Applied fixes and corrections captured in NCB-PCI_Express_Base_4.0r1.0_August-28-2017.fdf (Revision 8): https://members.pcisig.com/wg/PCIe-Protocol/document/10770 Updated contributor list in Appendix section.	September 20, 2017
	Updated contributor list in Appendix section. Inserted correct Figure 6-2. Applied minor fixes and corrections captured in: NCB-PCI_Express_Base_4.0r1.0_September-20-2017 https://members.pcisig.com/wg/PCIe-Protocol/document/10770	September 27, 2017
	“-c” version: Changes to match -b version of the Final NCB PDF approved by PWG and EWG on September 29, 2017. See change bars. Details include: EWG Changes: -Typo in Equation 8-3; changed 1.6.0 GT/s to 16.0 GT/s - § Section 8.4.2.1 ; corrected references from Table 8-11 to Table 8-10 - § Section 8.5.1.3.3 & § Section 8.5.1.4.3 (Figure 8-47); changed “median” to “mean” PWG Changes: -Sub-Sub-Bullet before Figure 4-27. Added “or higher” after 8.0 GT/s - § Section 5.12 Power Management Events; deleted last two paragraphs and Implementation Note. -Updated Acknowledgements section with additional contacts.	September 29, 2017
5.0	Version 0.3 Summary of intended changes for 5.0. This was a short document, referencing the PCI Express Base Specification but not including it.	2017-06-01
	Version 0.5 Further details on intended changes for 5.0. This was a short document, referencing the PCI Express Base Specification but not including it.	2017-11-02
	Version 0.7	2018-06-07

Revision	Revision History	Date
	This was the first release of Base 5.0 based on the 4.0 Specification text. The 4.0 specification was converted into HTML format during this process. This conversion process was imperfect but does not impact the new 5.0 material.	
	Version 0.9 This includes: • Additional details regarding operating at 32.0 GT/s • Corrections to match published Base 4.0 • Redrawing of some figures • PCIe_Base_r4_0_Errata_2018-10-04a.pdf • ECN-Thermal-Reporting_2017May18.pdf • ECN-Link-Activation-07-Dec-2017.pdf	2018-10-18
	Version 1.0 This includes: • Corrections and clarification for support of the 32.0 GT/s operation • Editorial Changes: ◦ Rewrite misleading / confusing text ◦ Update terminology for consistency and accuracy ◦ Update grammar for readability ◦ Add many hotlinks / cross references • Implement all 4.0 Errata • Incorporate Expansion ROM Validation ECN Expansion ROM Validation ECN.pdf • Incorporate Enhanced PCIe Precision Time Measurement (ePTM) ECN ECN_ePTM_10_January_2019.pdf • Incorporate Root Complex Event Collector Bus Number Association ECN ECN_EventCollector_13Sept2018a.pdf • Incorporate PCIe Link Activation ECN ECN_Link_Activation_07_Dec_2017.pdf • Incorporate Advanced Capabilities for Conventional PCI ECN (updated for PCIe) ECN_Conventional_Adv_Caps_27Jul06.pdf • Incorporate Async Hot-Plug Updates ECN ECN_Async_Hot-Plug_Updates_2018-11-29.pdf • Incorporate ACS Enhanced Capability ECN ECN_ACS_25_Apr_2019_Clean.pdf • Incorporate the Subsystem ID and Subsystem Vendor ID Capability, from the PCI-to-PCI Bridge Architecture Specification, Revision 1.2 (updated for PCIe) ppb12.pdf	2019-05-16
6.0	Version 0.3 Initial Release of Base 6.0, Standalone Document	2019-10-04

Revision	Revision History	Date
	Version 0.5, Standalone Document • Add L0p • Add Shared Flow Control • Update Physical Layer / Logical Sublayer material • Add Deprecation items: ◦ MR-IOV ◦ Lightweight Notification (LN) • Update new TLP Header material • Update Electrical Layer material	2020-01-30
	Version 0.7, Integrated Document First version relative to Base 5.0 Specification text. • Incorporate Base 5.0 Errata Matching Errata document to be published. • Incorporate Approved Base 5.0 ECNs: ◦ ACS Enhanced Capability ◦ ATS Memory Attributes Shadow Functions ◦ CMA ◦ DOE ◦ PTM Byte Order Adaptation ◦ DMWr ◦ PASID for Untranslated Addr • Support Flit Mode changes in Chapter 2 ◦ 14 bit Tag support ◦ Flit Mode TLP Format changes, including translation rules • Support Shared Flow Control, L0p, NULL2 DLLP in Chapter 3 • Integrate Flit Mode material into Chapter 4 • Flit Mode changes to Error Handling in Chapter 6 ◦ TLP Translation Blocked error • Integrate PAM4 Electrical changes into Chapter 8 • Refactor SR-IOV Registers from Chapter 9 into Chapter 7 Moves “VFs do things this way” material next to the original. • Define Shared Flow Control Supported and Shared Flow Control Enable bits in Chapter 7 (removed in Version 0.9) ◦ <u>Virtual Channel Extended Capability</u> ◦ <u>Multi-Function Virtual Channel Extended Capability</u> • Define new capabilities associated with 64.0 GT/s and Flit Mode in Chapter 7 ◦ <u>Physical Layer 64.0 GT/s Extended Capability</u> ◦ <u>Flit Logging Extended Capability</u> ◦ Device 3 Extended Capability Structure	2020-10-08

Revision	Revision History	Date
	◦ <u>Flit Performance Measurement Extended Capability</u> ◦ <u>Flit Error Injection Extended Capability</u> • Deprecate material: ◦ MR-IOV ◦ Lightweight Notification (LN)	
	Version 0.71, Integrated Document • Incorporate Combined Power ECN • Incorporate IDE ECN, plus Partial Header Encryption • Incorporate Errata: B90a, B90b, B90c, B90d, B90g, B90h, B90j, B90k, B90l, B90m, B90n, B90o, B90w, B90x, B90y, B90z, B113, B114 • L0p updates • Shared Flow control Updates • PAM4, 64.0 GT/s electrical updates • Physical Layer, Logical Sub-block updates ◦ Rework Flit Ack/Nak protocol ◦ Training Set changes • Max Payload Supported changes • TLP Layout changes • 14 bit tag updates • SR-IOV updates • Data Link Feature DLLP updates • DRS is <i>MUST@FLIT</i>, explain how software should use • Specify additional error behavior	2021-06-24
	Version 0.9, Integrated Document • Implement Errata B23, B64, B66-67, B74-75, B77, B78, B80-81, B85-88, B89a, B90f, B93-B95, B98-101, B103-105, B107-111, B112, B115-118, B120-121, B123-125, B127-128, B132, B133a-B133c, B133e • Simplified Shared Flow Control (always on in Flit Mode) and improvements to text • Numerous Phy Logical updates and issue fixes • Updated and Improved Flit Mode TLP type earmarking • Updated and Improved OHC content • Improve Segment-related content • Improve Max Payload Size content • <i>MUST@FLIT</i> changes, especially for Completion Timeout mechanism • Numerous editorial improvements	2021-09-23
	Version 1.0, Published Document • Incorporate Relaxed Detect Timing ECN	2021-12-16

Revision	Revision History	Date
	- Numerous editorial improvements - Update errata B117 - Add missing artwork - Fix and add cross references - Electrical section clarifications - Add Remote L0p Supported bit in Device Status 3 - Define Null Capability and Null Extended Capability - Update Reference Documents - Update encoding of Flit Mode Local TLP Prefix (was inconsistent in 0.9)	

Critical Errata

Revision	Errata	Description
6.0.1	A20	Partial Header Encryption Corrections: - Define Partial Header Encryption Mode field for Link IDE - Correct Partial Header Encryption Algorithm to correct functional issues and to use Byte level granularity instead of bit granularity as required in AES-GCM. - Update § Figure L-1 to include Attr[2] (a.k.a. A2 or ID0). - Add § Section L.2 .
6.0.1	A27	Clarify Flit Mode Receiver parsing requirements regarding of the Length field.
6.0.1	A35	Shared Flow Control Corrections: - Split Table 3-2 into § Table 3-2 and § Table 3-3 - Correct encoding of Symbol +0 bit 3 in § Figure 3-9, § Figure 3-10, and § Figure 3-11 to match § Table 3-3 - Remove obsolete “Shared Flow Control Flit_Marker” text from the Recommended Priority of Scheduled Transmissions Implementation Note.
6.0.1	A35a	Shared Flow Control Usage Limit is <u>RsvdP</u> for Non-Flit Mode components.
6.0.1	A37	Corrections to the Flit Sequence Number and Retry Mechanism (see § Section 4.2.3.4.2.1)
6.0.1	A38	L0p Corrections: - Preserve LFSR relationship between Lanes on width increase - Update scrambler behavior during L0p width increase - Update Link Management DLLP Behavior - Update LFSR advancement rules for L0p
6.0.1	A53	Define the <u>Flit Error Counter Interrupt Enable</u> bit in the <u>Flit Error Counter Control Register</u> .

Revision	Errata	Description
6.0.1	A62	Clarify OHC rules.
6.0.1	A67	Training Set Corrections: Clarify equalization fields in <u>TS0 Ordered Sets</u> (see <u>§ Table 4-40</u>).
6.0.1	A73	Modified TS1/TS2 corrections
6.1	A392	Clarify End-to-End TLP Prefix Support indications to software.
6.1	A393	Corrections to Flow Control rules - To clarify that when [Merged] is used, distinct UpdateFC DLLPs are sent for Posted and Completion credits. This provides buffer consumption visibility to the transmitter allowing a vendor-specific Quality of Service (QoS) mechanism to be implemented. - Clarify definition of <u>FC Unit Size</u> - Require consistency across VCs for shared credits – for example, if one VC advertises [Infinite.3], all VCs must also advertise it - Update minimum flow control credit rules. - Clarify scale factor rules when [Zero] or [Merged] are used. - Update <u>§ Equation 2-8</u> - Add <u>§ Equation 2-9</u> - Update <u>§ Equation 2-10</u> - Add <u>§ Equation 2-11</u> - Update <u>§ Equation 2-16</u> - Add <u>§ Equation 2-17</u> and <u>§ Equation 2-18</u>
6.2	A531	Correct <u>§ Figure 6-64</u> “IDE TLP Prefix (NFM)” and clarify text.
6.2	A540	Corrections to Equalization Phase 1 rules.
6.2	A544	Corrections to Flit Ack and Nak rules.
6.2	A577	Define EIE pattern for Retimers for 1b/1b encoding.
6.2	A578	Required modifications for Retimer timeouts for 64.0 GT/s.
6.2	A587	Correct inconsistencies between decimal and binary values listed in <u>§ Table 2-5</u> “Flit Mode TLP Header Type Encodings”.
6.3	A656	Sending NOP Flits with explicit sequence number set to NEXT_TX_FLIT_SEQ_NUM – 1 during Replay could cause issues at the far end receiver, especially if next Flit has an implicit sequence number.
6.3	A687	IMPLICIT_RX_FLIT_SEQ_NUM Rules must ignore invalid Flits.

Important Errata and Approved ECNs

Revision	Errata / ECN	Description
6.0.1	B5	Clarify value of the <u>Alternate Protocol Count</u> field in the <u>Alternate Protocol Capabilities Register</u> .
6.0.1	B7	Incorrect capability name used in the description of the Capability ID field of <u>Lane Margining at the Receiver Extended Capability Header</u> .
6.0.1	B8	Clarify that behavior is undefined when <u>Message Data Register</u> for MSI low bits conflict with the MME setting.
6.0.1	B10	A single PCI Express Function is permitted to contain multiple VSEC structures.
6.0.1	B11	Clarify Retimer Loopback behavior.
6.0.1	B12	Recommend that parity errors in Control SKP OS be checked only when immediately preceded by a Data Block.
6.0.1	B13	Clarify <u>Alternate Protocol Negotiation Status</u> error values.
6.0.1	B14	Define what it means for Scaled Flow Control to be “activated”.
6.0.1	B15	<u>Margining Ready</u> applies to speeds above 16.0 GT/s.
6.0.1	B16	Update Link Number encoding in § Table 4-39.
6.0.1	B17	Add Implementation Note: <u>Implementation Note: Delays in Data Link Layer Link Active Reflecting Link Control Operations</u>
6.0.1	B18	Clarify Receiver Impedance Propagation Rules.
6.0.1	B21	Update recommended behavior on transitions from Hot Reset to <u>Detect.Active</u> .
6.0.1	B22	Clarify Flit Mode poison rules.
6.0.1	B23	- Shrink figures § Figure 3-9 and § Figure 3-10 so they're not cut off. - Redraw § Figure 4-27 so it's readable. Figure 1 Old Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Transmit side §

Revision	Errata / ECN	Description
		Figure 2 New Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Transmit side § Redraw § Figure 4-28 so it's readable. Figure 3 Old Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Receive side §

Revision	Errata / ECN	Description
		<i>Figure 4 New Figure: Flit Mode and Non-Flit Mode processing with 8b/10b and 128b/130b encoding on the Receive side</i>
6.0.1	B24	Lane Margining at the Receiver Corrections
6.0.1	B25	Define FLIT in Terms and Acronyms
6.0.1	B26	Completion Timeout Ranges Supported, Completion Timeout Disable Supported, and Completion Timeout Value only apply to Non-Posted Requests
6.0.1	B28	Requester must accept any Segment number in completions.
6.0.1	B29	TC field must be 000b in § Figure 2-72, § Figure 2-74, § Figure 2-76, and § Figure 2-78. Type field was incorrect in § Figure 2-74 and § Figure 2-78.
6.0.1	B30	Routing field was incorrect in § Table 2-36 IDE Messages.
6.0.1	B31	Clarify origins of T_{rst} value of 1 ms.
6.0.1	B34	§ Table 4-36 TS1 Ordered Set in 8b/10b and 128b/130b Encoding: Symbol 4, Bit 7 had the wrong LTSSM state for the <u>speed_change</u> usage.
6.0.1	B40	Update the exponents of α in § Figure 4-56. α^{85} and α^{84} changed to α^{84} and α^{83} respectively. Update exponents of α in Column B ⁰ in § Figure 4-57. α^{85} through α^{92} changed to α^{84} through α^{91} respectively.

Revision	Errata / ECN	Description																																																																																																																														
		<table><tr><td></td><td>B₀</td><td>...</td><td>B₈₂</td><td>B₈₃</td><td>B₈₄</td><td>B₈₅</td></tr><tr><td>0</td><td>α⁸⁵</td><td>...</td><td>α²</td><td>α</td><td>1</td><td>p₀</td></tr><tr><td>1</td><td>α⁸⁶</td><td>...</td><td>α³</td><td>α²</td><td>α</td><td>p₁</td></tr><tr><td>2</td><td>α⁸⁷</td><td>...</td><td>α⁴</td><td>α³</td><td>α²</td><td>p₂</td></tr><tr><td>3</td><td>α⁸⁸</td><td>...</td><td>α⁵</td><td>α⁴</td><td>α³</td><td>p₃</td></tr><tr><td>4</td><td>α⁸⁹</td><td>...</td><td>α⁶</td><td>α⁵</td><td>α⁴</td><td>p₄</td></tr><tr><td>5</td><td>α⁹⁰</td><td>...</td><td>α⁷</td><td>α⁶</td><td>α⁵</td><td>p₅</td></tr><tr><td>6</td><td>α⁹¹</td><td>...</td><td>α⁸</td><td>α⁷</td><td>α⁶</td><td>p₆</td></tr><tr><td>7</td><td>α⁹²</td><td>...</td><td>α⁹</td><td>α⁸</td><td>α⁷</td><td>p₇</td></tr></table> Check Bits Row Parity [N-3:0] Figure 5 Old Figure: Powers of alpha for the check bits for Bytes 0 to 84 § <table><tr><td></td><td>B₀</td><td>...</td><td>B₈₂</td><td>B₈₃</td><td>B₈₄</td><td>B₈₅</td></tr><tr><td>0</td><td>α⁸⁴</td><td>...</td><td>α²</td><td>α</td><td>1</td><td>p₀</td></tr><tr><td>1</td><td>α⁸⁵</td><td>...</td><td>α³</td><td>α²</td><td>α</td><td>p₁</td></tr><tr><td>2</td><td>α⁸⁶</td><td>...</td><td>α⁴</td><td>α³</td><td>α²</td><td>p₂</td></tr><tr><td>3</td><td>α⁸⁷</td><td>...</td><td>α⁵</td><td>α⁴</td><td>α³</td><td>p₃</td></tr><tr><td>4</td><td>α⁸⁸</td><td>...</td><td>α⁶</td><td>α⁵</td><td>α⁴</td><td>p₄</td></tr><tr><td>5</td><td>α⁸⁹</td><td>...</td><td>α⁷</td><td>α⁶</td><td>α⁵</td><td>p₅</td></tr><tr><td>6</td><td>α⁹⁰</td><td>...</td><td>α⁸</td><td>α⁷</td><td>α⁶</td><td>p₆</td></tr><tr><td>7</td><td>α⁹¹</td><td>...</td><td>α⁹</td><td>α⁸</td><td>α⁷</td><td>p₇</td></tr></table> Check Bits Row Parity [N-3:0] Figure 6 New Figure: Powers of alpha for the check bits for Bytes 0 to 84 §		B ₀	...	B ₈₂	B ₈₃	B ₈₄	B ₈₅	0	α ⁸⁵	...	α ²	α	1	p ₀	1	α ⁸⁶	...	α ³	α ²	α	p ₁	2	α ⁸⁷	...	α ⁴	α ³	α ²	p ₂	3	α ⁸⁸	...	α ⁵	α ⁴	α ³	p ₃	4	α ⁸⁹	...	α ⁶	α ⁵	α ⁴	p ₄	5	α ⁹⁰	...	α ⁷	α ⁶	α ⁵	p ₅	6	α ⁹¹	...	α ⁸	α ⁷	α ⁶	p ₆	7	α ⁹²	...	α ⁹	α ⁸	α ⁷	p ₇		B ₀	...	B ₈₂	B ₈₃	B ₈₄	B ₈₅	0	α ⁸⁴	...	α ²	α	1	p ₀	1	α ⁸⁵	...	α ³	α ²	α	p ₁	2	α ⁸⁶	...	α ⁴	α ³	α ²	p ₂	3	α ⁸⁷	...	α ⁵	α ⁴	α ³	p ₃	4	α ⁸⁸	...	α ⁶	α ⁵	α ⁴	p ₄	5	α ⁸⁹	...	α ⁷	α ⁶	α ⁵	p ₅	6	α ⁹⁰	...	α ⁸	α ⁷	α ⁶	p ₆	7	α ⁹¹	...	α ⁹	α ⁸	α ⁷	p ₇
	B ₀	...	B ₈₂	B ₈₃	B ₈₄	B ₈₅																																																																																																																										
0	α ⁸⁵	...	α ²	α	1	p ₀																																																																																																																										
1	α ⁸⁶	...	α ³	α ²	α	p ₁																																																																																																																										
2	α ⁸⁷	...	α ⁴	α ³	α ²	p ₂																																																																																																																										
3	α ⁸⁸	...	α ⁵	α ⁴	α ³	p ₃																																																																																																																										
4	α ⁸⁹	...	α ⁶	α ⁵	α ⁴	p ₄																																																																																																																										
5	α ⁹⁰	...	α ⁷	α ⁶	α ⁵	p ₅																																																																																																																										
6	α ⁹¹	...	α ⁸	α ⁷	α ⁶	p ₆																																																																																																																										
7	α ⁹²	...	α ⁹	α ⁸	α ⁷	p ₇																																																																																																																										
	B ₀	...	B ₈₂	B ₈₃	B ₈₄	B ₈₅																																																																																																																										
0	α ⁸⁴	...	α ²	α	1	p ₀																																																																																																																										
1	α ⁸⁵	...	α ³	α ²	α	p ₁																																																																																																																										
2	α ⁸⁶	...	α ⁴	α ³	α ²	p ₂																																																																																																																										
3	α ⁸⁷	...	α ⁵	α ⁴	α ³	p ₃																																																																																																																										
4	α ⁸⁸	...	α ⁶	α ⁵	α ⁴	p ₄																																																																																																																										
5	α ⁸⁹	...	α ⁷	α ⁶	α ⁵	p ₅																																																																																																																										
6	α ⁹⁰	...	α ⁸	α ⁷	α ⁶	p ₆																																																																																																																										
7	α ⁹¹	...	α ⁹	α ⁸	α ⁷	p ₇																																																																																																																										
6.0.1	B44	Extended Synch is not used for L0s in Flit Mode, but is still relevant for other purposes.																																																																																																																														
6.0.1	B45	Add 64.0 GT/s encoding to Enable SKP OS Generation Vector.																																																																																																																														
6.0.1	B48	Clarify 1b/1b encoding <u>EIOSQ</u> terminology in Compliance and Modified Compliance Patterns.																																																																																																																														
6.0.1	B49	<u>Execute Requested</u> is reserved in all Memory Requests other than Untranslated Memory Read Requests.																																																																																																																														
6.0.1	B50	Typo in units column of § Table 8-16, Parameter <u>T_{TX-CH-UPW-RJ-64G}</u>																																																																																																																														

Revision	Errata / ECN	Description
6.0.1	B51	- Typo in units column of § Table 8-18, Parameter <u>Z_{C-DC}</u> - Locate artwork for § Figure 8-86
6.0.1	B54	Don't send Selective IDE Stream IDE Sync and IDE Fail messages unless the V bit is Set for that <u>RID Base</u> .
6.0.1	B55	The “subsequent IDE TLPs” error reporting language applies regardless of how Insecure was reached.
6.0.1	B56	<u>Selective IDE Stream Enable</u> and <u>Link IDE Stream Enable</u> are edge triggered.
6.0.1	B58	The <u>Selective IDE Stream Control Register</u> , <u>Default Stream</u> field only applied to Selective IDE Streams.
6.0.1	B59	Routed to Root Complex is permitted only when the Partner Port is a Root Port, as indicated by the Default Stream bit being Set.
6.0.1	B60	Clarify rules for selecting the Selective IDE Stream for a given TLP.
6.0.1	B64	Clarify Segment number processing rules in Flit Mode Completer Transaction ID processing.
6.0.1	B66	The <i>Data Link Feature Exchange Enable</i> field was renamed to <u>Data Link Feature Exchange Is Enabled</u>. This errata also applies to [PCIE-5.0].
6.0.1	B68	Correct Type field value in § Figure 10-13 (was 0000 0000b should be 0000 0011b).
6.0.1	B69	In Shared Flow Control Usage Limit Enable, clarify wording to indicate which Shared Flow Control Usage Limit is affected (duplicate field names in <u>VC Resource Control Register</u> and <u>MFVC VC Resource Control Register</u>).
6.0.1	B70	DC Balance correction in 1b/1b encoding of <u>TS0</u> , <u>TS1</u> , and <u>TS2</u> .
6.0.1	B71	Consolidate description of 1b/1b behavior for <u>SKP Ordered Sets</u> .
6.0.1	B72	TS2 Symbol 6, operating at 2.5 or 5.0 GT/s and Requesting equalization 32.0 GT/s encoding is only for 32.0 GT/s (was “32.0 GT/s or higher ”)
6.0.1	B75	Clarify behavior in CMA-SPDM Rules.
6.0.1	B77	Update artwork for § Figure 6-79.
6.0.1	B79	In Flit Mode, the 16.0 GT/s Local Data Parity Mismatch Status Register, 16.0 GT/s First Retimer Data Parity Mismatch Status Register, and 16.0 GT/s Second Retimer Data Parity Mismatch Status Register are used at all 128b/130b and 1b/1b data rates.
6.0.1	B81	Clarify 128b/130b Control and Standard SKP Alternation Rule interactions with <u>L0p</u>
6.0.1	B82	Editorial: <u>SRIS_Mode_Enabled</u> is a variable, not a bit.
6.0.1	B83	Define CXL Src bit in ATS Translation Requests. This is bit 3 in byte 15 in § Figure 10-10, § Figure 10-11, § Figure 10-12, and § Figure 10-13. Editorial change in § Table 10-3.
6.0.1	B85	Loopback Follower changes
6.0.1	B86	§ Table 4-39 Equalization Byte 0 correction (Symbols 1,9)

Revision	Errata / ECN	Description
		• § Table 4-37 Symbol 6 8.0 GT/s should be 8.0 GT/s or higher • § Section 4.2.7.4.4, Width change rule change, change “128b/138b” to “128b/130b or 1b/1b” (3 places)
6.0.1	B87	Update <u>Recovery.RcvrLock</u> behavior when hardware autonomous equalization is not adopted
6.0.1	B88	Define 1b/1b Phy Payload type Alternation Rule
6.0.1	B89	• Change name of <u>L0p Supported</u> capability bit (was Receiver L0p Supported) • Clarify <u>Target Link Width</u> behavior • Clarify <u>L0p.Priority</u> behavior
6.0.1	B90	Clarify Receiver Framing Requirements (including <u>L0p</u> in Flit Mode)
6.0.1	B92	Clarify that the LTSSM must not advertise speeds above 32.0 GT/s in <u>Polling.Active</u> and <u>Polling.Configuration</u> . During those states, negotiation for greater than 32.0 GT/s is occurring.
6.0.1	B93	Add implementation note “Reject Coefficient Values with TS0 Ordered Sets” in § Section 4.2.7.4.2.2.1 .
6.0.1	B94	Attr[2] was incorrectly shown as Reserved in § Figure 10-6, § Figure 10-7, § Figure 10-10, and § Figure 10-11.
6.0.1	B95	Clarify historical timestamp invalidation behavior in § Section 6.21.3.2 .
6.0.1	B96	Correct the register width of the <u>Root Capabilities Register</u> , § Table 7-31. Was 32 bits, but the register is actually 16 bits wide.
6.0.1	B98	Update § Figure 7-229 and § Table 7-208 to better reflect that the Virtual Channel Extended Capability uses Capability ID 0002h or 0009h.
6.0.1	B99	Clarify that intermediate receivers should not check crossing 4 KB boundaries. Implementing this check could invalidate such silicon changes for future TLP Type definitions (e.g., such future TLP Types could be similar to the existing Atomic CAS where the relationship between the TLP Length field and the affected memory range is not 1:1).
6.0.1	B100	Clarify that <u>L0p</u> is supported at all data rates.
6.0.1	B102	Segment was erroneously left out of IDE RID Association mechanism
6.0.1	B103	Clarify IDE TLP Aggregation rules. See § Section 6.33.6 .
6.0.1	B104	Add implementation note <u>SKP Ordered Sets in a Data Stream in Flit Mode</u> to § Section 4.2.8.5 .
6.0.1	B106	Clarify the FM/NFM rules for PCRC.
6.0.1	B107	Clarify that IDE TLPs in FM always have OHC-C.
6.0.1	B108	CMA-SPDM state should not be affected by FLR.
6.0.1	B111	Clarify Retimer response for Access Retimer Register command.
6.0.1	B112	Add implementation note <u>Security Issues Associated with Non-Enabled Bytes</u> .
6.0.1	B113	The following figures incorrectly described reserved bits as RsvdP that should be <u>RsvdZ</u> . some of these were also incorrect in [PCIe-5.0] but were correct in [PCIe-4.0].

Revision	Errata / ECN	Description
		• § Figure 7-32 – Slot Status Register • § Figure 7-124 – Device Status 3 Register • § Figure 7-382 – Link IDE Stream Status Register • § Figure 7-385 – Selective IDE Stream Status Register
6.0.1	B114	Add new § Section 4.2.18.3 describing Lane Margining contents for Flit Mode Control SKP Ordered Sets at 8.0 GT/s.
6.0.1	B115	Clarify the Variant bit rules in § Section 2.7.1 .
6.0.1	B116	OHC-A1 is also used for Route to Root Complex Messages (e.g., Page Request Messages) and for Translation Requests that Set NW
6.0.1	B117	Clarify that "overflow" for the PR Received Counters (NPR/CPL) means the 64b counter itself overflows.
6.0.1	B118	Clarify that, with system support for DRS, Devices are permitted to take longer than 1 second to become Configuration-Ready.
6.0.1	B119	Clarify text in Segment Captured bit description.
6.0.1	B120	When the length of an ATS Request is improper, the result should be undefined, not a Malformed TLP.
6.0.1	B121	Clarify TLP Prefix Processing rules for Flit Mode.
6.0.1	B122	Clarify completion rules for TLPs with Reserved Type field values.
6.0.1	B123	Deprecate Flit Mode TLP Suffixes. They were unused and have latency concerns.
6.0.1	B124	Clarify that Equalization Bypass to Highest NRZ Rate Disable is optional.
6.0.1	B127	Clarify L0p Framing Error rule. Clarify L0p Retimer behavior.
6.0.1	B128	Update the Compliance Pattern for 1b/1b operation.
6.0.1	B130	Clarify Loopback.Active language.
6.0.1	B131	• Deprecate Report Longest Burst vs First Burst • Flit Error Counter does not roll over • Introductory text in § Section 7.7.8.7 describes all FBER Measurement Status Registers (i.e., FBER Measurement Status 1 Register through FBER Measurement Status 10 Register) • Remove Pseudo Port wording in Lane #0 Correctable Counter (also applies to the rest of the Lane counters).
6.0.1	B132	Correct Tag field usage in Messages and Vendor-Defined Messages. • Message byte 6 is Reserved unless specifically defined. • Reserved bits must be copied intact during translation to/from Flit Mode and Non-Flit Mode. • Change wording for byte 6 of Non-Flit Mode Messages § Figure 2-49.

Revision	Errata / ECN	Description
		- Change wording for byte 6 of Flit Mode Messages § Figure 2-50. - Low 7 bits of byte 6 are available for use in Vendor-Defined Messages. - Change wording for byte 6 of Non-Flit Mode Vendor-Defined Messages § Figure 2-53 - Change wording for byte 6 of Flit Mode Vendor-Defined Messages § Figure 2-54 This errata also applies to [PCIE-5.0].
6.1	B305	Clarify flit error counting Behavior Events to Count field of the Flit Error Counter Control Register § Section 4.2.3.4.2 § Section 4.2.5.1
6.1	B306	Electrical Idle exit timing changes
6.1	B308	Clarify implementation note Detection of Improper Reordering in IDE TLP Sub-Streams (§ Section 6.33.5)
6.1	B309	Support 64.0 GT/s in Loopback.Entry and Detect.Quiet
6.1	B310	Clarify Recovery.RcvrLock , Recovery.RcvrCfg , Recovery.Speed , and Recovery.Idle rules
6.1	B311	Text was misplaced in Recovery.RcvrCfg
6.1	B312	L0p rules
6.1	B314	Lane Numbering matching rules in 1b/1b
6.1	B315	Centralize text describing Default Lane numbers
6.1	B316	IDE Terminus rules for Switch / Root Port
6.1	B317	Add implementation note Determination of Slot Number Information
6.1	B319	- Update Symbol Time definition for 1b/1b encoding - Update Retimer Latency Limit values for SRIS Table 4-79
6.1	B320	Provide Test Keys for testing IDE in § Section L.5
6.1	B322	Recovery.RcvrLock to Recovery.Equalization transition
6.1	B323	SRIS Clocking bit is also meaningful in Flit Mode with 8b/10b and 128b/130b encoding
6.1	B325	Recovery.Idle to Loopback
6.1	B326	Configuration.Linkwidth.Start rules for 64.0 GT/s
6.1	B328	Clarify Requester Segment field behavior in OHC-C
6.1	B329a	Configuration Requests cannot map to the Default IDE Stream
6.1	B330	Clarify definition of Flit halves
6.1	B331	Corrections to EP and Length fields of the following messages:

Revision	Errata / ECN	Description
		• § Figure 10-22 • § Figure 10-24 • § Figure 10-26 • § Figure 10-28 • § Figure 2-72 • § Figure 2-74 • § Figure 2-76 • § Figure 2-78
6.1	B332	Function Level Reset clarification
6.1	B335	<u>Recovery.RcvrLock</u> transmitter equalization coefficients
6.1	B336	<u>EIEOS</u> inference behavior only occurs when the LTSSM rules already allow entry to <u>Recovery</u>
6.1	B338	Define terms <u>Endpoint Upstream Port</u> and <u>Switch Port</u> .
6.1	B340	Clarify rules for where a <u>PASID Extended Capability</u> is permitted and define associated behavior.
6.1	B341	Errata against TDISP ECN-Link-Activation-07-Dec-2017 Poisoned TLP rules in TDISP
6.1	B342	Returning TC0/VC0 to secure state in Link IDE
6.1	B344	TEE rules when not using TDISP
6.1	B345	Remove confusing and unnecessary text regarding CMA-SPDM
6.1	B346	Remove confusing and unnecessary text regarding CMA-SPDM
6.1	B347	Update § Figure 6-63 to recommend setting IDE Enable last when programming keys. Add text to strongly recommend this behavior while permitting otherwise.
6.1	B348	Errata against TDISP ECN DOE errors don't usually impact TDISP
6.1	B349	ERRATA against TDISP ECN Add examples applicable IDE errors.
6.1	B353	Translation Requests always contain <u>OHC-A1</u> as they always contain the NW bit.
6.1	B354	Correct naming of the Received IDE Fail Message in <u>Link IDE Stream Status Register</u> and <u>Selective IDE Stream Status Register</u>
6.1	B355	Clarify requirements on system software behavior during IDE enablement.
6.1	B356	Update precoding behavior
6.1	B357	Clarify Alternate Protocol Negotiation behavior
6.1	B358	Add <u>Implementation Note: Handling TLPs Spanning Multiple Flits</u>

Revision	Errata / ECN	Description
6.1	B359	Clarify wording for Gray coding of DC balance symbol
6.1	B360	Clarify K_SET_GO behavior
6.1	B361	Errata against TDISP ECN Reword § Section 11.2.1 to reflect T-Bit behavior required by TDISP
6.1	B363	<u>Flit Mode Status</u> has different behavior for Upstream and Downstream Ports
6.1	B365	Clarify precoding text for 64.0 GT/s
6.1	B367	Clarify All VCs Enabled wording in the <u>Port VC Control Register</u> and <u>MFVC Port VC Control Register</u>
6.1	B369	Add rules in § Section 6.33.8
6.1	B370	Clarify Latency Tolerance Reporting behavior when entering and exiting <u>D0</u>
6.1	B373	Clarify Retimer rules for Hot Reset
6.1	B374	Clarify rules for <u>Loopback.Entry</u>
6.1	B375	Clarify rules for <u>Loopback.Entry</u>
6.1	B376	Clarify wording in Phase 0 and Phase 1 of Transmitter Equalization
6.1	B377	Clarify rules for <u>Loopback.Entry</u>
6.1	B378	Clarify Ordered Set Error Injection behavior
6.1	B381	Clarify Local TLP Prefix rules in Flit Mode TLP Grammar
6.1	B382	Correct uniqueness rules for Transaction IDs
6.1	B384	<u>Recovery.Speed</u> corrections
6.1	B385	Correct Flit Ack, Nak, and Discard rules
6.1	B386	Add 64.0 GT/s to <u>Recovery.RcvrLock</u> rule
6.1	B388	Framing Error in the context of § Section 4.2.5.8 is only for Non-Flit Mode
6.1	B394	Update Configuration Request Routing Rules for IDE
6.1	B395	K_SET_GO or K_SET_STOP behavior with an invalid Stream_ID
6.1	B397	<u>L0p</u> behavior when an <u>EIEOSQ</u> is scheduled near an <u>SDS</u> Ordered Set
6.1	B398	Flit Sequence Number and Retry Mechanism
6.1	B405	<u>VF Enable</u> behavior
6.1	B406	Clarify rules for Local TLP Prefixes and Local TLPs
6.1	B407	Clarify <u>L0</u> behavior when receiving <u>EIOS</u> behavior (interaction with <u>L0p</u>)
6.1	B408	Flit Sequence Number rules
6.1	B409	In § Section 4.2.7.4.4, the LTSSM must not initiate a width change if <u>speed_change</u> is set to 1b
6.1	B412	Errata to TDISP ECN

Revision	Errata / ECN	Description
		FLR to Functions other than Function 0 do not affect SPDM or IDE streams.
6.1	B414	Errata to UIO ECN TC to VC Mapping rules
6.1	B415	Behavior for L0s in Flit Mode
6.1	B416	Clarify wording around Hot and Warm Reset
6.1	B418	Recommend software recovery procedure for use when the page request queue overflows This Errata is part 1 of 10 that constitute ATS 1.2
6.1	B419	U-bit will eventually be deprecated This Errata is part 2 of 10 that constitute ATS 1.2
6.1	B420	Address space overlap clarification This Errata is part 3 of 10 that constitute ATS 1.2
6.1	B421	ITag uniqueness rules This Errata is part 4 of 10 that constitute ATS 1.2
6.1	B422	ATS and PRI interactions with Shadow Functions This Errata is part 5 of 10 that constitute ATS 1.2
6.1	B423	ATS Translation Completion Status updates This Errata is part 6 of 10 that constitute ATS 1.2
6.1	B425	<u>Invalidate Queue Depth</u> , <u>Page Aligned Request</u> , and <u>Global Invalidate Supported</u> updates This Errata is part 7 of 10 that constitute ATS 1.2
6.1	B426	<u>Execute Requested</u> update This Errata is part 8 of 10 that constitute ATS 1.2
6.1	B427	<u>Execute Permission Supported</u> update This Errata is part 9 of 10 that constitute ATS 1.2
6.1	B428	ATS Enable update This Errata is part 10 of 10 that constitute ATS 1.2
6.1	B429	Scaled Flow Control updates
6.1	B430	Data Link Feature is required in Flit Mode
6.1	B431	Errata against UIO ECN:

Revision	Errata / ECN	Description
		Remove the sentence “Because UIO can only be enabled when end-to-end UIO support exists, it is never necessary for the Root Complex to take ownership of UIO Requests.” from the Implementation Note <u>Root Complex Support for peer-to-peer Non-Posted Memory Transactions that Traverse Hierarchies</u> .
6.1	B432	Mandate capacitor placement on the Transmit end of a Link.
6.1	B433	Clarify valid tag values for 14-bit tags.
6.1	B434	Clarify rules for Switch FM to NFM translation.
6.1	B435	Address Routed Message corrections in Flit Mode.
6.1	B436	Clarification of Tag keep-out ranges.
6.2	B500	Clarify rules for PASID application.
6.2	B501	Clarify rules for Address Routed Messages.
6.2	B502	Clarify <u>OHC-A5</u> requirements for Selective IDE Completions.
6.2	B503	Remove needless indirection regarding requirement that VFs must not implement PASID Extended Capability.
6.2	B506	Clarify rules for TDI regarding MSI/MSI-X.
6.2	B507	Errata B507: Remove Implementation Note “FUTURE TEE EXTENSIONS”: TEE extensions for sub-function I/O-virtualization techniques will be covered in a future revision of this specification.
6.2	B508	Remove unnecessary restriction on VC1 in <u>SVC</u> .
6.2	B510	Correct typo in <u>Recovery.RcvrCfg</u> .
6.2	B512	Clarify application of IDE Default Stream to Translation Requests and Untranslated Memory Requests.
6.2	B513	Clarify application of ordering rules for Flow-Through Selective IDE.
6.2	B515	Clarify Flit Performance Measurement mechanisms.
6.2	B515a	Clarify Segment rules for Flit Mode Requesters/Completers within an RC.
6.2	B517	Correct § <u>Figure 6-46</u> “CMA-SPDM as Part of a Layered Architecture”.
6.2	B518	Correct use of “Precision Time Measurement”.
6.2	B520	Clarify Phase 1 of Transmitter Equalization text.
6.2	B522	Correct paragraph alignment.
6.2	B524	Clarify rules for <u>EQ TS2</u> Ordered Set use.
6.2	B525	Add <u>L0p</u> register cross-references.
6.2	B529	Clarify earmarked TLP handling.
6.2	B535	FLIT_REPLAY_NUM clarifications.
6.2	B536	Clarify Segment Base programming for certain scenarios.

Revision	Errata / ECN	Description
6.2	B537	Clarify rules for Selective IDE for Configuration Requests.
6.2	B538	Clarify Privileged Mode Requested and Execute Requested/Execute Permitted rules corrupted during Revision 6.0 development.
6.2	B541	Clarify <u>L0p</u> request handling rules.
6.2	B546	Clarify figures and text regarding TLP Header Byte 6 for Message Requests. The following NFM Message Request Header figures in § Section 2.2.8 have byte 6 labeled as “Tag”. Change the label to “R” or “Reserved” (based on what fits). Note: Figures are automatically generated and these changes are not marked in the document. • § Figure 2-51 • § Figure 2-55 • § Figure 2-63 • § Figure 2-65 • § Figure 2-67 • § Figure 2-68
6.2	B547	Clarify <u>AT</u> field requirements for Messages Routed by Address.
6.2	B548	Clarify <u>MMB Capabilities Register</u> language.
6.2	B549	Clarify handling of received DMWr Requests.
6.2	B550	Clarify and cross-reference existing rules regarding Segment exceptions for Root Ports.
6.2	B551	Clarify <u>SKP Ordered Set</u> rules with <u>L0p</u> .
6.2	B552	Correct omission of <u>TS0</u> in rules for Phase 1 of Transmitter Equalization.
6.2	B553	Clarify text regarding Electrical Idle Sequences on entry to <u>Recovery.RcvrLock</u> .
6.2	B554	Clarify Autonomous Change / Selectable De-emphasis for <u>TS1/TS2 Ordered Set</u> with 1b/1b Encoding.
6.2	B555	Clarify content of <u>TS2 Ordered Set</u> with 1b/1b Encoding.
6.2	B556	Clarify handling of <u>NOP2 DLLP</u> during equalization.
6.2	B557	Clarify Flit Replay Transmit rules.
6.2	B559	UIO clarifications.
6.2	B561	Clarify and cross-reference DMWr capability reporting.
6.2	B562	<u>OHC-E</u> clarifications.
6.2	B564	Clarify <u>L0p</u> Link width rules.
6.2	B565	Clarify rules for handling an invalid Flit Sequence Number in a received Ack Flit or Nak Flit.
6.2	B566	Clarify Retimer data rate change and determination rules.
6.2	B568	Correct Bad DLLP entry in § Table 6-4 “Data Link Layer Error List”.
6.2	B569	Clarify handling of ID Routed Messages targeting a Function that is not implemented.

Revision	Errata / ECN	Description
6.2	B571	Correct figures to reduce the Key Sub-Stream field width to 3 bits. This affects: • § Figure 6-57 • § Figure 6-58 • § Figure 6-59 • § Figure 6-60 • § Figure 6-61
6.2	B572	Clarify rules for implementing <u>PTM Extended Capability</u> .
6.2	B573	Clarify Data Stream, and rules for choosing Flit Mode vs. Non-Flit Mode.
6.2	B575	Clarify <u>Half Scrambling</u>
6.2	B576	Add Implementation Note: Forwarding D21.3 Symbol with Incorrect Disparity.
6.2	B580	Clarify rules to detect an <u>SDS</u> .
6.2	B581	Add Implementation Note <u>Orthogonality of L0p and L1/L2</u>
6.2	B583	Clarify definition of UIO TLPs.
6.2	B583a	Clarify that Translation Requests are not defined for UIO Memory Reads.
6.2	B585	Correct section references.
6.2	B586	Clarify definition of Idle Flit, <u>Nop Flit</u> , and Payload Flit.
6.2	B588	DPC doesn't invent Completions for UIO Requests.
6.2	B595	Clarify Larger Tags rules.
6.2	B599	Clarify Precoding negotiaion.
6.3	B600	Effect of <u>Detect</u> on Precoding
6.3	B601	Population of TS1/TS2 Ordered Set fields during L0p
6.3	B602	Retimer changes to Enter Compliance Rules
6.3	B603	Correct Flit Layout figures to use Symbol Times instead of UI. With PAM4, UI was no longer accurate.
6.3	B604	Add Transmitter Preset rules for Phase 0 of Transmitter Equalization when sending TS0 Ordered Sets
6.3	B605	Change label for rightmost block in § Figure 4-22 from “PAM4 voltage Conversion to 2-bit aligned” to “2-bit aligned to PAM-4 voltage Conversion” Update fonts in § Figure 4-22 and § Figure 4-23
6.3	B606	Clarify TS0 rules for Phase 0 of Transmitter Equalization TS0s don't have the Reject Coefficient Values bit, so references to setting it need to be qualified by TS1 Ordered Sets, and the text needs to say what happens when transmitting TS0s.
6.3	B607	Clarify rules for determining behavior for coalesced UIO Completions with mixed Completions Status values

Revision	Errata / ECN	Description
6.3	B608	Correction of Electrical Idle time rules in <u>Polling.Compliance</u>
6.3	B616	Remove leftover text referring to Receiver Preset Hints (that are only present at 8.0 GT/s). Update text to use correct field names.
6.3	B617	Clarify SKP OS behavior in Rules for Transmitters
6.3	B618	Define Flit Mode behavior when TPH is used and Byte Enable values do not match the Non-Flit Mode implied values (see § Section 2.2.5.1).
6.3	B619	Clarify Alternate Protocol Negotiation when more than one alternate protocol is supported (i.e., PCIe plus two or more alternate protocols).
6.3	B621	Defined minimum Receiver rules for detecting Idle Flits.
6.3	B623	Clarify behavior when changing VF Enable. <u>Immediate Readiness</u> does not apply to VFs.
6.3	B624	Define the terms <u>RefClk</u> , <u>recommended</u> , and <u>strongly recommended</u>
6.3	B625	Clarify software rules for interpreting Flit Error Log registers.
6.3	B626	Add implementation note <u>Summary of L0p Transmitter/Receiver Behavior</u> .
6.3	B626a	Clarify <u>L0p Enable</u> rules. Note: This Erratum adds two sentences to the implementation note § Implementation Note: <u>Summary of L0p Transmitter/Receiver Behavior</u> located immediately before § Table 4-55. That change may be marked as part of Erratum B626.
6.3	B628	Cleanup terminology regarding Update Flow Control and <u>Optimized_Update_FC</u>
6.3	B630	Remove values for fields Fmt and Type in § Figure 10-6 and § Figure 10-7. MRd and MWrr are not the only types of Memory Requests.
6.3	B632	Flit Latency Tracking Counter unit changed to ns from µs. Incomplete integration error of earlier Erratum B515. This unfortunately creates a situation where a software workaround is needed for <u>Flit Latency Tracking Counter</u> use.
6.3	B633	Clarify rules for when N_FTS is meaningful in <u>TS1</u> , <u>TS2</u> , and <u>Modified TS1/TS2 Ordered Sets</u> .
6.3	B634	Relax definition of L0p Exit Latency encodings. Indicate that L0p Exit Latency is a hint (e.g., there is no compliance issue if latency is longer).
6.3	B635	PASID is permitted on all Memory Requests, including UIO and DMWrr.
6.3	B640	• Correct § Figure L-4 to match § Table L-2 • Correct § Figure L-8 to match § Table L-4 • Update both figures to reflect that Sub-Stream is now a 3 bit field.
6.3	B641	Increase Retimer Force Timeout value from 48 ms to 96 ms
6.3	B642	In 8b/10b and 128b/130b, the <u>Training Control</u> field is bit encoded (e.g., the Hot Reset bit). In 1b/1b, the <u>Training Control</u> field is encoded.

Revision	Errata / ECN	Description
		This erratum corrects wording elsewhere in the specification to reflect this difference (e.g., use Hot_Reset_Request which is indicated in two different ways based on data rate).
6.3	B643	Correct incorporation error in cross reference (there is no Section 6.99.4).
6.3	B647	Add VOL parameter to § Table 12-2 (PESTI DC Specifications)
6.3	B648	Add implementation note <u>Rationale for T2wrst</u> to § Section 12.5.4.2 .
6.3	B648a	In § Figure 12-15, update left end of the arrow for timing parameter $T_{i3c2smb}$.
6.3	B649	Update caption of § Table 10-2
6.3	B650	Clarify INTx/MSI/MSI-X interrupt wording.
6.3	B652	Clarify Precoding behavior in <u>Loopback.Active</u>
6.3	B653	Clarify wording for gray coding and precoding of Compliance Pattern in 1b/1b Encoding.
6.3	B655	Delete redundant bullet in § Section 4.2.6.7 .
6.3	B657	Define Flit Mode behavior when receiving TLPs with a Reserved Type encoding.
6.3	B658	IDE Fail and IDE Sync messages are not defined in the UIO VC
6.3	B659	Clarify Implementation Note <u>TS0 to TS1 Transitions</u> .
6.3	B661	NPEM registers are not affected by Function Level Reset
6.3	B662	Clarify text indicating that ACS Source Validation does not check Segment Numbers.
6.3	B665	PCI-PM is not required for VFs.
6.3	B666	Clarify behavior for reserved type encodings in § Table 2-5, Flit Mode TLP Header Type Encodings
6.3	B667	Clarify behavior when DOE Object Length is not as expected.
6.3	B669	Clarify definition of <u>data payload</u> .
6.3	B670	Clarify definition of <u>Physical Function</u> .
6.3	B671	Clarify definition of <u>Single-Function Device</u>
6.3	B672	Clarify definition of <u>Requester</u>
6.3	B673	Clarify software interpretation of the <u>Consecutive Flit Error after the Last Flit Error</u> field.
6.3	B674	Refer to tables instead of figures in § Section 2.2.4.2 .
6.3	B675	Remove outdated text in § Section 6.13
6.3	B676	Correct Poisoned and Nullified wording in § Section 4.2.3.4.1 .
6.3	B677	Remove PAM4 versions of Lane Margining parameters in § Table 8-13. Move rules into non-PAM4 parameters (i.e., there is only one set of parameters with different behaviors for PAM4 and NRZ).
6.3	B678	Clarifications in § Section 2.2.9.2 – Completion Rules for Flit Mode

Revision	Errata / ECN	Description
6.3	B679	In § Section 6.6.2 , remove redundant text that explicitly lists Link Equalization Request 16.0 GT/s and 16.0 GT/s Lane Equalization Control Register. Those registers are covered by the subsequent existing text “All registers in the <u>Physical Layer 16.0 GT/s Extended Capability structure</u> ”.
6.3	B680	Correct cross references in § Section 7.7.3.3
6.3	B681	Define behavior when Extended Synch is Set and L0p is activating Lanes.
6.3	B682	Simplify access rules for MMIO Register Blocks in § Section 6.35 .
6.3	B683	Clarify <u>Downstream Component Presence</u> behavior.
6.3	B684	Clarify Root Complex behavior regarding <u>AtomicOp Requester Enable</u> and <u>AtomicOp Egress Blocking</u> .
6.3	B685	Indicate which MSI interrupt vector should be used for <u>Link Equalization Request Interrupt Enable</u> .
6.3	B686	Correct inconsistent crosslink behavior between Flit Mode and non-Flit Mode
6.3	B688	Clarify Ordered Set modification rules for Retimers.
6.3	B689	Correct reference in § Table 7-315 – SFI DRS Received description.
6.3	B690	Define term <u>Role-Based Error Reporting</u> . Use it in the definition of the Role-Based Error Reporting bit in the <u>Device Capabilities Register</u> .
6.3	B691	Define terms <u>DSP</u> and <u>USP</u> . Global replace DP with <u>DSP</u> and UP with <u>USP</u> .
6.3	B692	Update § Figure 2-53 to make symbol 6, bit 7 Reserved. Bits 6:0 remain “{For Vendor Definition}”. Only the low 7 bits get copied during translation. This makes it consistent with the existing text: The low 7 bits of byte 6 is available for vendor definition. Byte 6, bit 7 is Reserved in Non-Flit Mode and is the EP bit in Flit Mode. Update § Figure 2-56 to include the EP bit in symbol 6, bit 7. Update § Figure 2-27 through § Figure 2-31 to define EP in Symbol 6, bit 7
6.3	B693	Use <u>TS0</u> and <u>TS1</u> in text that previously only used <u>TS1</u> .
6.3	B694	Clarify L0p rules regarding abandoned requests.
6.3	B695	UIO Requests always use 14-bit tags. Clarify conflicting text.
6.3	B696	Add OHC-C content to § Figure L-3 and § Figure L-7 Update Requester Segment value in § Figure L-8.
6.3	B697	Endpoint behavior when receiving an ID Routed Message for an unimplemented Function.
6.3	B698	Update pseudocode for RX replay buffer controls in § Section 4.2.3.4.2.1.5
6.3	B699	Remove text describing what happens when receiving a <u>TS1</u> in a case where <u>TS1s</u> can't occur.
6.3	B701	Clarify Switch and Root Complex rules regarding support for Flow-Through Selective IDE.
6.3	B704	Protocol Multiplexing is not supported in Flit Mode.
6.3	B705	Clarify Flit Error Counter interrupt behavior.

Revision	Errata / ECN	Description
6.3	B705a	Clarify <u>Flit Error Counter Enable</u> affects on <u>Flit Error Counter</u> .
6.3	B706	Recommend Lane Numbering behavior in the <u>Flit Logging Extended Capability</u> .
6.3	B707	Clarify Type 1 Config Request behavior when Selective IDE Stream are in use.
6.3	B710	Clarify discard rules for Link IDE.
6.3	B713	Clarify <u>Matching Link and Lane Numbers for 1b/1b Encoding</u> behavior for TS2s in <u>Recovery.RcvrCfg</u> .
6.3	B714	In § Figure 4-33, replace “MAX_UNACKNOWLEDGED_FLITS” with “511” in the two checks so that they read: “(TX_ACKNAK_FLIT_SEQ_NUM – IMPLICIT_RX_FLIT_SEQ_NUM) mod 1023 < 511”. This makes the figure agree with the pseudo code.
6.4	B803	Correct VPD Checksum Computation rules.
6.4	B807	Clarify Tag rules between UIO and non-UIO channel operation.
6.4	B811	Clarify IDE Byte Enable field rules. Clarify IDE Translation Request rules.
6.4	B812	Remove left over sentence from review draft of Base 6.0. “Note that in Flit Mode, the Byte Count and Lower Address Fields are not present.” Those fields are present in Flit Mode.
6.4	B814	Unused bits in <u>SVC Port Status Register</u> and <u>SVC Resource Status Register</u> are RsvdZ.
6.4	B816	Clarify forwarding ruled for Retimers.
6.4	B817	Clarify Transactoin Pending behavior when UIO Requests are outstanding.
6.4	B818	Clarify ACS behavior in conjunction with UIO Requests.
6.4	B819	No RO-enabled PR-PR Passing only applies to non-UIO, non-IDE TLPs.
6.4	B820	Clarify Precode Request behavior.
6.4	B821	Clarify TS[2:0] = 001b behavior.
6.4	B822	Clarify error behavior for non-IDE TLPs received in an IDE Stream.
6.4	B823	Explicitly describe Downstream and Upstream Port functionality of certain control bits.
6.4	B824	Clarify the behavior for <u>ECRC Check Failed</u> by adding a bulleted list.
6.4	B825	Define and use term <u>Physical Lane Number</u> .
6.4	B826	Correct wording of <u>Debug Opcodes Requested</u> .
6.4	B827	Add caution for using <u>L0p</u> at 16.0 GT/s with multiple Retimers.
6.4	B828	• Clarify Segment Number rules. • Removing DPC does not enter <u>Detect</u> with Link is disabled. • Add default values for <u>SFI Status Register</u> bits.
6.4	B829	Clarify Hot Reset and Link Disable behavior.
6.3	B831	Clarify Loopback scrambling and precoding behavior.

Revision	Errata / ECN	Description
6.3	B833	Add rules for Switch and Root Port Virtual Bridge Behavior in non-D0 State. Previously, these were included by reference from <u>PCI-to-PCI-Bridge</u> .
6.3	B834	Include UIO Requests in many places that used to only include Non-Posted Requests.
6.4	B836	Clarify ACS redirect behavior
6.4	B837	Clarify <u>Nak Ignore Window</u> and Ack/Nak Latency rules
6.4	B839	Clarify ACS Source Validation error reporting behavior
6.4	B840	Recommend behavior to avoid configuring with less than maximum link width
6.4	B841	Clarify IDE PCRC Check Failed behavior
6.4	B842	Clarify <u>CREDIT_LIMIT</u> and <u>SHARED_CREDIT_LIMIT</u> update rules.
6.4	B843	Add Implementation Note: Use Caution When Injecting Too Many Uncorrectable Errors Into A Flit.
6.4	B845	Clarify SKP Ordered Set insertion rules, especially for L0p.
6.4	B858	Bytes 8 and 9 of Non-Flit Mode Invalidation Request and Invalidation Completion Messages should be called Destination ID, not Device ID. This affects <u>§ Figure 10-18</u> and <u>§ Figure 10-21</u> . Flit Mode equivalents were already correct.
6.4	B859	Define Retimer Margin CRC checking rules for 1b/1b.
6.1	ECN: TDISP	Incorporate <u>TEE Device Interface Security Protocol (TDISP)</u>, Approved July 22, 2022 Minor issues corrected during TDISP incorporation (not individually marked): - Update reserved field width in <u>§ Table 11-25</u> to account for IS_NON_TEE_MEM Range Attribute. - In <u>§ Section 11.5.7</u>, table reference corrected to <u>§ Table 11-34</u>. - In <u>§ Section 11.2</u> “An example list of architectural configurations registers that should be locked and tracked is shown in <u>§ Section 11.2.6</u>.”
6.1	ECN: UIO	Incorporate <u>Unordered I/O (UIO)</u> , Approved March 16, 2023
6.1	ECN: Alt-Protocol-DLLP	Incorporate <i>Alternate Protocol DLLP Reservation</i>, Approved December 14, 2022 Section references in Note 3 of <u>§ Figure 3-5</u> were corrected from the published ECN.
6.1	ECN: DOE 1.1	Incorporate <i>Data Object Exchange (DOE)</i> , <i>Revision 1.1</i> Approved September 29, 2022
6.2	ECN: 12V-2x6	Incorporate 12V-2x6 Connector Updates to PCIe Base 6.0 (12V-2x6), Approved August 9, 2023
6.2	ECN: MMPT	Incorporate Management Message Passthrough via MMIO Mailbox (MMPT), Approved 2023-09-21 Errata to the ECN since publication are also incorporated: - Added a note that <u>MMB Registers</u> offset 018h-019h are reserved for CXL compatibility. - Added reserved bit definitions to the <u>MMB Capabilities Register</u>, the <u>MMB Control Register</u>, and the <u>MMB Status Register</u> where fields are reserved for CXL compatibility. - Updated the section numbers to align with PCIe Base Spec v6.1. - Updated CXL to <u>[CXL]</u> for notes about legacy compatibility.

Revision	Errata / ECN	Description
6.2	ECN: Architectural Out-of-Band Management	Incorporate Architectural Out-of-Band Management, Approved 2023-12-12
6.2	ECN: Revised CMA-SPDM	Incorporate Component Measurement and Authentication (CMA-SPDM), Revised December 2022
6.3	OHC-E Capability Enumeration	Add a three-bit field OHC-E Support in <u>Device Capabilities 3 Register</u> to indicate to software a Function's level of OHC-E support.
6.3	Removing Prefetchable Terminology	Through misunderstanding and misuse, “Prefetchable” & “Non-Prefetchable” have become damaged words – a source of more trouble than good. This ECN reworks the PCIe Base Specification to more accurately reflect modern device & system requirements, with the ultimate goal of significantly reducing industry confusion and correctly focusing development efforts.
6.3	NOP Flit Extensions	Defines two new standardized NOP Flit types: 1. <u>NOP.Debug</u> (Debug information) 2. <u>NOP.Vendor</u> (Vendor-defined content) Existing NOP Flit definition is renamed as <u>NOP.Empty</u> Flit type. Any NOP Flit type is permitted when a NOP Flit is transmitted. NOP.Debug and NOP.Vendor Flits are not subject to retransmission or flow control. Transmission is on a best effort basis. A NOP Flit Counter mechanism is defined to detect missing NOP Flits (in most cases). A NOP Stream ID indicates the origin of the NOP Flit and enables forwarding use cases of NOP Flits.
6.3	SNDR _{TX} and R _{LM-TX} Measurement	Tx Signal-to-Noise Distortion Ratio (SNDR) and Ratio of Level Mismatch (R_{LM-TX}) Measurement Methodologies at 64.0 GT/s Describes ways to measure Signal-to-Noise Distortion Ratio (SNDR_{TX}) and Ratio of Level Mismatch (R_{LM-TX}) at 64.0 GT/s using Multi Pulse Response Fit (MPRF)-based methodologies. The new MPRF-based SNDR_{TX} method mitigates the impact of duty cycle error on the SNDR_{TX} measurement. The new MPRF-based R_{LM-TX} method replaces the method where the 61_{st} UI from each consecutive run of 64 or more UIs of the same symbol {0, 1, 2, 3}, is used to establish a mean voltage for the PAM4 signal levels (V₀, V₁, V₂, and V₃) transmitted for PAM4 2-bit symbols for the R_{LM-TX} equation. The new R_{LM-TX} method will be applied on a compliance pattern waveform captured on a real-time oscilloscope. Pulse responses are established based on each PAM4 signal level. The peak of these pulse responses become the input to the R_{LM-TX} expression. The lower data rates are unaffected as SNDR_{TX} and R_{LM-TX} are only defined at 64.0 GT/s.
6.4	Optical Aware Retimer	The ECN provides a retimer-based mechanism to extend PC Express Links partially containing optical technologies. The definition is optical technology neutral so as to avoid bias against any specific optical technology.
6.4	PCI Express Management Interface (PCIe-MI)	This ECN builds upon the Chapter 12. Architectural Out-of-Band Management ECN to define an MCTP-based out-of-band management interface for managing PCIe MI Components (e.g., components or adapters). Out-of-band (OOB) management refers to the ability to perform management operations (e.g., inventorying, monitoring, and configuration) on <u>PCIe-MI</u>

Revision	Errata / ECN	Description
		Components without interacting with software/firmware running on a host CPU. PCIe-MI provides a standardized interface for BMCs to manage PCIe-MI Components over any transport with an MCTP binding (e.g., SMBus/I2C, I3C, PCIe VDM, USB, and MMBI). The PCIe-MI architecture in this ECN provides a framework for out-of-band management of a PCIe-MI Component that is easily extensible to new standardized commands and vendor-specific commands, along with a base set of commands to perform inventorying, monitoring, and configuration operations.
6.4	Scalable I/O Virtualization (SIOV)	This ECN introduces Scalable I/O Virtualization (SIOV), a successor to SR-IOV, intended for devices that expose more virtual device interfaces than is currently practical with SR-IOV. A comprehensive description of the use cases and usage model will be found in The Scalable I/O Virtualization project of the Open Compute Project [OCP]. Scalable IOV relaxes many of the requirements of SR-IOV, with the expectation that virtualization software (hypervisors, VMMs, container hosting systems, operating systems which map devices into usermode processes) supplies what the SIOV hardware interface does not.
6.4	TDISP eXtended TEE (XT) Extensions	TDISP eXtended TEE (XT) extensions require changes to the following areas of the PCIe base specification: 1. Integrity & Data Encryption (IDE) sections ◦ Introduce a new XT bit in the IDE TLPs and corresponding changes. 2. Address Translation Services (ATS) sections ◦ Introduce a new TE bit in the Translation Completions and corresponding changes. 3. TEE Device Interface Security Protocol (TDISP) sections ◦ Introduce TLP and security rules associated with the XT/TE bits and corresponding changes. The following figures were updated to add the XT bit: • § Figure 2-13 • § Figure 6-64 • § Figure L-2 • § Figure L-3 • § Figure L-4 • § Figure L-6 • § Figure L-7 • § Figure L-8
6.4	Enhanced SFI (eSFI)	System Firmware Intermediary (SFI) functionality was introduced with the Async Hot-Plug Updates ECN against [PCIe-4.0]. It primarily provided: • Optional functionality for the Firmware-First model, where SFW serves as the intermediary • Blocking rogue Config Reads before the device becomes ready following reset • SFW intermediary handling of PD, DLL Link Active, Device Readiness Status (DRS)

Revision	Errata / ECN	Description
		Hiding & preparatory config of a newly added adapter by SFW, transparent to the OS Enhanced SFI (eSFI) adds the following functionality: • Control of the <u>SFI Extended Capability</u> structure by the <u>BMC</u> via <u>PCIe-MI</u>, and lock-out of host software control. • Advanced traffic blocking/filtering for added protection against malicious devices • Triggerable quarantine mode upon device removal to avoid swapping w/o SFW/BMC supervision • Making the SFI Configuration Access Mechanism (CAM) optional for RPs
6.4	RC ECS Handling	Enhanced ERR_COR functionality that introduced the ERR_COR Subclass (ECS) field was included in the Async Hot-Plug Updates ECN against [PCIe-4.0]. It defined the ECS field in ERR_COR, which identifies and distinguishes the following subclasses: ECS SIG_SFW for signaling SFW, ECS SIG_OS for signaling OS SW, and ECS Legacy for Requesters that don't support ECS. RC ECS Handling provides “missing” functionality in RP/RCEC AER for handling received ERR_CORs with different ECS field values. Architecting this functionality enables OSs to use ERR_COR concurrently with ERR_COR being used by SFW.
6.4	Tx PS21 Methodology at 32.0 & 64.0 GT/s	Describes a method to measure TX PS21 parameter at 32.0 & 64.0 GT/s more accurately by using brickwall filter. The relationship of Nyquist to 3rd harmonic can vary in different transmitter designs. This was proposed to remove the impact of the 3rd harmonic on the Nyquist measurement.
6.4	LTR Enhancements for MFD	A Multi-Function Device with latency sensitive applications on different Functions can experience interruption or degradation due to an FLR on Function 0. With this ECN, LTR settings for the Device will be maintained without interruption, even after an FLR is performed on any individual function.

Objective of the PCI Express® Architecture

This document defines the “base” specification for the PCI Express architecture, including the electrical, protocol, platform architecture and programming interface elements required to design and build devices and systems. A key goal of the PCI Express architecture is to enable devices from different vendors to inter-operate in an open architecture, spanning multiple market segments including clients, servers, embedded, and communication devices. The architecture provides a flexible framework for product versatility and market differentiation.

This specification describes the PCI Express® architecture, interconnect attributes, fabric management, and the programming interface required to design and build systems and peripherals that are compliant with the PCI Express Specification.

The goal is to enable such devices from different vendors to inter-operate in an open architecture. The specification is intended as an enhancement to the PCI™ architecture spanning multiple market segments; clients (desktops and mobile), servers (standard and enterprise), and embedded and communication devices. The specification allows system OEMs and peripheral developers adequate room for product versatility and market differentiation without the burden of carrying obsolete interfaces or losing compatibility.

PCI Express Architecture Specification Organization

The PCI Express specifications are organized as a base specification and a set of companion documents.

The *PCI Express Base Specification* contains the technical details of the architecture, protocol, Data Link Layer, Physical Layer, and software interface. The *PCI Express Base Specification* (this document) is applicable to all variants of PCI Express.

The companion specifications define a variety of form factors, including mechanical and electrical chapters covering topics including auxiliary signals, power delivery, and the Adapter interconnect electrical budget.

Documentation Conventions

Capitalization

Some terms are capitalized to distinguish their definition in the context of this document from their common English meaning. Words not capitalized have their common English meaning. When terms such as “memory write” or “memory read” appear completely in lower case, they include all transactions of that type.

Register names and the names of fields and bits in registers and headers are presented with the first letter capitalized and a mixture of capitalization for the remainder.

Numbers and Number Bases

Hexadecimal numbers are written with a lower case “h” suffix, e.g., FFFh and 80h. Hexadecimal numbers larger than four digits are represented with a space dividing each group of four digits, as in 1E FFFF FFFFh. Binary numbers are written with a lower case “b” suffix, e.g., 1001b and 10b. Binary numbers larger than four digits are written with a space dividing each group of four digits, as in 1000 0101 0010b.

All other numbers are decimal.

Implementation Notes

IMPLEMENTATION NOTE:

§

Implementation Notes should not be considered to be part of this specification. They are included for clarification and illustration only.

Notes

NOTE §

Notes pertain to the specification itself as opposed to implementations of the specification. For example, they are used to describe the document process. They are also used for forward looking information describing anticipated changes for a future version of this specification.

Issues

ISSUE 1

Issues are outstanding items in the specification. They indicate things like missing, outdated, or lower quality artwork, anticipated changes that are being deferred to a subsequent version of the specification, potential errata items noticed during the editing process, etc.

PCI-SIG's goal is to resolve all notes before the 1.0 published release of the specification (where “resolving” includes deferring an item to later, or determining that the item is not needed or is incorrect).

Implementation Notes §, Notes §, and Issues § can also be inline.

Terms and Acronyms

8b/10b

The data encoding scheme ¹ used in the PCI Express Physical Layer for 5.0 GT/s and below.

128b/120b

The data encoding scheme used in the PCI Express Physical Layer for 8.0, 16.0, and 32.0 GT/s.

10-Bit Tags

A Tag's capability that provides a total of 10 bits for the Tag field. See Tag.

14-Bit Tags

A Tag's capability that provides a total of 14 bits for the Tag field. See Tag.

Access Control Services, ACS

A set of capabilities and control registers used to implement access control over routing within a PCI Express component.

ACS Violation

An error that applies to a Posted or Non-Posted Request when the Completer detects an access control violation.

Adapter

Used generically to refer to an add-in card or module.

Advanced Error Reporting, AER

Advanced Error Reporting (see § [Section 7.8.4](#)).

Alternative Routing-ID, ARI

Alternative Routing-ID Interpretation. Applicable to Requester IDs and Completer IDs as well as Routing IDs.

ARI Device

A Device associated with an Upstream Port, whose Functions each contain an [ARI Extended Capability](#) structure.

ARI Downstream Port

A Switch Downstream Port or Root Port that supports ARI Forwarding.

ARI Forwarding

Functionality that enables the Downstream Port immediately above an ARI Device to access the Devices extended Functions. Enabling ARI Forwarding ensures the logic that determines when to turn a Type 1 Configuration Request into a Type 0 Configuration Request no longer enforces a restriction on the traditional Device Number field being 0.

Asserted

The active logical state of a conceptual or actual signal.

Async Removal

Removal of an adapter or cable from a slot without lock-step synchronization with the operating system (e.g., in an asynchronous manner without button presses).

1. IBM Journal of Research and Development, Vol. 27, #5, September 1983 "A DC-Balanced, Partitioned-Block 8B/10B Transmission Code" by Widmer and Franaszek.

Atomic Operation, AtomicOp

One of three architected Atomic Operations where a single PCI Express transaction targeting a location in Memory Space reads the location's value, potentially writes a new value to the location, and returns the original value. This read-modify-write sequence to the location is performed atomically. AtomicOps include FetchAdd, Swap, and CAS.

Attribute

Transaction handling preferences indicated by specified Packet header bits and fields (e.g., non-snoop).

Authentication

A process for determining that an entity is what it appears to be (its identity) using defined data objects and digital signatures.

Base Address Register, BAR

Base Address Registers exist within Configuration Space and are used to determine the amount of system memory space needed by a Function and to provide the base address for a mapping to Function memory space. A Base Address Register may map to memory space or I/O space.

Beacon

An optional 30 kHz to 500 MHz in-band signal used to exit the L2 Link Power Management state. One of two defined mechanisms for waking up a Link in L2 (see Wakeup).

BMC

Baseboard Management Controller

Bridge

One of several defined System Elements. A Function that virtually or actually connects a PCI/PCI-X segment or PCI Express Port with an internal component interconnect or with another PCI/PCI-X bus segment or PCI Express Port. A virtual Bridge in a Root Complex or Switch must use the software configuration interface described in this specification.

by-1, x1

A Link or Port with one Physical Lane.

by-8, x8

A Link or Port with eight Physical Lanes.

by-N, xN

A Link or Port with "N" Physical Lanes.

Compare and Swap, CAS

An AtomicOp where the value of a target location is compared to a specified value and, if they match, another specified value is written back to the location. Regardless, the original value of the location is returned.

Character

An 8-bit quantity treated as an atomic entity; a byte.

Clear

A bit is Clear when its value is 0b.

Cold Reset

A Fundamental Reset following the application of main power.

Completer

The Function that terminates or "completes" a given Request, and generates a Completion if appropriate. Generally, the Function targeted by the Request serves as the Completer. For cases when an uncorrectable error prevents the Request from reaching its targeted Function, the Function that detects and handles the error serves as the Completer.

Completer Abort, CA

1. A status that applies to a Posted or Non-Posted Request that the Completer is permanently unable to complete successfully, due to a violation of the Completer's programming model or to an unrecoverable error associated with the Completer.
2. A status indication returned with a Completion for a Non-Posted Request or a UIO Request that suffered a Completer Abort at the Completer.

Completer ID

The combination of a Completer's Bus Number, Device Number, and Function Number that uniquely identifies the Completer of the Request within a Hierarchy. With an ARI Completer ID, bits traditionally used for the Device Number field are used instead to expand the Function Number field, and the Device Number is implied to be 0.

Completion

A Packet used to terminate, or to partially terminate, a transaction sequence. A Completion always corresponds to a preceding Request, and, in some cases, includes data.

component

A physical device (a single package).

Configuration Software

The component of system software responsible for accessing Configuration Space and configuring the PCI/PCIe bus.

Configuration Space

One of the four address spaces within the PCI Express architecture. Packets with a Configuration Space address are used to configure Functions.

Configuration-Ready

A Function is Configuration-Ready when it is guaranteed that the Function will respond to a valid Configuration Request targeting the Function with a Completion indicating Successful Completion status.

Containment Error Recovery, CER

A general error containment and recovery approach supported by Downstream Port Containment (DPC), where with suitable software/firmware support, many uncorrectable errors can be handled without disrupting applications.

Conventional PCI

Behaviors or features originally defined in the PCI Local Bus Specification. The PCI Express Base 4.0 and subsequent specifications incorporate the relevant requirements from the PCI Local Bus Specification.

Conventional Reset

A Hot, Warm, or Cold Reset. Distinct from Function Level Reset (FLR).

Data Link Layer

The intermediate Layer that is between the Transaction Layer and the Physical Layer.

Data Link Layer Packet, DLLP

A Packet generated in the Data Link Layer to support Link management functions.

data payload

Information following the header in some packets that is destined for consumption by the targeted Receiver of the Packet (e.g., Write Requests or Read Completions).

deasserted

The inactive logical state of a conceptual or actual signal.

Deferrable Memory Write, DMWr

A Memory Write where the Requester attempts to write to a given location in Memory Space using the Non-Posted DMWr TLP Type. A Completer that supports this TLP Type can accept or decline the Request, indicating this by means of the Completion status returned. See § Section 6.32 .

Design for Testability, DFT

Design for Testability.

Device (uppercase 'D')

A collection of one or more Functions within a single hierarchy identified by common Bus Number and Device Number. An SR-IOV Device may have additional Functions accessed via additional Bus Numbers and/or Device Numbers configured through one or more SR-IOV Extended Capability structures. An SIOV Device may have SDIs accessed via additional Bus Numbers and/or Device Numbers configured through one or more SIOV Extended Capability structures.

device (lowercase 'd')

1. A physical or logical entity that performs a specific type of I/O.
2. A component on either end of a PCI Express Link.
3. A common imprecise synonym for Function, particularly when a device has a single Function.

Device Readiness Status, DRS

A mechanism for indicating that a Device is Configuration-Ready (see § Section 6.22.1).

DLP

In Flit Mode, the Data Link Layer Payload within a Flit.

Downstream

1. The relative position of an interconnect/System Element (Port/component) that is farther from the Root Complex. The Ports on a Switch that are not the Upstream Port are Downstream Ports. All Ports on a Root Complex are Downstream Ports. The Downstream component on a Link is the component farther from the Root Complex.
2. A direction of information flow where the information is flowing away from the Root Complex.

Downstream Path

The flow of data through a Retimer from the Upstream Pseudo Port Receiver to the Downstream Pseudo Port Transmitter.

Downstream Port Containment, DPC

The automatic disabling of the Link below a Downstream Port following an uncorrectable error, which prevents TLPs subsequent to the error from propagating Upstream or Downstream.

DSP

Downstream Port

DWORD, DW

Four bytes. Used in the context of a data payload, the 4 bytes of data must be on a naturally aligned 4-byte boundary (the least significant 2 bits of the byte address are 00b).

Egress Port

The transmitting Port; that is, the Port that sends outgoing traffic.

Electrical Idle

A Link state used in a variety of defined cases, with specific requirements defined for the Transmitter and Receiver.

End-End TLP Prefix

A TLP Prefix that is carried along with a TLP from source to destination. See § Section 2.2.10.4 .

Endpoint

One of several defined System Elements. A Function that has a Type 00h Configuration Space header.

Endpoint Upstream Port

An Upstream Port that contains Endpoint Functions exclusively.

Enhanced SFI, eSFI

A defined set of enhancements for SFI, primarily providing improved security and improved support for a BMC serving as the intermediary. Support for Enhanced SFI is indicated by the Enhanced SFI Supported bit being Set.

error detection

Mechanisms that determine that an error exists, either by the first agent to discover the error (e.g., Malformed TLP) or by the recipient of a signaled error (e.g., receiver of a poisoned TLP).

error logging

A detector setting one or more bits in architected registers based on the detection of an error. The detector might be the original discoverer of an error or a recipient of a signaled error.

error reporting

In a broad context, the general notification of errors. In the context of the Device Control register, sending an error Message. In the context of the Root Error Command register, signaling an interrupt as a result of receiving an error Message.

error signaling

One agent notifying another agent of an error either by (1) sending an error Message, (2) sending a Completion with UR/CA Status, or (3) poisoning a TLP.

Extension Device

A component whose purpose is to extend the physical length of a Link.

Extended Function

Within an ARI Device, a Function whose Function Number is greater than 7. Extended Functions are accessible only after ARI-aware software has enabled ARI Forwarding in the Downstream Port immediately above the ARI Device.

FetchAdd, Fetch and Add

An AtomicOp where the value of a target location is incremented by a specified value using two's complement arithmetic ignoring any carry or overflow, and the result is written back to the location. The original value of the location is returned.

Flit

Flow Control Unit

Flow Control

The method for communicating receive buffer status from a Receiver to a Transmitter to prevent receive buffer overflow and allow Transmitter compliance with ordering rules.

Flow Control Packet, FCP

A DLLP used to send Flow Control information from the Transaction Layer in one component to the Transaction Layer in another component.

Flow-Through

Refers to the behavior, by a Switch or Root Complex supporting peer-to-peer, of passing an IDE TLP associated with a Selective IDE Stream from Ingress Port to Egress Port without modification.

FRU

Field Replacable Unit

FRU Information

Information used to provide inventory and capability information about a board on which the FRU Information Device is located.

FRU Information Device

A storage component such as a real or firmware emulated EEPROM that stores FRU Information, compatible with components such as the AT 24C256.

Function

Within a Device, an addressable entity in Configuration Space associated with a single Function Number. Used to refer to one Function of a Multi-Function Device, or to the only Function in a Single-Function Device. Specifically included are special types of Functions defined in § Chapter 9. , notably Physical Functions and Virtual Functions.

Function Group

Within an ARI Device, a configurable set of Functions that are associated with a single Function Group Number. Function Groups can optionally serve as the basis for VC arbitration or access control between multiple Functions within the ARI Device.

Function Level Reset, FLR

A mechanism for resetting a specific Endpoint Function (see § Section 6.6.2).

Function Readiness Status, FRS

A mechanism for indicating that a Function is Configuration-Ready (see § Section 6.22.2).

Fundamental Reset

A hardware mechanism for setting or returning all Port states to the initial conditions specified in this document (see § Section 6.6).

GT/s

The number of encoded bits transferred in a second on a direction of a Lane. Short for Giga Transfers per Second.

header

A set of fields that appear at or near the front of a Packet that contain the information required to determine the characteristics and purpose of the Packet.

Hierarchy

A PCI Express I/O interconnect topology, wherein the Configuration Space addresses, referred to as the tuple of Bus/Device/Function Numbers (or just Bus/Function Numbers, for ARI cases), are unique. These addresses are used for Configuration Request routing, Completion routing, some Message routing, and for other purposes. In some contexts a Hierarchy is also called a Segment, and in Flit Mode, the Segment number is sometimes also included in the ID of a Function.

hierarchy domain

The part of a Hierarchy originating from a single Root Port.

Host Bridge

Part of a Root Complex that connects a host CPU or CPUs to a Hierarchy.

Hot Reset

A reset propagated in-band across a Link using a Physical Layer mechanism.

IDE Partner Port

The remote IDE Terminus for an IDE Stream.

IDE Stream

A Port to Port connection established using the mechanisms defined by Integrity and Data Encryption (IDE) to secure TLP traffic between the two Ports. The connection may be in the form of a Selective IDE Stream, in which case it is possible for IDE TLPs to flow through Switches without affecting their security, or in the form of a Link IDE Stream, in which case the two Ports must be connected without intervening Switches.

IDE Terminus

A Port acting as the originator or ultimate destination for IDE TLPs associated with one or more IDE Streams.

IDE TLP

A TLP associated with an IDE Stream and secured using Integrity and Data Encryption (IDE).

in-band management

Management operations (e.g., inventorying, monitoring, and configuration) that are performed on a peripheral subsystem (e.g., a PCIe-MI Component) initiated by software/firmware running on a host CPU. See § Chapter 12.

in-band signaling

A method for signaling events and conditions using the Link between two components, as opposed to the use of separate physical (sideband) signals. All mechanisms defined in this document can be implemented using in-band signaling, although in some form factors sideband signaling may be used instead.

Ingress Port

Receiving Port; that is, the Port that accepts incoming traffic.

Internal Error

An error associated with a PCI Express interface that occurs within a component and which may not be attributable to a packet or event on the PCI Express interface itself or on behalf of transactions initiated on PCI Express.

I/O Space

One of the four address spaces of the PCI Express architecture.

isochronous

Data associated with time-sensitive applications, such as audio or video applications.

invariant

A field of a TLP header or TLP Prefix that contains a value that cannot legally be modified as the TLP flows through the PCI Express fabric.

Lane

A set of differential signal pairs, one pair for transmission and one pair for reception. A by-N Link is composed of N Lanes.

Layer

A unit of distinction applied to this specification to help clarify the behavior of key elements. The use of the term Layer does not imply a specific implementation.

Link

The collection of two Ports and their interconnecting Lanes. A Link is a dual-simplex communications path between two components.

Link IDE Stream

An IDE Stream applied to all TLPs, except those associated with Selective IDE Stream(s), where the two Ports are connected without intervening Switches, although extension devices may be present on the Link.

Link Segment

The collection of a Port and a Pseudo Port or two Pseudo Ports and their interconnecting Lanes. A Link Segment is a dual simplex communications path between a Component and a Retimer or between two Retimers (two Pseudo Ports).

Lightweight Notification, LN

This protocol is now deprecated. It was a lightweight protocol that supported notifications to Endpoints via a hardware mechanism when cache lines of interest were updated.

Local TLP Prefix

A TLP Prefix that is carried along with a TLP on a single Link. See [§ Section 2.2.10.2](#).

Logical Bus

The logical connection among a collection of Devices that have the same Bus Number in Configuration Space.

Logical Idle

A period of one or more Symbol Times when no information (TLPs, PMUX Packets, DLLPs, or any special Symbol) is being transmitted or received. Unlike Electrical Idle, during Logical Idle the Idle data Symbol is being transmitted and received.

LTR

Abbreviation for Latency Tolerance Reporting

Malformed TLP

A TLP that violates specific TLP formation rules as defined in this specification.

Measurement

A process for calculating a cryptographic hash value of firmware or other configuration state, applying a digital signature, and returning this information.

Memory Space

One of the four address spaces of the PCI Express architecture.

Message

A TLP used to communicate information outside of the Memory, I/O, and Configuration Spaces.

Message Signaled Interrupt, MSI/MSI-X

Two similar but separate mechanisms that enable a Function to request service by writing a system-specified DWORD of data to a system-specified address using a Memory Write Request. Compared to MSI, MSI-X supports a larger maximum number of vectors and independent message address and data for each vector.

Message Space

One of the four address spaces of the PCI Express architecture.

MMIO

Memory-mapped I/O space. Synonymous with the term Memory Space.

MPS

Abbreviation for [Max_Payload_Size](#).

Multicast, MC

A feature and associated mechanisms that enable a single Posted Request TLP sent by a source to be distributed to multiple targets.

Multicast Group, MCG

A set of Endpoints that are the target of Multicast TLPs in a particular address range.

Multicast Hit

The determination by a Receiver that a TLP will be handled as a Multicast TLP.

Multicast TLP

A TLP that is potentially distributed to multiple targets, as controlled by Multicast Capability structures in the components through which the TLP travels.

Multicast Window

A region of Memory Space where Posted Request TLPs that target it will be handled as Multicast TLPs.

Multi-Function Device, MFD

A Device that has multiple Functions.

Multi-Root I/O Virtualization, MR-IOV

A Function that supports the MR-IOV capability. See [MR-IOV] for additional information.

MUST@FLIT

MUST@FLIT features are mandatory for components that support Flit Mode (the Flit Mode Supported bit is Set).

MUST@FLIT features are strongly recommended for all other components.

naturally aligned

A data payload with a starting address equal to an integer multiple of a power of two, usually a specific power of two. For example, 64-byte naturally aligned means the least significant 6 bits of the byte address are 00 0000b.

NPEM

Native PCIe Enclosure Management

OBFF

Optimized Buffer Flush/Fill

Operating System

Throughout this specification, the terms operating system and system software refer to the combination of power management services, device drivers, user-mode services, and/or kernel mode services.

orderly removal

A hot-plug removal model where the OS is notified when a user/operator wishes to remove an adapter, and the OS has the opportunity to prepare for the event (e.g., quiescing adapter activity) before granting permission for removal.

out-of-band management

Management operations (e.g., inventorying, monitoring, and configuration) that are performed on a peripheral subsystem (e.g., a PCIe-MI Component) without interacting with software/firmware running on a host CPU (e.g., management operations initiated by a BMC). See § Chapter 12. .

P2P

Peer-to-peer.

Path

The flow of data through a Retimer, in either the Upstream Path or the Downstream Path.

Packet

A fundamental unit of information transfer consisting of an optional TLP Prefix, followed by a header and, in some cases, followed by a data payload.

Parts per Million, ppm

Applied to frequency, the difference, in millionths of a Hertz, between a stated ideal frequency, and the measured long-term average of a frequency.

PCI Bridge

See Type 1 Function.

PCI Software Model

The software model necessary to initialize, discover, configure, and use a PCI-compatible device, as specified in [PCI], [PCI-X], and [Firmware].

PCIe®

PCI Express®

PCI Express Management Interface™, PCIe-MI™

An optional architecture and command set for message-based management of a PCIe-MI Component (e.g., an adapter or component) by entities such as in-band management system software running on a host CPU or out-of-band management firmware running on a BMC (see § Chapter 12.).

PCIe-MI Command

A PCIe-MI Message that is transmitted from a PCIe-MI Commander to a PCIe-MI Completer that specifies a management operation to be performed.

PCIe-MI Commander

An entity on a host (e.g., system software running on a host CPU or a BMC) that transmits a PCIe-MI Command to a PCIe-MI Completer.

PCIe-MI Completer

An entity (e.g., an MCTP endpoint or a Function supporting MMPT) on a PCIe-MI Component that processes PCIe-MI Commands and transmits PCIe-MI Completions to a PCIe-MI Commander.

PCIe-MI Completer Reset

A reset that causes the processing of any PCIe-MI Commands in progress or pending to be aborted and may cause the transmission of any PCIe-MI Completions in progress to be aborted (see § Section 12.7.1.2).

PCIe-MI Completion

A PCIe-MI Message that is transmitted from a PCIe-MI Completer to the PCIe-MI Commander that transmitted the corresponding PCIe-MI Command in order to provide the results of that PCIe-MI Command.

PCIe-MI Component

A collection of one or more non-IOV Functions and/or Physical Functions (e.g., a component or adapter) that implements the PCI Express Management Interface on one or more PCIe-MI Completers.

PCIe-MI Message

A PCIe-MI Command or PCIe-MI Completion.

PCIe-MI Port

A PCIe Upstream Port, PCIe Downstream Port, 2-Wire interface port, or USB port on a PCIe-MI Component.

Phantom Function Number, PFN

An unclaimed Function Number that may be used to expand the number of outstanding transaction identifiers by logically combining the PFN with the Tag identifier to create a unique transaction identifier.

Physical Function, PF

A Function that contains an SR-IOV Extended Capability structure and/or contains an SIOV Extended Capability structure, defined in § Chapter 9. .

Physical Lane

See Lane.

Physical Lane Number

Lane Number which is assigned in an implementation specific manner and is invariant to Link width and Lane reversal negotiation. Physical lane numbers must be unique and sequential. For example, each Lane of a x16 Link must be assigned a unique Lane number from 0 to 15.

Physical Layer

The Layer that directly interacts with the communication medium between two components.

Port

1. Logically, an interface between a component and a PCI Express Link.
2. Physically, a group of Transmitters and Receivers located on the same chip that define a Link.

Power Management

Software or Hardware mechanisms used to minimize system power consumption, manage system thermal limits, and maximize system battery life. Power management involves tradeoffs among system speed, noise, battery life, and AC power consumption.

PMUX Channel

A multiplexed channel on a PMUX Link that is configured to transport a specific multiplexed protocol. See [§ Appendix G. .](#)

PMUX Link

A Link where Protocol Multiplexing is supported and enabled. See [§ Appendix G. .](#)

PMUX Packet

A non-PCI Express Packet transported over a PCI Express Link. See [§ Appendix G. .](#)

Precision Time Measurement, PTM

An optional capability for communicating precise timing information between components.

Process Address Space ID, PASID

The Process Address Space ID, in conjunction with the Requester ID, uniquely identifies the address space associated with a transaction.

Programmed I/O, PIO

A transaction sequence that's initiated by a host processor, often as the result of executing a single load or store instruction that targets a special address range, but can be generated by other mechanisms such as the PCI-Compatible Configuration Mechanism. Notably, host processor loads or stores targeting an ECAM address range generate Configuration Space transactions. Other memory-mapped ranges typically exist to generate Memory Space and I/O Space transactions.

Pseudo Port

1. Logically, an interface between a Retimer and a PCI Express Link Segment.
2. Physically, a group of Transmitters and Receivers located on the same Retimer chip that define a Link Segment.

Quality of Service, QoS

Attributes affecting the bandwidth, latency, jitter, relative priority, etc., for differentiated classes of traffic.

QWORD, QW

Eight bytes. Used in the context of a data payload, the 8 bytes of data must be on a naturally aligned 8-byte boundary (the least significant 3 bits of the address are 000b).

RCiEP

Root Complex Integrated Endpoint.

Read Side Effect

An observable change in system state due to a read. A classic Read Side Effect is a read that returns an element from a FIFO, removing that element from the FIFO. This specification does not define or require the implementation of Read Side Effects.

recommended

Among several possibilities one is particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required.

Receiver, Rx

The component that receives Packet information across a Link.

Receiving Port

In the context of a specific TLP, PMUX Packet, or DLLP, the Port that receives the Packet on a given Link.

Re-driver

A non-protocol aware, software transparent, Extension Device.

Refclk

An abbreviation for Reference Clock.

repeater

An imprecise term for Extension Device.

Reported Error

An error subject to the logging and signaling requirements architecturally defined in this document.

Request

A Packet used to initiate a transaction sequence. A *Request* includes operation code and, in some cases, address and length, data, or other information.

Requester

The Function or system element that first introduces a transaction sequence into the PCI Express domain.

Requester ID

The combination of a Requester's Bus Number, Device Number, and Function Number that uniquely identifies the Requester within a Hierarchy. With an ARI Requester ID, bits traditionally used for the Device Number field are used instead to expand the Function Number field, and the Device Number is implied to be 0.

Reserved

The contents, states, or information are not defined at this time. Using any Reserved area (e.g., packet header bit-fields, configuration register bits) is not permitted. Reserved register fields must be read-only and must return 0 (all 0's for multi-bit fields) when read. Reserved encodings for register and packet fields must not be used. Any implementation dependence on a Reserved field value or encoding will result in an implementation that is not PCI Express-compliant. The functionality of such an implementation cannot be guaranteed in this or any future revision of this specification.

Retimer

A Physical Layer protocol aware, software transparent, Extension Device that forms two separate electrical Link Segments.

Role-Based Error Reporting

A set of error handling semantics based on the role of the agent that detects the error, the type of TLP involved in the error, and the error handling settings of the agent. See Implementation Note: Use of ERR_COR, ERR_NONFATAL, and ERR_FATAL.

Root Complex, RC

A defined System Element that includes at least one Host Bridge, Root Port, or Root Complex Integrated Endpoint.

Root Complex Component

A logical aggregation of Root Ports, Root Complex Register Blocks, Root Complex Integrated Endpoints, and Root Complex Event Collectors.

Root Port, RP

A PCI Express Port on a Root Complex that maps a portion of a Hierarchy through an associated virtual PCI-PCI Bridge.

Routing Element

A term referring to a Root Complex, Switch, or Bridge in regard to its ability to route, multicast, or block TLPs.

Routing ID, RID

Either the Requester ID or Completer ID that identifies a PCI Express Function or an SIOV SDI.

RP PIO

Root Port Programmed I/O. See § Section 6.2.11.3.

Rx_MPS_Limit

The computed data payload size limit for a Function receiving a TLP, which is determined by the Rx_MPS_Fixed bit value and Max_Payload_Size setting in one or more Functions. See § Section 2.2.2 for details.

Scalable IOV, SIOV

A strategy for device virtualization relying on the Virtualization Intermediary (VI) to supply many aspects of a virtual device interface, relying on hardware for data-path operations and software for setup, teardown, error handling in order to simplify each device interface.

Scalable Device Interface, SDI

A device interface exposed by a Physical Function employing Scalable IOV. It has a unique Routing ID, though no Configuration Space or BARs.

Segment

See Hierarchy

Selective IDE Stream

An IDE Stream applied selectively to TLPs based on ranges of Memory Addresses and RIDs, and where it is possible for secured TLPs to flow through Switches without affecting their security.

Set

A bit is Set when its value is 1b.

Shadow Function

An otherwise unimplemented Function, where its Transaction ID space is used by a Function that implements the Shadow Functions Extended Capability structure.

sideband signaling

A method for signaling events and conditions using physical signals separate from the signals forming the Link between two components. All mechanisms defined in this document can be implemented using in-band signaling, although in some form factors sideband signaling may be used instead.

Single-Function Device, SFD

A Device that has a single Function.

Single Root I/O Virtualization, SR-IOV

A strategy for device virtualization relying on the device hardware or firmware to supply both control-path and data-path aspects of a virtual device interface.

Single Root PCI Manager, SR-PCIM

Software responsible for configuration and management of the SR-IOV Extended Capability and PF/VF, the SIOV Extended Capability and PF/SDI, as well as dealing with associated error handling. Multiple implementation options exist; therefore, SR-PCIM implementation is outside the scope of this specification.

SIOV Device

A Device containing one or more Functions that have an SIOV Extended Capability structure.

SR-IOV Device

A Device containing one or more Functions that have an SR-IOV Extended Capability structure.

SSD

Solid State Drive

strongly recommended

This item is not mandatory, but in the absence of compelling reasons, it is the most desired choice.

Swap, Unconditional Swap

An AtomicOp where a specified value is written to a target location, and the original value of the location is returned.

Switch

A defined System Element that connects two or more Ports to allow Packets to be routed from one Port to another. To configuration software, a Switch appears as a collection of virtual PCI-to-PCI Bridges [PCI-to-PCI-Bridge].

Switch Port

A Port that contains a Switch Downstream Port Function or at least one Switch Upstream Port Function.

Symbol

A 10-bit quantity when using 8b/10b encoding. An 8-bit quantity when using 128b/130b or 1b/1b encodings.

Symbol Time

The period of time required to place a Symbol on a Lane (10 times the Unit Interval when using 8b/10b encoding, 8 times the Unit Interval when using 128b/130b encoding, and 4 times the Unit Interval when using 1b/1b encoding).

System Element

A defined Device or collection of Devices that operate according to distinct sets of rules. The following System Elements are defined: Root Complex, Endpoint, Switch, and Bridge.

System Image, SI

A software component running on a virtual system to which specific Functions, PFs, and VFs can be assigned. Specification of the behavior and architecture of an SI is outside the scope of this specification. Examples of SIs include guest operating systems and shared/non-shared protected domain device drivers.

System Software

Includes System Firmware (BIOS, UEFI), Operating System, VMM, management software, platform vendor's add-on to the Operating System.

Tag

A number assigned to a given Non-Posted Request or UIO Request to distinguish Completions for that Request from other Requests.

TEE Device Interface (TDI)

The unit of assignment for an IO-virtualization capable device. For example, a TDI may be an entire Device, a non-IOV Function, a PF (and possibly its subordinate VFs or SDIs), a VF, or an SDI.

TEE-I/O

A conceptual framework for establishing and managing Trusted Execution Environments (TEEs) that include a composition of resources from one or more devices (see § Chapter 11.).

TLP Prefix

Additional information that may be optionally prepended to a TLP. TLP Prefixes are either Local or End-End. A TLP can have multiple TLP Prefixes. See § Section 2.2.10.

TPH

Abbreviation for TLP Processing Hints

Transaction Descriptor

An element of a Packet header that, in addition to Address, Length, and Type, describes the properties of the Transaction.

Transaction ID

A component of the Transaction Descriptor including Requester ID and Tag.

Transaction Layer

The Layer that operates at the level of transactions (for example, read, write).

Transaction Layer Packet, TLP

A Packet generated in the Transaction Layer to convey a Request or Completion.

transaction sequence

A single Request and zero or more Completions associated with carrying out a single logical transfer by a Requester.

Transceiver

The physical Transmitter and Receiver pair on a single chip.

Translated Request

A Request using a Translated Memory Address, as indicated by the AT field.

Transmitter, Tx

The component sending Packet information across a Link.

Transmitting Port

In the context of a specific TLP, PMUX Packet, or DLLP, the Port that transmits the Packet on a given Link.

Trusted Computing Base (TCB)

A combination of hardware, firmware, and software, responsible for enforcing a security policy. Bugs or vulnerabilities occurring inside the TCB might jeopardize the security properties of the entire system. By contrast, parts of a computer system outside the TCB shall not be able to create a condition that would allow any more privileges than are granted to them in accordance with the security policy.

Trusted Execution Environment, TEE

Refers to an environment, which may include only a portion of a device, the whole of a device, or a composition of multiple devices, within which some level of “trust” is established, such that operations (including code execution) that occur within this environment are considered “trustworthy”. It is generally the case that one TEE is isolated from other TEEs that are intended to be distinct, and that all TEEs are isolated from untrusted environments.

Trusted Execution Environment Virtual Machine (Trusted Execution Environment VM, TVM)

A Trusted Execution Environment Virtual Machine as defined in the TEE Device Interface Security Protocol (TDISP) reference architecture (see § Chapter 11.).

Tx_MPS_Limit

The computed data payload size limit for a Function transmitting a TLP, which is determined by the Max_Payload_Size setting in one or more Functions. See § Section 2.2.2 for details.

Type 0 Function

Function with a Type 0 Configuration Space Header (see § Section 7.5.1.2).

Type 1 Function

Function with a Type 1 Configuration Space Header (see § Section 7.5.1.3).

Unconditional Swap, Swap

An AtomicOp where a specified value is written to a target location, and the original value of the location is returned.

Unit Interval, UI

Given a data stream of a repeating pattern of alternating 1 and 0 values, the Unit Interval is the value measured by averaging the time interval between voltage transitions, over a time interval long enough to make all intentional frequency modulation of the source clock negligible (see RX: UI and TX: UI).

Unordered I/O, UIO

A set of Request/Completion Types used with supporting Virtual Channel(s) to support a Requester-managed ordering model. See § Section 6.34 .

Unsupported Request, UR

1. A status that applies to a Posted or Non-Posted Request that specifies some action or access to some space that is not supported by the Completer.
2. A status indication returned with a Completion for a Non-Posted Request or a UIO Request that suffered an Unsupported Request at the Completer.

Upstream

1. The relative position of an interconnect/System Element (Port/component) that is closer to the Root Complex. The Port on a Switch that is closest topologically to the Root Complex is the Upstream Port. The Port on a component that contains only Endpoint or Bridge Functions is an Upstream Port. The Upstream component on a Link is the component closer to the Root Complex.
2. A direction of information flow where the information is flowing towards the Root Complex.

Upstream Path

The flow of data through a Retimer from the Downstream Pseudo Port Receiver to the Upstream Pseudo Port Transmitter.

USP

Upstream Port

variant

A field of a TLP header that contains a value that is subject to possible modification according to the rules of this specification as the TLP flows through the PCI Express fabric.

Virtual Function, VF

A Function that is associated with a Physical Function. A VF shares one or more physical resources, such as a Link, with the Physical Function and other VFs that are associated with the same PF.

Virtualization Intermediary, VI

A software component supporting one or more SIs-colloquially known as a hypervisor or virtual machine monitor. Specification of the behavior and architecture of the VI is outside the scope of this specification.

wakeup

An optional mechanism used by a component to request the reapplication of main power when in the L2 Link state. Two such mechanisms are defined: Beacon (using in-band signaling) and WAKE# (using sideband signaling).

Warm Reset

A Fundamental Reset without cycling main power.

Zero

The numerical value of zero in a bit, field, or register, of appropriate width for that bit, field, or register.

Reference Documents

PCI

PCI-3.0

[PCI Local Bus Specification, Revision 3.0](#)

PCIe

PCIe-6.4

[PCI Express Base Specification, Revision 6.4](#)

PCIe-6.3

[CI Express Base Specification, Revision 6.3](#)

PCIe-6.2

[PCI Express Base Specification, Revision 6.2](#)

PCIe-6.1

[PCI Express Base Specification, Revision 6.1](#)

PCIe-6.0.1

[PCI Express Base Specification, Revision 6.0.1](#)

PCIe-6.0

[PCI Express Base Specification, Revision 6.0](#)

PCIe-5.0

[PCI Express Base Specification, Revision 5.0](#)

PCIe-4.0

[PCI Express Base Specification, Revision 4.0](#)

PCIe-3.1

PCIe-3.1a

[PCI Express Base Specification, Revision 3.1a](#)

PCIe-3.0

[PCI Express Base Specification, Revision 3.0](#)

PCIe-2.1

[PCI Express Base Specification, Revision 2.1](#)

PCIe-2.0

[PCI Express Base Specification, Revision 2.0](#)

PCIe-1.1

[PCI Express Base Specification, Revision 1.1](#)

PCIe-1.0

PCIe-1.0a

[PCI Express Base Specification, Revision 1.0a](#)

[PCI Express Card Electromechanical Specification, Revision 6.0 – Work in Progress](#)

CEM**CEM-5.0****CEM-5.1**

PCI Express Card Electromechanical Specification, Revision 5.1 plus PCI Express CEM Specification, Revision 5.1 Errata

CEM-4.0

PCI Express Card Electromechanical Specification, Revision 4.0

CEM-3.0

PCI Express Card Electromechanical Specification, Revision 3.0

CEM-2.0

PCI Express Card Electromechanical Specification, Revision 2.0

PCIe-to-PCI-PCI-X-Bridge

PCI Express to PCI/PCI-X Bridge Specification, Revision 1.0

PCI-to-PCI-Bridge

PCI-to-PCI Bridge Architecture Specification Revision 1.2

PCI-X

PCI-X Addendum to the PCI Local Bus Specification, Revision 2.0a

Mini-Card

PCI Express Mini Card Electromechanical Specification, Revision 2.1

OCuLink

PCI Express OCuLink Specification, Revision 1.1

M.2

PCI Express M.2 Specification, Revision 4.0

MR-IOV

MR-IOV Specification, Revision 1.0

U.2**SFF-8639**

PCI Express SFF-8639 Module Specification, Revision 4.0, Version 1.0

SNIA SFF-TA-1009

SNIA SFF-TA-1009 Enterprise and Datacenter Standard Form Factor Pin and Signal Specification
<https://www.snia.org/sff>

Ext-Cabling

PCI Express External Cabling Specification, Revision 3.0a

ExpressModule

PCI Express ExpressModule Electromechanical Specification, Revision 1.0

PCI-Hot-Plug**PCI-Hot-Plug-1.1**

PCI Hot-Plug Specification, Revision 1.1

PCI-PM

PCI Bus Power Management Interface Specification, Revision 1.2

PCI-Code-and-IDPCI Code and ID Assignment Specification, Revision 1.11 (or later)**Firmware**PCI Firmware Specification, Revision 3.2**USB****USB-2.0**usb-2.0 Universal Serial Bus Specification, Revision 2.0 <https://www.usb.org>**ACPI**Advanced Configuration and Power Interface Specification, Revision 6.2**UEFI**Unified Extensible Firmware Interface (UEFI) Specification, Version 2.8**SMBIOS**System Management BIOS (SMBIOS) Reference Specification, Version 3.6.0 or later**EUI-48****EUI-64**Guidelines for Use of Extended Unique Identifier (EUI), Organizationally Unique Identifier (OUI), and Company ID (CID)**JEDEC-JESD22-C101**JEDEC JESD22-C101F: Field-Induced Charged-Device Model Test Method for Electrostatic Discharge Withstand Thresholds of Microelectronic Components**JEDEC-JEP155-JEP157**JEDEC JEP155: Recommended ESD Target Levels for HBM/MM Qualification and JEP157 Recommended ESD-CDM Target Levels**ESDA-JEDEC-JS-001-2010**ESDA/JEDEC JS-001-2010: Joint JEDEC/ESDA Standard for Electrostatic Discharge Sensitivity Test – Human Body Model (HBM) – Component Level**ITU-T-Rec-X-667**ITU T-Rec. X.667: Information technology – Procedures for the operation of object identifier registration authorities: Generation of universally unique identifiers and their use in object identifiers**ISO-IEC-9834-8**ISO/IEC 9834-8: Information technology -- Procedures for the operation of object identifier registration authorities -- Part 8: Generation of universally unique identifiers (UUIDs) and their use in object identifiers**RFC-4122**IETF RFC-4122: A Universally Unique IDentifier (UUID) URN Namespace**DNS****RFC-1034**IETF RFC-1034: DOMAIN NAMES – CONCEPTS AND FACILITIES**PICMG**PICMG**PLUG-PLAY-ISA-1.0a**Plug and Play ISA Specification, Version 1.0a, May 5, 1994

PC-Card

PC-Card

MCTP-VDM**DSP0238**

Management Component Transport Protocol (MCTP) PCIe VDM Transport Binding Specification – <https://www.dmtf.org/dsp/DSP0238>.

SPDM**DSP0274**

DMTF Security Protocol & Data Model (SPDM) Specification – <https://www.dmtf.org/dsp/DSP0274>. IDE_KM requires SPDM Version 1.1 or above. TDISP requires version 1.2 or above.

SPDM-MCTP**DSP0275**

Security Protocol and Data Model (SPDM) over MCTP Binding Specification – <https://www.dmtf.org/dsp/DSP0275>.

AES-GCM

NIST Special Publication 800-38D Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC – <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38d.pdf>

Secured SPDM**DSP0277**

Secured Messages using SPDM Specification (IDE requires version 1.0 or above) – <https://www.dmtf.org/dsp/DSP0277>

Secured MCTP**DSP0276**

Secured Messages using SPDM over MCTP Binding Specification (version 1.0 or above) – <https://www.dmtf.org/dsp/DSP0276>

CXL**CXL-3.0**

Compute Express Link Specification – <https://www.computeexpresslink.org>

SMBus**SMBus Version 3.2**

SMBus Specification – <http://smbus.org/specs/>

I2C

I2C Specification – <https://www.nxp.com/docs/en/user-guide/UM10204.pdf>

I3C**I3C-Basic****I3C-Basic-1.1.1**

MIPI I3C Basic Specification – <https://resources.mipi.org/mipi-i3c-basic-download>

I3C-DCR

I3C Device Characteristic Register – https://www.mipi.org/mipi_i3c_device_characteristics_register

NVMe**NVM-Express**

NVM Express Specification – <https://nvmexpress.org/specifications>

NVMe-MI

NVM Express Management Interface Specification – <https://nvmexpress.org/specification/nvme-mi-specification>

PLDM-Firmware-Update**DSP0267**

PLDM for Firmware Update Specification – <https://www.dmtf.org/dsp/DSP0267>), or

PLDM-Platform-Monitoring-Control**DSP0248**

PLDM for Platform Monitoring and Control Specification – <https://www.dmtf.org/dsp/DSP0248>

MMBI**DSP0282**

Memory-Mapped BMC Interface (MMBI) Specification – <https://www.dmtf.org/dsp/DSP0282>

DMTF-DSP0233

MCTP I3C Binding Specification <https://www.dmtf.org/dsp/DSP0233>

DMTF-DSP0236

MCTP Base Specification <https://www.dmtf.org/dsp/DSP0236>

DMTF-DSP0237

MCTP SMBus/I2C Binding Specification <https://www.dmtf.org/dsp/DSP0237>

DMTF-DSP0256

MCTP Host Interface Specification <https://www.dmtf.org/dsp/DSP0256>

DMTF-DSP0283

MCTP USB Binding Specification <https://www.dmtf.org/dsp/DSP0283>

DMTF-DSP0284

MCTP MMBI Binding Specification <https://www.dmtf.org/dsp/DSP0284>

DMTF-DSP0291

PCIe-MI over MCTP Binding Specification <https://www.dmtf.org/dsp/DSP0291>

DMTF-DSP2015**Platform Management Communications Infrastructure (PMCI) Architecture Whitepaper <https://www.dmtf.org/dsp/DSP2015>****IPMI-FRU**

IPMI Platform Management FRU Information Storage Definition – <https://www.intel.com/content/dam/www/public/us/en/documents/specification-updates/ipmi-platform-mgt-fru-info-storage-def-v1-0-rev-1-3-spec-update.pdf>

CopprLink™**CopprLink-Internal™**

CopprLink Internal Cable Specification for PCI Express 5.0 and 6.0 – Work in Progress: Draft 0.9 is <https://members.pcisig.com/wg/PCI-SIG/document/20254>

Open Compute Project

The OCP Community forges new technology norms, fostering an ecosystem where industry players collaborate in a safe framework, shaping a versatile and diverse supply chain. See <https://www.opencompute.org>

OCP Attestation of System Components v1.0 Requirements and Recommendations

<https://www.opencompute.org/documents/attestation-v1-0-20201104-pdf>

OCP Server/MHS

OCP DC/MHS

Open Compute Project Server Modular Hardware System (MHS) Subproject <https://www.opencompute.org/wiki/Server/DC-MHS>

TCG Reference Integrity Manifest (RIM) Information Model

<https://trustedcomputinggroup.org/resource/tcg-reference-integrity-manifest-rim-information-model>